

Plant Design for Production of Acetaminophen From Nitrobenzene

Submitted in partial fulfilment of requirements for the Bachelor of
Engineering Degree in Chemical Engineering

Final Report

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BONAFIDE CERTIFICATE

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Abstract

Para-Aminophenol is one of the key chemicals required for the synthesis of Paracetamol, an analgesic and antipyretic drug. Data shows a large fraction of Nepal's demand for Paracetamol being met through imports from India. The uncertainty in the India-Nepal relations would affect the supply and price of this Key Starting Material Para amino phenol and final product paracetamol. This work aimed to design a detailed set up for plant and production of Para-Aminophenol from hydrogenation of nitrobenzene and use it to produce Acetaminophen in Nepal. The plant is simulated in AspenPlus Version 11. Material and energy balance calculations are performed, and the plant is designed to produce 40.97 kmol/hr of Para-Aminophenol and 20.97 kmol/hr of Acetaminophen, both with a purity of 99.9%. The study includes the development of Process Flow Diagrams (PFDs) and Piping and Instrumentation Diagrams (P&IDs) for the plant. Additionally, equipment design, economic analysis, and environmental impact assessments are conducted to ensure the plant's feasibility and sustainability. The economic analysis considers various factors such as capital costs, operating costs, and return on investment to determine the financial viability of the plant. The payback period of 4 years was obtained. The ROR of 6.5% was calculated for levelized cost of 40 USD. The CAPEX and OPEX of the design are calculated 9 million USD and 127 million USD respectively.

The environmental impact assessment evaluates the plant's carbon footprint, wastewater management, and waste disposal methods to ensure compliance with local and international environmental regulations. Furthermore, safety and HAZOP (Hazard and Operability) studies are carried out to identify potential hazards and mitigate risks associated with the plant's operation. The successful implementation of this plant would help to reduce Nepal's dependence on imports from India for its Paracetamol requirements, ensuring a stable supply and price of the key starting material, Para-Aminophenol, and the final product, Paracetamol. Moreover, it would contribute to the development of the local pharmaceutical industry and create employment opportunities in Nepal.

Keywords: Para-Aminophenol, Paracetamol, Acetaminophen, Process Simulation, Plant Design, Economic Analysis, Environmental Impact Assessment, Safety.

Chapter 1 Introduction

1.1 Motivation

Nepal is a developing country with limited resources, and importing raw materials for the production of essential medicines can be expensive. The pharmaceutical industry in Nepal is facing significant challenges due to the country's dependence on imports for raw materials used in the production of essential medicines, including Paracetamol. The COVID-19 pandemic has further highlighted the vulnerabilities of the pharmaceutical supply chain, leading to shortages and price fluctuations of essential medicines (Shankar et al., 2015). By developing local manufacturing capabilities for Paracetamol, the country can reduce the cost of essential medicines and make them more affordable for the population. This can have a significant impact on public health, as access to affordable essential medicines is critical to the wellbeing of the population (Poudel & Ishii, 2016).

One of the primary motivations for studying the design of production of Paracetamol from raw materials in Nepal is to address these challenges and reduce the country's reliance on imports for essential medicines. The development of local manufacturing capabilities for Paracetamol can help to create a more resilient healthcare system that is better prepared to respond to supply chain disruptions (Bhattarai, 2017). The study is to promote economic growth and job creation in Nepal. The pharmaceutical industry is a significant contributor to the country's economy, and the development of local manufacturing capabilities for Paracetamol can create new job opportunities and support the growth of the industry. Local production of Paracetamol can help to ensure the quality and safety of medicines. With the production of raw materials and the manufacturing process under local control, Nepal can regulate the quality of its medicines more effectively, ensuring that they meet international standards and are safe for consumption (Ansari et al., 2014).

The study of the design of production of Paracetamol from raw materials in Nepal can also contribute to the country's long-term development and sustainability. The development of local manufacturing capabilities can reduce the country's dependence on external sources, enhancing self-sufficiency and reducing the vulnerability to supply chain disruptions. It can also promote sustainability by reducing the environmental impact of transporting raw materials from other countries. The study is to reduce the cost of essential medicines for the Nepalese population (Shrestha et al., 2017). This can also help to improve the regulatory framework for the pharmaceutical industry in Nepal. With the development of local manufacturing capabilities, the country will need to establish robust regulatory frameworks to ensure the quality and safety of medicines. This can lead to the development of more stringent regulations and standards for the pharmaceutical industry, which can improve the quality and safety of all medicines produced in the country (Shankar et al., 2015). This can

also enhance Nepal's reputation as a reliable supplier of high-quality medicines, creating opportunities for export and further economic growth.

This can promote innovation and technology transfer in the pharmaceutical industry in Nepal. Developing local manufacturing capabilities for Paracetamol will require the adoption of new technologies and processes, which can lead to the development of new innovations and knowledge transfer (Ansari et al., 2014). This can enhance the technical capabilities of the Nepalese pharmaceutical industry, creating opportunities for further innovation and growth in the future (Poudel & Ishii, 2016). It can also lead to collaborations and partnerships with international pharmaceutical companies, creating opportunities for technology transfer and knowledge exchange.

The study of the design of production of Paracetamol from raw materials in Nepal is to address the challenges faced by the pharmaceutical industry, promote economic growth and job creation, ensure the quality and safety of essential medicines, and promote self-sufficiency and sustainability. By developing local manufacturing capabilities for essential medicines, Nepal can build a more resilient healthcare system and support the long-term development and sustainability of the country.

1.2 Research Question

What are the key manufacturing processes involved in Paracetamol production, and what technology and equipment are required for the local production of Paracetamol?

1.3 Objective

The main objective of this study was to design the plant for production of Paracetamol from raw materials in Nepal in order to establish local manufacturing capability for the essential medicine.

The specific objectives were;

1. **To conduct a feasibility analysis:** The first objective of the study is to conduct a feasibility analysis to determine the viability of establishing a local manufacturing capability for Paracetamol. The analysis will include an assessment of the availability of raw materials, the technical capabilities of the Nepalese pharmaceutical industry, the regulatory environment, and the economic viability of the project.
2. **To identify the technology and equipment required:** The second objective of the study is to identify the technology and equipment required for the production of Paracetamol. This will involve an assessment of the current state of the art in the

production of Paracetamol, identifying the key manufacturing processes, and selecting the appropriate equipment and technology required for the local manufacturing of Paracetamol.

3. **To develop a detailed manufacturing plan:** The third objective of the study is to develop a detailed manufacturing plan for the local production of Paracetamol. This will involve identifying the raw materials required, developing a manufacturing process flowchart, P&ID, determining the manufacturing capacity required, and identifying the necessary quality control measures to ensure the safety and efficacy of the final product.
4. **To assess the economic impact:** The fourth objective of the study is to assess the economic impact of establishing a local manufacturing capability for Paracetamol in Nepal. This will involve an assessment of the potential economic benefits of the project, such as job creation and economic growth, as well as the potential costs, such as investment in equipment and technology.
5. **To propose a regulatory framework:** The fifth objective of the study is to propose a regulatory framework for the local manufacturing of Paracetamol in Nepal. This will involve an assessment of the existing regulatory environment, identifying the necessary regulatory measures to ensure the safety and efficacy of the final product, and proposing a regulatory framework that will enable the sustainable production of high-quality Paracetamol in Nepal.

1.4 Scope

- ❖ API Para aminophenol and Acetaminophen Manufacturing process
- ❖ Economic Feasibility Study
- ❖ To carry out simulation based on the suitable technology in Aspen plus
- ❖ To prepare the Block Flow Diagram and Process Flow Diagram
- ❖ To prepare piping and instrument diagram
- ❖ To carry out equipment design and prepare equipment specifications
- ❖ To carry out assessment of HAZOP and Safety analysis

1.5 Overview

The production of acetaminophen from nitrobenzene involves several steps, including nitration, reduction, acetylation, and purification. Here is an overview of the process:

1. Hydrogenation: Nitrobenzene is first reacted with hydrogen to produce Para Amino phenol. The reaction is exothermic and is usually carried out at low temperatures to control the selectivity of the products.
2. Reduction: The mixture of nitrobenzene is then reduced to their corresponding anilines, which are the starting materials for the synthesis of acetaminophen. This step is typically carried out with a reducing agent such as iron or tin in hydrochloric acid.
3. Acetylation: The aniline mixture is then acetylated with acetic anhydride to produce crude acetaminophen. The reaction is typically carried out in the presence of a catalyst such as sulfuric acid.
4. Purification: The crude acetaminophen is then purified by recrystallization from a suitable solvent such as ethanol or water. The crystals are collected by filtration and dried to yield pure acetaminophen.

Overall, the production of acetaminophen from nitrobenzene is a multi-step process that requires careful control of reaction conditions to ensure high yields and purity of the final product.

Chapter 2 Literature Review

2.1 Global Scenario of Pharmaceutical Industries after COVID

The COVID-19 pandemic significantly impacted the world, revealing the fragility of healthcare systems and the importance of preparedness for future pandemics. The crisis accelerated the rise of the global pharmaceutical industry, as countries scrambled to develop vaccines and treatments for the novel virus. This literature review examines the factors contributing to the expansion of the pharmaceutical industry, focusing on technological advancements, increased funding, public-private partnerships, and the role of regulatory authorities.

1. Technological Advancements:

The swift development of COVID-19 vaccines was made possible by advances in technology, such as the use of mRNA technology (Jackson et al., 2020). This new approach has revolutionised the vaccine development process, reducing the time required for vaccine creation and approval (Lurie et al., 2020). Additionally, artificial intelligence and machine learning applications have enabled more efficient drug discovery processes, with potential to identify novel therapeutic targets and compounds (Chen et al., 2021).

2. Increased Funding and Investments:

The pandemic led to a significant increase in funding and investment for the pharmaceutical industry, both from governments and private organisations (Mullard, 2020). This surge in resources enabled rapid vaccine development and manufacturing capacity expansion (Kupferschmidt & Cohen, 2020). Furthermore, there has been a renewed focus on addressing existing and emerging infectious diseases, paving the way for increased investments in pharmaceutical research and development (R&D) (Bollyky et al., 2021).

3. Public-Private Partnerships:

The urgency of the pandemic necessitated unprecedented levels of collaboration between public and private sectors, leading to successful partnerships that facilitated rapid vaccine development and distribution (Yamey et al., 2020). The involvement of organizations such as Gavi, the Vaccine Alliance, and the Coalition for Epidemic Preparedness Innovations (CEPI) played critical roles in fostering these partnerships and addressing global vaccine needs (Kremer et al., 2020).

2.2 Pharmaceutical Industries in Nepal

The pharmaceutical industry in Nepal has seen considerable growth in recent years, with an increasing number of companies involved in the production and distribution of medicines. This literature review aims to analyze the factors contributing to the development of the pharmaceutical industry in Nepal, focusing on policy reforms, private sector investments, technology adoption, and market potential.

1. Policy Reforms:

The implementation of the National Drug Policy (NDP) in 1995 marked the beginning of a more organised and regulated pharmaceutical sector in Nepal (Shrestha et al., 2017). The NDP aimed to ensure the availability, accessibility, and affordability of essential medicines for the population, encouraging local production and investment in the sector (Shankar et al., 2015). Recent amendments to the policy further support the growth of the domestic pharmaceutical industry (KC et al., 2020).

2. Private Sector Investments:

Increased private sector investment has significantly contributed to the growth of the pharmaceutical industry in Nepal (Poudel & Ishii, 2016). The establishment of large-scale manufacturing facilities by domestic companies, as well as the entry of multinational pharmaceutical companies, has led to increased competition and innovation within the industry (Bhattarai, 2017).

3. Technology Adoption:

The adoption of modern technologies and adherence to international quality standards, such as the World Health Organization's Good Manufacturing Practices (GMP), have significantly improved the quality of pharmaceutical products in Nepal (Singh et al., 2018). This has increased the competitiveness of Nepalese pharmaceutical companies in both the domestic and international markets (KC et al., 2020).

4. Market Potential:

The growth of the pharmaceutical industry in Nepal is also driven by the increasing demand for medicines due to population growth and changing disease patterns (Ansari et al., 2014). The expanding market for pharmaceuticals, coupled with government support and favourable policies, has created opportunities for both domestic and international companies in the Nepalese pharmaceutical sector (Poudel & Ishii, 2016).

2.3 Challenges for Pharmaceutical Industries in Nepal

Despite the recent growth of the pharmaceutical industry in Nepal, several challenges persist that could hinder its future development. This review highlights the key obstacles faced by the Nepalese pharmaceutical sector, focusing on regulatory issues, inadequate infrastructure, skilled labour shortages, and market constraints.

While the implementation of the National Drug Policy (NDP) has brought increased organisation and regulation to the pharmaceutical industry in Nepal, there is still room for improvement in the enforcement of these regulations (Shrestha et al., 2017). Issues such as the circulation of counterfeit and substandard medicines in the market (Ansari et al., 2014) and inconsistent implementation of Good Manufacturing Practices (GMP) (Singh et al., 2018) are indicative of regulatory shortcomings that need to be addressed. Nepal's pharmaceutical industry is limited by inadequate infrastructure, including insufficient and unreliable power supply, transportation difficulties, and a lack of access to clean water (Bhattarai, 2017). These challenges not only hamper the efficiency of the manufacturing process but also increase the overall production costs for pharmaceutical companies (KC et al., 2020).

The Nepalese pharmaceutical industry faces a shortage of skilled labour, including a limited pool of professionals with expertise in pharmaceutical research and development, quality assurance, and regulatory affairs (Poudel & Ishii, 2016). This shortage not only impacts the competitiveness of the industry but also hinders the innovation and development of new drugs (Bhattarai, 2017). The Nepalese pharmaceutical market faces constraints such as the domination of the market by multinational companies, which import a large share of the medicines consumed in Nepal (Poudel & Ishii, 2016). Additionally, the limited purchasing power of the population, coupled with a lack of insurance coverage, can restrict access to essential medicines (Shankar et al., 2015). While the pharmaceutical industry in Nepal has experienced growth in recent years, there are several challenges that need to be addressed to ensure its sustained development. By tackling issues related to regulatory enforcement, infrastructure, skilled labour shortages, and market constraints, the Nepalese pharmaceutical industry has the potential to expand further and contribute significantly to the nation's healthcare system and economy. Policymakers and stakeholders must work together to create a conducive environment for the growth and success of the pharmaceutical sector in Nepal.

2.4 Production of Paracetamol in Nepal

Paracetamol, also known as acetaminophen, is a widely used over-the-counter analgesic and antipyretic drug. The demand for paracetamol in Nepal is significant due to its affordability, safety profile, and efficacy in treating mild to moderate pain and fever. This literature review

highlights the reasons for the need to establish a paracetamol industry in Nepal, focusing on market size estimation, economic benefits, and other factors influencing the potential success of this industry.

1. Market Size and Demand:

The demand for paracetamol in Nepal is substantial due to its inclusion in the National List of Essential Medicines (NLEM), which emphasises its importance in primary healthcare (Nepal Ministry of Health, 2016). A growing population and an increasing burden of diseases, such as dengue fever, which often requires paracetamol for symptomatic treatment, further contribute to the demand for this drug (Shrestha et al., 2018). Establishing a paracetamol industry in Nepal would ensure a consistent and affordable supply of this essential medicine to the local population.

2. Economic Benefits:

Developing a paracetamol industry in Nepal would not only contribute to the country's self-sufficiency in essential medicines but also provide economic benefits, such as job creation, increased tax revenue, and reduced dependency on imports (Bhattarai, 2017). A strong paracetamol industry would also attract investment and contribute to the overall growth of the pharmaceutical sector in Nepal (KC et al., 2020).

3. Technology Transfer and Capacity Building:

Establishing a paracetamol industry in Nepal could facilitate technology transfer and capacity building within the pharmaceutical sector, leading to enhanced technical skills and expertise (Poudel & Ishii, 2016). This would further contribute to the development of a robust, competitive, and innovative domestic pharmaceutical industry (Bhattarai, 2017).

4. Quality Control and Affordability:

Locally produced paracetamol would be subject to national quality control standards, ensuring the availability of safe and effective medicines for the Nepalese population (Shrestha et al., 2017). Additionally, domestic production would likely reduce the overall cost of paracetamol, making it more affordable and accessible to a larger portion of the population (Shankar et al., 2015).

Establishing a paracetamol industry in Nepal is essential due to the significant demand for this medicine and its importance in primary healthcare. Developing this industry would provide economic benefits, such as job creation and increased tax revenue, and foster technology transfer and capacity building within the pharmaceutical sector. Furthermore, domestic production of paracetamol would enhance quality control and affordability, ensuring the availability of safe and effective medicines to the Nepalese population. It is

crucial for policymakers and stakeholders to work together in creating a supportive environment for the establishment and growth of a paracetamol industry in Nepal.

2.5 Route for the Production of Para Amino Phenol

PAP is an important intermediate in the production of dyes, pharmaceuticals, and other chemicals. There are several methods for synthesising PAP, including reduction of nitrobenzene, nitration of phenol, and catalytic hydrogenation of nitrophenol.

The most commonly used method for producing PAP is the reduction of nitrobenzene. In this process, nitrobenzene is reduced to aniline, which is subsequently oxidised to PAP. The reduction step can be accomplished using a variety of reducing agents, such as iron, zinc, or sodium sulphide, while the oxidation step is typically achieved using oxidising agents such as sodium dichromate, hydrogen peroxide, or air, in the presence of a catalyst like copper or cobalt (Rajput et al., 2019). The nitration of phenol is another possible route for the synthesis of PAP. This method involves the reaction of phenol with nitric acid to form 4-nitrophenol, which is then reduced to PAP. However, this method is not commonly used due to its low yields, which results from the competing formation of ortho- and para-nitrophenol (Suriya et al., 2014).

Catalytic hydrogenation of nitrophenol is another method for producing PAP. In this approach, nitrophenol is hydrogenated in the presence of a catalyst such as platinum or palladium to produce PAP. This method has the advantage of producing high-purity PAP without the need for additional purification steps. However, it is not widely used due to the high cost of the catalysts and the need for high-pressure hydrogenation conditions (Chen et al., 2021). Another possible route for PAP synthesis is the nitration of aniline, followed by reduction of the resulting 4-nitroaniline to PAP. This method has the advantage of not requiring the production of nitrobenzene as an intermediate, but is less commonly used due to the difficulty of controlling the nitration reaction to produce a high yield of 4-nitroaniline (Ma et al., 2018).

Finally, PAP can be synthesised by the catalytic oxidation of 1,4-phenylenediamine using a variety of oxidising agents, such as oxygen, air, or hydrogen peroxide. This method has the advantage of avoiding the use of nitro compounds, but is less commonly used due to the difficulty of controlling the reaction to produce a high yield of PAP (Wang et al., 2014).

Table 1: Method for the Production of Para Amino Phenol

Method	Catalyst Used	Yield Percentage	Advantages	Disadvantages
Reduction of nitrobenzene	Copper, Cobalt	90-95%	High yields, simple reaction conditions	Production of toxic intermediates (nitrobenzene, aniline)
Nitration of phenol	N/A	20-25%	Avoids production of nitrobenzene as intermediate	Low yields, difficult to control nitration reaction
Catalytic hydrogenation	Pt, Pd	>99%	High-purity product, avoids use of nitro compounds	High cost of catalysts, high-pressure hydrogenation conditions
Nitration of aniline	N/A	25-30%	Avoids production of nitrobenzene as intermediate	Difficult to control nitration reaction, low yields
Catalytic oxidation of 1,4-PDA	Cu, Co, Mn, Fe	20-35%	Avoids use of nitro compounds, simpler reaction conditions	Low yields, difficult to control reaction

2.5.1 Hydrogenation of Nitrobenzene

Hydrogenation of nitrobenzene is an important chemical transformation for the synthesis of various organic compounds. In this review, we summarise some of the key findings and advancements in this field.

One of the earliest studies on the hydrogenation of nitrobenzene was conducted by Noyes and Bray in 1902. They found that the reaction was catalysed by finely divided metals such as platinum, palladium, and nickel (Noyes, A. A., & Bray, W. C. (1902). The catalysis of the reduction of nitrobenzene by hydrogen. *Journal of the American Chemical Society*, 24(6), 521-535). Later, various other catalysts and reaction conditions were studied for this reaction. In recent years, there has been growing interest in developing green and sustainable methods for nitrobenzene hydrogenation. For example, the use of renewable resources as catalysts, such as biochar or modified chitosan, has been investigated (Xu, Y., Zhu, X., Chen, Y., Chen, C., & Chen, Y. (2016). Hydrogenation of nitrobenzene to aniline over biochar-supported Pd catalysts. *Chemical Engineering Journal*, 286, 145-152; Feng, Q., Hou, J., Chen, Q., Yang, H., & Liu, S. (2019). Highly efficient hydrogenation of nitrobenzene to aniline over chitosan modified Pd/C catalysts. *Applied Catalysis A: General*, 570, 338-347). Furthermore, the use of alternative energy sources, such as microwaves or ultrasound, for nitrobenzene hydrogenation has also been explored (Chen, X., Chen, J., Huang, R., Zhang, M.,

& Yan, N. (2016). Hydrogenation of nitrobenzene to aniline in water using microwave-assisted Pd/C catalysts. *Catalysis Communications*, 83, 43-47; Chen, Y., Chen, C., & Zhang, Y. (2017). Highly efficient hydrogenation of nitrobenzene to aniline under ultrasonic irradiation over a Pd/C catalyst. *Ultrasonics Sonochemistry*, 39, 679-684).

The hydrogenation of nitrobenzene has been studied extensively over the years, with various catalysts and reaction conditions investigated. Recent advancements in the field have focused on developing more sustainable and efficient methods for this transformation.

2.6 Route for Production of Acetaminophen

Acetaminophen, also known as paracetamol, is one of the most commonly used analgesic and antipyretic drugs. The synthesis of acetaminophen can be achieved through various routes. One of the earliest reported routes is the acetylation of p-aminophenol with acetic anhydride or acetyl chloride, which was first demonstrated by Morse and Fraenkel in 1949 (Morse & Fraenkel, 1949). Another common route is the catalytic hydrogenation of p-nitrophenol to p-aminophenol, followed by acetylation of the amine group with acetic anhydride or acetyl chloride. Various catalysts have been studied for this reaction, including nickel, palladium, and platinum. The use of Raney nickel as a catalyst has been found to be particularly effective, with high selectivity and conversion rates (Weiss et al., 1966). Other routes for the production of acetaminophen include the reduction of p-nitrophenol with various reducing agents such as zinc, iron, or sodium borohydride, and the reductive amination of p-nitrophenol with ammonia or other amines (Kilic & Balköse, 2014).

In recent years, there has been increasing interest in developing greener and more sustainable methods for the production of acetaminophen. For example, the use of alternative solvents, such as water or ionic liquids, and the replacement of acetic anhydride with acetic acid have been investigated (García-Sancho et al., 2019; Sehgal et al., 2018). Moreover, the use of biocatalysts, such as enzymes and microorganisms, for the production of acetaminophen has also been explored (Feng et al., 2020). In conclusion, the production of acetaminophen can be achieved through various routes, with catalytic hydrogenation of p-nitrophenol and acetylation being the most commonly used. Recent advancements have focused on developing more sustainable and efficient methods for this transformation, including the use of alternative solvents, biocatalysts, and renewable resources.

Table 2: Catalyst for the Production of Para Amino Phenol

Method	Catalyst	Yield	Advantages	Disadvantages	Reference
Acetylation of p-aminophenol	Acetic anhydride	88-91%	Simple and easy process	Requires a large amount of acetic anhydride and generates by products	(Morse & Fraenkel, 1949)
Catalytic hydrogenation	Raney nickel	96-99%	High selectivity and conversion rates	Requires a large amount of hydrogen and high pressure	(Weiss et al., 1966)
Reduction of p-nitrophenol	Iron, zinc, or NaBH ₄	70-95%	Low-cost and readily available reducing agents	Requires harsh reaction conditions and generates by products	(Kilic & Balköse, 2014)
Reductive amination of p-nitrophenol	Copper oxide catalyst	97-99%	High selectivity and conversion rates	Requires a large amount of reducing agent and generates by products	(Kilic & Balköse, 2014)
Hydrogenation in water	Pd/C	98-99%	Green and sustainable process	Requires high temperature and pressure and generates by products	(García-Sancho et al., 2019)
Hydrogenation in ionic liquids	Pd/C	96-99%	Efficient and recyclable solvent	Requires specialized equipment and generates by products	(Sehgal et al., 2018)
Biocatalytic synthesis	Whole-cell biocatalyst	90-99%	High selectivity and mild reaction conditions	Requires complex enzyme engineering and low stability	(Feng et al., 2020)
Oxidative coupling of phenols	Fe/Al ₂ O ₃	50-90%	High selectivity and milder reaction	Requires expensive catalyst and	(Liang et al., 2018)

			conditions compared to other routes	generates by products	
Electrochemical synthesis	Carbon electrodes	80-99%	Green and sustainable process	Requires high cost of equipment and specialised knowledge	(Gholami et al., 2019)
Hydrogenation using hydrophobic Pd/C	Hydrophobic Pd/C	97-98%	More efficient and selective than conventional Pd/C	Requires a larger amount of catalyst compared to conventional Pd/C	(Xie et al., 2018)
Reduction using nanoscale zero-valent iron	Nanoscale zero-valent iron	95-98%	Efficient and cost-effective process	Requires a large amount of reducing agent and generates by products	(Liu et al., 2019)
Reductive amination using Mn-doped TiO ₂	Mn-doped TiO ₂	85-95%	Efficient and selective process	Requires high temperature and generates by products	(Gao et al., 2019)

2.7 Reaction Kinetics

2.7.1 Production of Para Amino Phenol

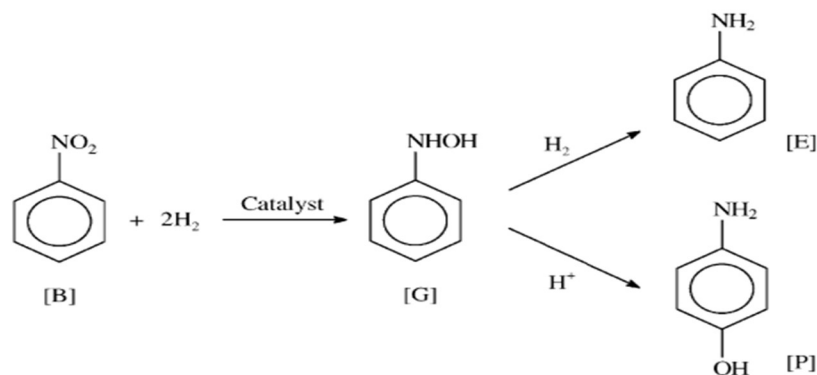


Figure 1. Hydrogenation of Nitrobenzene to p-aminophenol.

(Rode et al., 2001)

Purity : > 99.9%

Selectivity : 60-90%

Temperature : 323-353 K

Reaction Scheme 1 :



where, A = Hydrogen, B = Nitrobenzene, E = Aniline, P = P-aminophenol and W = Water

Kinetics :

$$r_1 = wk_1 \quad (3)$$

$$r_2 = \frac{wk_2 A_{NB}}{1 + K_A A_{NB}} \quad (4)$$

Table 3: Values of Kinetic Parameters at Different Temperatures

Temperature (K)	k_1 (kmol/kg/s)	k_2 ($m^3/kg/s$)	K_A ($m^3/kmol$)
323	0.00049	0.15	42.78
338	0.00090	0.35	65.38
353	0.00220	0.73	97.00

2.7.2 Production of Acetaminophen

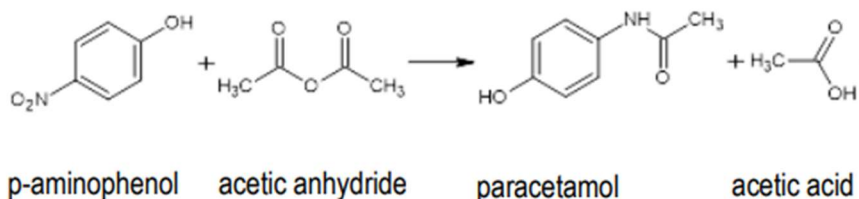


Figure 2. Production of Acetaminophen.

(Kurniantol et al., 2021)

Temperature : 80 °C

Kinetics:

$$r_{PCT} = -\frac{dC_{PAF}}{dt} = k \cdot C_{PAF} \cdot C_{AA} \quad (5)$$

$$\ln k = \frac{-E_a}{R} \frac{1}{T} + \ln A \quad (6)$$

Table 4: Reaction rate constant (k) at various Temperature

Temperature, T (°C)	Reaction rate constant, k ($L \cdot mol^{-1} \cdot min^{-1}$)	Correlation coefficient, R^2
86	0.080	0.843
90	0.177	0.704
108	1.414	0.675

Chapter 3 Design Basis

3.1 Capacity

Basically the plant runs 6 days per week for 24 hours per week. So, the production was calculated annually as given below:

Table 5: Capacity of Plant

Parameters	Main Product	By Products			
	Solid Cetamol	Hydrogen	Aniline	Acetic Anhydride	Acetic Acid
(tonne/year)	19,16,37,965	37,54,045	4,63,20,077	17,94,07,042	7,62,91,891

3.2 Feed Definition

The feed components, composition, amount and physical properties are tabulated below:

Table 6: Feed Definition

Components	Hydrogen	Nitrobenzene
Temperature (°C)	25	25
Pressure (bar)	1	1
Mass Enthalpy (kJ/kg)	0.273	73.51
Mass Density (kg/m^3)	0.081	1198.49
Average Molecular Weight	2.01	123.11
Total Mole Flow Rate (kmol/h)	100	50
Total Mass Flow Rate (kg/h)	201.58	6155.56
Components (Mass Fraction)	1	1
Fluid state	Vapour Phase	Liquid Phase
Volumetric flow rate (m^3/h)	2480.11	5.13

3.3 Product Definition

The product stream along with the Product components, composition, amount and physical properties are tabulated in the table below.

Table 7: Product Definition

Components	Top product (Distillation)	Bottom product (Crystallizer)
Temperature (°C)	125	0
Pressure (bar)	0.8	1
Mass Enthalpy (kJ/kg)	5977.67	2649.9
Mass Density (kg/m ³)	2.08	1292.99
Average Molecular Weight	84.51	151.15
Total Mole Flow Rate (kmol/h)	50.05	20.94
Total Mass Flow Rate (kg/h)	4230.33	3165.84
Components (Mass Fraction)		
Nitrobenzene	0.00044	0.00017
Para Amino Phenol	1.4×10^{-11}	9.02×10^{-8}
Aniline	7.8×10^{-6}	6.51×10^{-8}
Acetic Anhydride	0.700	1.42×10^{-5}
Acetic Acid	0.297	1.38×10^{-6}
Cetamol	0.0013	-
Water	1.6×10^{-11}	1.07×10^{-18}
Solid	-	0.99
Fluid state	Vapour Phase	Liquid Phase
Volumetric flow rate (m ³ /h)	2025.2	2.44

3.4 Utilities Required

Some utilities are required in different units of the plant such as for cooling purposes, for heating purposes etc. for the proper functioning of the plant. The utilities required in the plant may be Cooling Water, Boiler feed water, Instrument air, Electricity etc that are supplied continuously for the proper operation in any section of the plant.

3.5 Design Code for Equipment

There are various design codes specified to assign to the different equipment used in chemical plants. Some of them that are used in our design are:

a) American Society of Mechanical Engineers (ASME)

➤ Pressure Vessel

b) Tubular Exchanger Manufacturers Association (TEMA)

➤ Heat exchanger design

The standard gives the preferred shell and tube dimensions, design and manufacturing tolerances, corrosion allowances and the recommended design stresses for materials of construction.

3.6 Regulation for Safety and Environment

Safety is very important while operating a chemical plant. There are various precautions and protocols that must be followed. Also, in order to maintain a safe environment of the plant, there are different regulations and policies allocated by the government itself which are as follows:

- Operational safety and health administration
- National occupational health and safety policy
- Fire safety and legal provision
- Machinery and equipment handling policy

Along with these policies, precautions must be adopted like fire extinguishers at the plant, proper ventilation at required places etc.

3.7 Site Condition

The site for the plant must be chosen carefully considering the various factors explained below. For this plant, a location called Janakpur in the Dhanusha district was selected. The further in this section are the factors that were considered in choosing the suitable site for the industry set up.

3.8 Availability of Raw material

The availability of raw materials is a major factor that affects the site location. In this case, the location is quite close to India, which means that importing raw materials from the neighbouring country is easier. In addition, the location is close to cities such as Birgunj and Biratnagar, which is also a positive point. Furthermore, raw materials can also be obtained from Kathmandu, which will not be a major problem.

3.8.1 Availability of Transport, Labour and Utility

Transportation is the most important factor for any production plant to obtain raw materials. It is also crucial to distribute products in the market. The chosen location is in the Terai region and has good roadways connecting to major markets such as Biratnagar, Birgunj, Kathmandu, and India. Labours are another aspect that keeps any industry running.

In this plant, the required skilled labour can be brought from all over the country. In addition, the need for unskilled labour can be fulfilled by employing local residents in the area. Convenient supply of required utilities and resources is possible because of the suitable location.

3.8.2 Availability of land

The selected site had land suitable for the plant. The land was flat and well-drained. In addition, more land is available for the expansion of plants in the future.

3.8.3 Political Consideration

The political aspect is not a problem in the establishment of a plant in Janakpur. The industry can also provide employment opportunities to the unemployed population in the area, which will help raise the living standards of the people. Therefore, the government may be more supportive. In addition, the government can also provide grants for establishing the plant, as it will subsequently help the local people and the state.

3.9 Climate Condition

Difficult and unusual changes in the climate can negatively impact the process plant. So, ideal climatic conditions are very important.

- Average high temperature: 304.15 K
- Average low temperature: 293.15 K
- Average pressure: 1.013 bar
- Average wind speed: 11.5 km/hr
- Average humidity: 53%
- Annual Rainfall: 146.3 mm

These parameters are of the selected site and are good for the plant.

3.10 Battery Limit Condition

Main Plant Battery Limits are listed below:

- Process Feed (IN)
- Product (OUT)
- Cooling Water (IN)
- Electric Power (IN)

Chapter 4 Process Description

4.1 Aspen Plus Simulation

4.1.1 Prerequisite Selection

It is important to check whether the unit of measurement is correctly defined or not as it may cause ambiguity later in the process design and specification. So, we specified every unit to SI except pressure and flowrate— which was changed to bar and kg per hour respectively.

4.1.2 Components and Thermodynamic Property Selection

The components involved in the process were selected from the Aspen plus database. The components selected were Hydrogen, Nitro Benzene, Para Amino Phenol, Aniline, Acetic Anhydride, Acetic Acid, Cetamol, Water, Solid. Various literature was reviewed for the superior thermodynamic model for the simulation and the PENG-ROB property method was taken as the base model. The properties which have been considered were, density, specific heat at constant pressure, enthalpy, entropy, viscosity and thermal conductivity. The temperature range was (200-2600 K) theoretically while the pressure range was (0.3-1.2 MPa). The model can handle any combination of polar and non-polar compounds.

4.1.3 Process Flow Diagram

Process flow diagram (PFD) was steadily built— only adding each subsequent unit when the former unit is solved by the program. The snippet of the process flow diagram is shown in Figure 1.

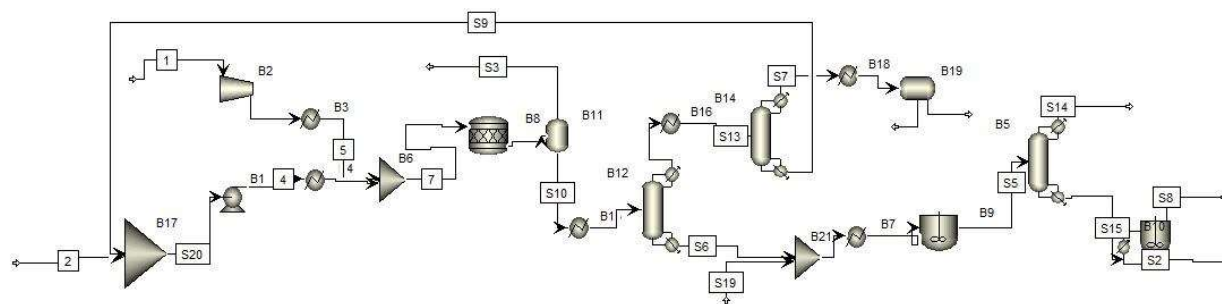


Figure 3. Process Flow Diagram of Aspen Simulation

The description of units in the figure are:

1. Compressor (B2)

Hydrogen is fed to the compressor(B2) having mass fraction 1. The rate of feed for Hydrogen is 201.58 kg/hr and compressed such that the output pressure of the compressor should be 4 bar.

2. Pump (B1)

Nitrobenzene is fed to the pump(B1) at the rate of 6155.56 kg/hr and compressed to 4 bar. The pump is used for the compression purpose as Nitrobenzene in liquid state cannot be compressed using a compressor. The compression is done to meet the pressure requirement for the reaction to occur.

3. Heaters (B3, B4, B15, B16, B18 and B7)

The heat exchangers are taken normal heaters in this simulation to analyse the heat exchange required to increase the temperature of the reactants. The heat exchangers B3 and B4 have inlets Hydrogen and Nitrobenzene respectively compressed at 4 bar pressure from the compressor(B2) and pump(B1) respectively. The outlet temperature from both the heaters is set at 358.15 K.

4. Mixer (B6 and B21)

The mixer takes Hydrogen and Nitrobenzene as inlet to pass to the reactor for reaction. The both feed to the mixer has temperature 358.15 K and pressure 4 bar which is assumed to be the base requirement for reaction. The mixer outlet is taken to the reactor as feed.

5. Reactor (B8 and B9)

The stoichiometric reactor with constant thermal fluid temperature was chosen for the simulation. The water was chosen as fluid for cooling fluid for the reactor. The CSTR reactor was chosen with a constant thermal fluid temperature of 381.15 K and a constant pressure of 1 bar.

6. Flash (B11)

The flash takes Hydrogen, Nitrobenzene, Para Amino Phenol, Aniline, Water as an input. The input temperature and pressure are 358.15 K and 4 bar respectively. The output temperature and pressure of both the outlets are 298.15 K and 1 bar respectively.

7. Distillation Column (B12, B14 and B5)

The Distillation Column takes Hydrogen, Nitrobenzene, Para Amino Phenol, Aniline, Water as an input. The input temperature and pressure are 393.15 K and 1 bar respectively. The output temperature and pressure of the 1st outlet is 545.09 K and 0.8 bar respectively and for the second outlet the temperature and pressure are 440.93 K and 0.8 bar respectively.

8. Decanter (B19)

The Decanter used in our simulation is used for the separation of Aniline. The feed for the Decanter is Hydrogen, Nitrobenzene, Para Amino Phenol, Aniline and Water. The temperature and pressure for both the inlet and outlet are 298.15 K and 1 bar respectively.

9. Crystallizer (B10)

The main role of Crystallizer in our simulation was to obtain Solid Cetamol. The feed for the Crystallizer is Nitrobenzene, Para Amino Phenol, Aniline Acetic Anhydride, Acetic Acid, Cetamol and Water. The temperature and pressure for the inlet are 518.87 K and 0.8 bar respectively. The temperature and pressure for the outlet are 273.15 K and 1 bar respectively.

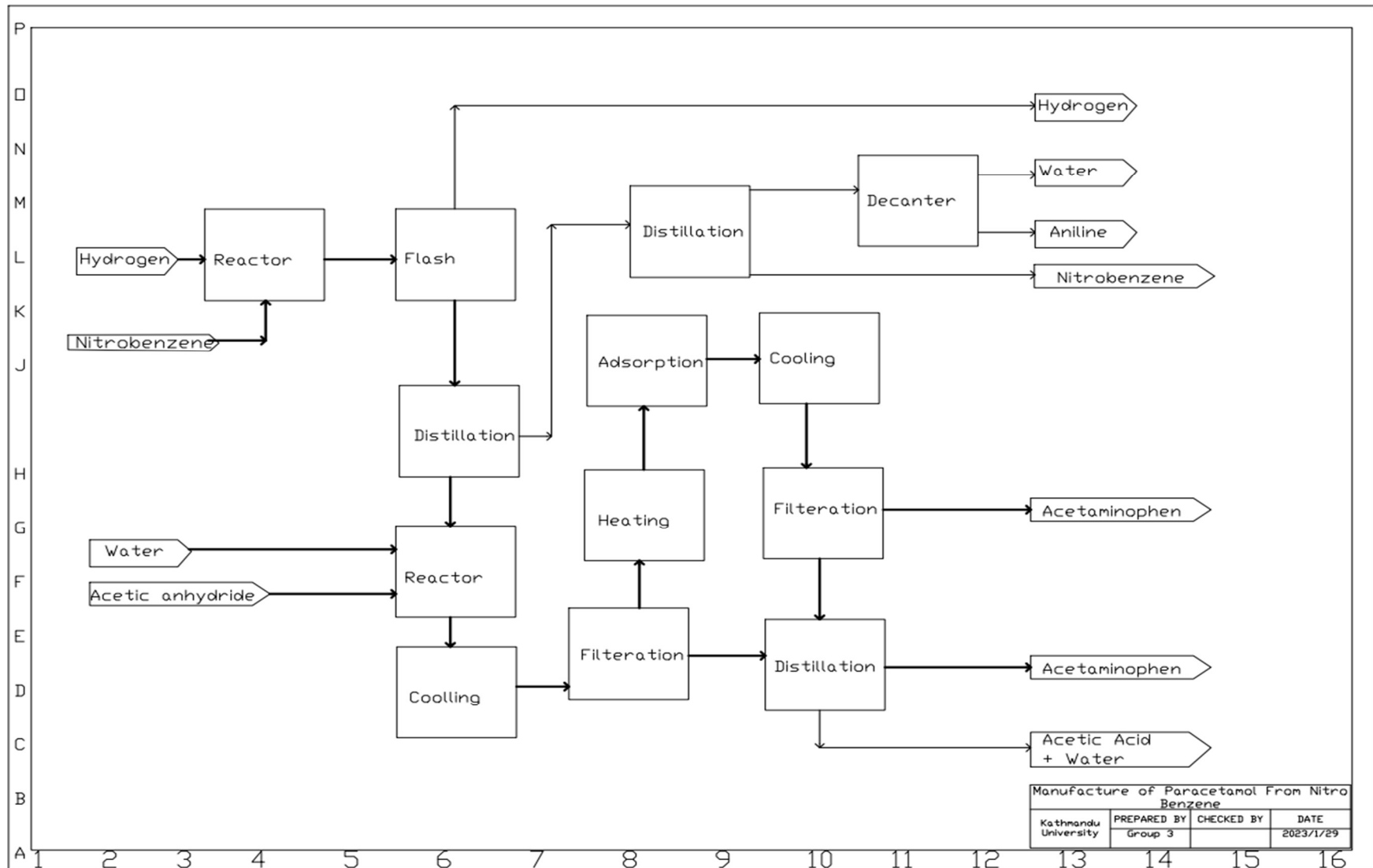
4.1.4 Results and Discussion

Table 8: Heater Results From Aspen Plus

Heater						
Name	B3	B4	B7	B15	B16	B18
Property method	PENG-ROB	PENG-ROB	PENG-ROB	PENG-ROB	PENG-ROB	PENG-ROB
Henry's component list ID						
Electrolyte chemistry ID						
Use true species approach for electrolytes	YES	YES	YES	YES	YES	YES
Free-water phase properties method	STEAM-TA	STEAM-TA	STEAM-TA	STEAM-TA	STEAM-TA	STEAM-TA
Water solubility method	3	3	3	3	3	3
Specified pressure [bar]	4	4	1	1	1	1
Specified temperature [C]	85	85	108	120	120	25
Specified vapour fraction						
Specific heat duty [cal/sec]						
EO Model components						
Calculated pressure [bar]	4	4	1	1	1	1

Calculated temperature [C]	85	85	108	120	120	25
Calculated vapour fraction	1	0	0	0.17788 1407	0.31260 7179	2.411E -05
Calculated heat duty [cal/sec]	- 27379. 34	- 14019. 82	- 3870.9 5	151098. 02	- 277680. 84	- 24291 4.48
Temperature change [C]						
Degrees of superheating [C]						
Degrees of subcooling [C]						
Pressure-drop correlation parameter						
Net duty [cal/sec]	- 27379. 34	- 14019. 82	- 3870.9 5	151098. 02	- 277680. 84	- 24291 4.48
First liquid / total liquid		1	1	1	1	1
Total feed stream CO2e flow [kg/hr]	0	0	0	0	0	0
Total product stream CO2e flow [kg/hr]	0	0	0	0	0	0
Net stream CO2e production [kg/hr]	0	0	0	0	0	0
Utility CO2e production [kg/hr]	0	0	0	0	0	0
Total CO2e production [kg/hr]	0	0	0	0	0	0
Utility usage						
Utility cost						
Utility ID						

4.2 Block Flow Diagram



4.3 Process Flow Diagram

The process begins with the compression and transfer of hydrogen to the reactor, where Nitro Benzene and Hydrogen react to form P Aminophenol and Aniline. The resulting mixture is then transferred to the first distillation column, where the bottom product consists of Water, Aniline, and unreacted Nitrobenzene. The bottom product is then sent to the second distillation column, where unreacted Nitrobenzene is extracted as the bottom product. The top product of the second distillation column is cooled and sent to a decanter, where the separation of Water and Aniline takes place, with Aniline being an organic compound.

The bottom product of the first distillation column is mixed with Acetic Anhydride and heated. The mixture is then transferred to a reactor where Acetic Anhydride and Para aminophenol react to form Acetaminophen (also known as paracetamol) and Acetic Acid. The top product of the distillation column is separated into Acetic Acid and Acetic Anhydride, while the bottom product of the column is Paracetamol.

After the Paracetamol is obtained as the bottom product of the distillation column, the next step in the process is to crystallise it. This is typically done by transferring the liquid Paracetamol to a crystallizer, where the temperature and other conditions are controlled to allow the solid Paracetamol to form. The resulting crystals are then separated from the remaining liquid, washed and dried, to obtain the final product. The crystalized form of Paracetamol is often used in tablets, capsules, and other solid dosage forms, as it is more stable and easier to handle compared to the liquid form.

The end product obtained through this process is Paracetamol in solid form from Nitrobenzene and Hydrogen.

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Chapter 5 Piping & Instrumentation Diagram

The P & ID diagram shows how all the process equipment and instrumentation are interconnected, as well as the process flow and the location of all valves and instrumentation, for the production of Para acetaminophen from Nitrobenzene.

In the P&ID for the production of Para acetaminophen from Nitrobenzene, we have used three distillation columns with a regenerative heating system. The steam used in the condenser is reheated in a furnace and used to provide heat in the reboiler. The steam out of the reboiler is then cooled and used for condensation.

We have also included two jacketed reactors for two reactions, and they are fitted with level sensors and controllers to maintain the fluid level inside the reactor. The temperature in the reactor is measured and controlled by controlling the flow of hot fluid into the heating jacket of the reactor.

Additionally, every control valve has an alternative path that is normally closed so that in case control valves fail, the process won't stop. We have used parallel pumps with Y-type strainer, and the pump motor is controlled instead of controlling the valve outflow of the pump. This design ensures that there will not be any pressure in the pump when the valve is closed, and the pump is running in its full capacity.

We have used pressure relief valves in each pressure vessel and also in pipes to prevent the system from failing in case the pressure becomes excessively high or more than the design pressure. There are six heat exchangers in the process, and all of these are shell and tube heat exchangers where the chemicals flow in the shell sides. This design allows easy removal of corrosion or fouling that may occur.

To connect the pipes with the control valves, we have used reducers, and isolation valves are used alongside the control valve after the reducer. The control valves have a 1-inch less diameter than the main pipe.

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Chapter 6 Equipment Design

6.1 Design of Pipe

6.1.1 Calculation of Pipe Sizes

Pipeline design includes a selection of the route traversed by the pipe, determination of the throughput (*i.e.*, the amount of fluid or solids transported) and the operational velocity, calculation of pressure gradient, selection of pumps and other equipment, determination of pipe thickness and material (*e.g.*, whether to use steel, concrete, cast iron or PVC pipe), and an engineering economic analysis and a market analysis to determine the optimum system based on alternate designs. In each design, careful consideration must be given to safety, leak and damage prevention, government regulations, and environmental concerns.

Table 9: Design Parameters for Pipe 1

S.N.	Design Parameters	Values
1	Temperature (°C)	25
2	Pressure (Bar)	1
3	Density (kg/m ³)	0.0812
4	Mass Flow Rate (Kg/s)	0.0559
5	Volume Flow (m ³ /s)	0.68892
6	Linear velocity(m/s)	30.48

$$\text{Volumetric flow rate } (\dot{v}) = \frac{\text{Mass Flow Rate}}{\text{Density}} = 0.6884 \text{ m}^3/\text{s}$$

$$\text{Cross Sectional Area } (A_c) = \frac{\text{Volumetric flow rate}}{\text{velocity}} = 0.0225 \text{ m}^2$$

$$\text{Internal diameter of pipe}(d_i) = \sqrt{\frac{A_c * 4}{\pi}} = 0.1692 \text{ m} = 6.66 \text{ in}$$

Corresponding standard stainless steel pipe;

Inside diameter = 7.981 in

Wall thickness = 0.322 in

Schedule number = 40s

Outside diameter = 8.625 in

Nominal diameter = 8 in

6.2 Design of Heat Exchangers

The shell-and-tube heat exchanger is probably the most common type found in industry. It is widely used in the process industries as well as in many types of HVAC equipment. Uses for these workhorses of heat transfer include heat removal in chillers, condensers, and reboilers, process stream cooling, and as critical parts of evaporative cooling and refrigeration systems.

Shell-and tube units consist of round tubes mounted in cylindrical shells. Components include the head, tube bundle, and shell. They can be built to any diameter or length. Tube bundles are typically hydrostatically tested. Many are ASME-designated as pressure vessels. The shell is a cylinder of seamless or rolled and welded pipe with a bolting flange at each end. Carefully placed holes in the tube sheets accommodate the tubes. Baffles help create the turbulence needed for heat transfer.

Shell-and-tube heat transfer technology has remained essentially the same over the years. Most recent developments include enhanced surface tubes that let units achieve approach temperatures of as small as 2-3 °F. True counterflow construction is now commonly used to avoid heat-transfer limiting pinch points. Finally, some units feature double-wall construction, essentially a second tube within each tube. Double-wall construction offers significant leak protection and safety, though at higher cost.

Heat Exchanger B3

Step 1: Find mean temperature difference (ΔT_m)

From Aspen,

$$\text{Log mean temperature difference (LMTD)} = 424.8 \text{ K}$$

$$\text{Correction factor } (F_T) = 1 \text{ (As countercurrent)}$$

$$\text{So, } \Delta T_m = \text{LMTD} \cdot F_T = 424.8 \text{ K}$$

Step 2: Find heat duty (Q)

From Aspen,

$$Q = 114632 \text{ Watt}$$

Step 3: Assume a reasonable value of overall heat transfer coefficient ($U_{o,assume}$)

Overall heat transfer coefficient is assumed as,

$$U_{o,assume} = 750 \frac{\text{Watt}}{\text{m}^2 \cdot \text{K}}$$

Step 4: Decide tentative number of shell and tube passes (n_p)

$$n_p = 2$$

Step 5: Calculate heat transfer area (A) required

$$\text{Heat transfer area (A)} = \frac{Q}{U_o \cdot LMTD} = 0.359 \text{ m}^2$$

Step 6: Select tube material, decide the tube diameter (ID = d_i , OD = d_o), its wall thickness (in terms of BWG or SWG) and tube length (L). Calculate the number of tubes (n_T).

Tube side material is Hydrogen.

Since a significant amount of Hydrogen is present. Hydrogen is a combustible gas. Thus, Stainless steel 304 (as it is cheaper than 316) is used as the tube material.

Let,

$$\text{Tube length (L)} = 1.8228 \text{ m (6 ft is a standard length)}$$

$$\text{Outside diameter (}d_o\text{)} = 19.05 \text{ mm (For the process industry, 19.05 is most common)}$$

$$\text{Let, Birmingham Wire Gauge (BWG)} = 16 = 1.651 \text{ mm}$$

$$\text{Internal diameter (}d_i\text{)} = d_o - \text{BWG} = 17.399 \text{ mm}$$

$$\text{So, Area of 1 tube} = \pi \cdot d_o \cdot L = \pi \cdot 19.05 \times 10^{-3} \cdot 1.8228 = 0.1091 \text{ m}^2$$

$$\text{Thus, Number of tubes (}N_T\text{)} = \frac{A}{\text{Area of 1 tube}} = \frac{0.359}{0.1091} = 3.29 \approx 4$$

Step 7: Decide type of shell and tube exchanger (fixed tube sheet, U-tube etc.). Select the tube pitch (P_T), determine inside shell diameter (D_s) that can accommodate the calculated number of tubes (n_T)

As the heat exchanger is subjected to a temperature of (425 K), U-tube heat exchanger can be selected.

Triangular pitch is better for heat transfer and surface area per unit length (greatest tube density). Thus, triangular pitch is used.

$$\text{Tube pitch (}P_T\text{)} = 1.25 \cdot d_o = 0.0238 \text{ m}$$

Now,

$$\text{Bundle diameter (BD)} = d_o \cdot \left(\frac{N_T}{K_1} \right)^{\frac{1}{n_1}}$$

For single pass and triangular pitch,*

$$n_1 = 2.142 \text{ and } K_1 = 0.319$$

So,

$$BD = 0.0190 \cdot \left(\frac{4}{0.319} \right)^{\frac{1}{2.142}} = 0.0618 \text{ m} = 61.87 \text{ mm}$$

And,

Bundle diameter clearance (BDC) = 90 mm for pull through U – Tube.

$$\text{Inside shell diameter } (D_s) = (61.87 + 90) \text{ mm} = 151.87 \text{ mm}$$

Step 8: Assign fluid to shell side or tube side. Select the type of baffle, its size (i.e. percentage cut), spacing and number.

Tube side material is Hydrogen . Meanwhile, the shell side material is water.

Baffle type is cut-segmental baffle.

$$\text{Baffle spacing } (B_s) = 0.2 \cdot 151.87 = 30.37 \text{ mm}$$

Percentage cut of baffles = 25% (The most widely used)

$$\text{Number of baffles} = \frac{\text{Inside shell diameter}}{\text{Baffle spacing}} = \frac{151.87}{30.37} = 5$$

Step 9: Comparison of calculated and assumed overall heat transfer coefficient (U)

A. Shell side coefficient calculation

$$h_o = \frac{j_n}{D_e} \cdot k \cdot Re \cdot Pr^{\frac{1}{3}}$$

Now,

$$Re = \frac{G_s \cdot D_e}{\mu}$$

Where,

$$\text{Mass velocity } (G_s) = \frac{\text{Mass flow rate}}{\text{Area of Shell}} = \frac{0.306}{\frac{(P_T - d_o) \cdot D_s \cdot B_s}{P_T}} = \frac{0.306}{0.00092} = 331.68 \frac{\text{kg}}{\text{m}^2 \cdot \text{s}}$$

$$D_e = \frac{1.1}{d_o} (P_T^2 - 0.917 \cdot d_o^2) = 0.0135 \text{ m}$$

$$\mu = 5.85 \times 10^{-4} \text{ Pa} \cdot \text{s}$$

So,

$$Re = 7648.93$$

And,

$$Pr = \frac{\mu \cdot C_p}{k} = 3.88$$

Here,

$$C_p = 4140 \frac{J}{kg \cdot K}$$

$$k = 0.625 \frac{W}{m \cdot K}$$

Again,

$$h_o = 77921.67 \frac{W}{m^2 \cdot K}$$

B. Tube side calculation

$$h_i = \frac{j_n}{D_e} \cdot k \cdot Re \cdot Pr^{\frac{1}{3}}$$

$$T_{mean} = 428.69 K$$

Here,

$$Re = \frac{4 \cdot \text{mass flow rate} \cdot \frac{n_p}{N_T}}{\pi \cdot d_i \cdot \mu} = 82820.54$$

Here,

$$\mu = 2.445 \times 10^{-5} Pa \cdot s$$

$$C_p = 11455 \frac{J}{kg \cdot K}$$

$$k = 0.186 \frac{W}{m \cdot K}$$

$$Pr = \frac{\mu \cdot C_p}{k} = \frac{0.0005745 \cdot 4180}{0.64} = 1.5$$

So,

$$h_i = \frac{0.15}{0.0136} \cdot 0.186 \cdot 82778.55 \cdot (1.5)^{\frac{1}{3}} = 196314.97 \frac{W}{m^2 \cdot K}$$

Thus,

$$U_{o,calc} = \frac{1}{\frac{1}{h_o} + \frac{1}{h_{do}} + \frac{d_o \cdot \ln\left(\frac{d_o}{d_i}\right)}{2 \cdot K_W} + \frac{d_o}{d_i \cdot h_i} + \frac{d_o}{d_i \cdot h_{di}}}$$

From literature,

$$h_{do} = 4500 \frac{W}{m^2 \cdot K}$$

$$h_{di} = 3500 \frac{W}{m^2 \cdot K}$$

So,

$$U_{o,calc} = \frac{1}{3.06 \times 10^{-5} + 2.22 \times 10^{-4} + 5.60 \times 10^{-4} + 5.64 \times 10^{-6} + 3.13 \times 10^{-4}}$$

$$\text{or, } U_{o,calc} = 897.28 \frac{W}{m^2 \cdot K}$$

$$\text{Since } \frac{U_{o,calc} - U_{o,assume}}{U_{o,assume}} \cdot 100\%$$

= 20 %, which is less than 30%, our assumptions for design are valid.

Step 10: Over design calculation

Over design percentage is zero according from Aspen

Step 11: Pressure drop

Pressure drop is neglected in our design

6.2.1 Design of Condenser B12

Physical Properties

Heat Duty = 1969.09 kg/h

Pressure = 0.8 bar

Vapour Enter = 120 °C

Condensation Complete = 168 °C

Average Molecular Weight of Vapours = 60

Enthalpy of Vapour = -1535.79 kJ/kg

Enthalpy of condensate = 247.0 kJ/kg

Cooling Water Available = 30 °C

Temperature limited = 40 °C

O.D. of tubes = 20 mm

I.D. of tubes = 16.8 mm

Length = 4.88 m (16 ft)

The physical properties of the mixture will be taken as the mean, at the average temperature.

$$\text{Heat transferred from vapour} = \frac{1969.09}{3600} (1535.79 - 247.0) = 704.92 \text{ kW}$$

$$\text{Cooling water flow} = \frac{704.92}{(40 - 30) * 4.18} = 16.86 \text{ kg/s}$$

$$\text{Assumed overall coefficient (Table 12.1)} = 900 \text{ W/m}^2\text{°C}$$

Mean temperature difference: the condensation range is small and the change in saturation temperature will be linear, so the corrected logarithmic mean temperature difference can be used.

$$R = \frac{R = (168 - 120)(40 - 30) = 4.8(T_1 - T_2)}{(t_2 - t_1)}$$

$$R = \frac{(168 - 120)}{(40 - 30)} = 4.8$$

$$S = \frac{(t_2 - t_1)}{(T_1 - t_1)}$$

$$S = \frac{(40 - 30)}{(168 - 30)} = 0.072$$

Try a horizontal exchanger, condensation in the shell, four tube passes. For one shell pass, four tube passes, from Figure 12.19, $F_t = 0.92$

$$\Delta T_{lm} = \frac{(168 - 40) - (120 - 30)}{\ln \frac{(168 - 40)}{(120 - 30)}} = 107.88 \text{ } ^\circ\text{C}$$

$$\Delta T_m = 0.92 * 107.88 = 99.24 \text{ } ^\circ\text{C}$$

$$\text{Trial area} = \frac{704.92 * 10^3}{900 * 99.24} = 7.89 \text{ m}^2$$

$$\begin{aligned} \text{Surface area of one tube} &= 20 * 10^{-3} * 3.14 * 4.88 \\ &= 0.305 \text{ m}^2 \text{ (ignore tube sheet thickness)} \end{aligned}$$

$$\text{Number of tubes} = \frac{7.89}{0.305} = 26$$

Use square pitch, $P_t = 1.25 * 20 \text{ mm} = 25 \text{ mm}$.

Tube bundle diameter

$$D_b = d_o \left(\frac{N_t}{K_1} \right)^{1/n_1}$$

$$D_b = 20 \left(\frac{26}{0.158} \right)^{1/2.263} = 190.72 \text{ mm}$$

$$\text{Number of tubes in centre row } N_r = \frac{D_b}{P_t} = \frac{190.72}{25} = 7.62$$

Shell-Side Coefficient

Estimate tube wall temperature, T_w ; assume condensing coefficient of $1500 \text{ W/m}^2\text{ } ^\circ\text{C}$,

Mean temperature

$$\text{Shell} - \text{Side} = \frac{168 + 120}{2} = 144 \text{ } ^\circ\text{C}$$

$$\text{Tube} - \text{Side} = \frac{40 + 30}{2} = 35 \text{ } ^\circ\text{C}$$

$$(144 - T_w)1500 = (144 - 35)900$$

$$T_w = 78 \text{ } ^\circ\text{C}$$

$$\text{Mean temperature condensate} = \frac{144 + 78}{2} = 111 \text{ } ^\circ\text{C}$$

Physical properties at 47 °C

$$\mu_L = 0.0087 \text{ mNs/m}^2$$

$$\rho_L = 113.03 \text{ kg/m}^3$$

$$k_L = 0.136 \text{ W/m}^\circ\text{C}$$

vapour density at mean vapour temperature

$$\rho_v = \frac{60}{22.4} * \frac{273}{(273 + 144)} * \frac{10}{1} = 17.53 \text{ kg/m}^3$$

$$\Gamma_h = \frac{W_c}{LN_t} = \frac{1969.09}{3600} * \frac{1}{4.88 * 26} = 4.31 * 10^{-3} \text{ kg/s m}$$

$$N_r = \frac{2}{3} * 7.62 = 5.08$$

Using Kern's method, the mean coefficient for a tube bundle is given by:

$$(h_c)_b = 0.95 k_L \left[\frac{\rho_L (\rho_L - \rho_v) g}{\mu_L \Gamma_h} \right]^{1/3} N_r^{-1/6}$$

$$\begin{aligned} h_c &= 0.95 * 0.136 \left[\frac{113.03 * (113.03 - 17.53) * 9.81}{0.0087 * 10^{-3} * 4.31 * 10^{-3}} \right]^{1/3} * 5.08^{-1/6} \\ &= 1392.86 \text{ W/m}^2\text{C} \end{aligned}$$

Close enough to assumed value of 1500 W/m²°C, so no correction to T_w needed.

Tube-Side Coefficient

$$\text{Tube cross - sectional area} = \frac{3.14}{4} (16.8 * 10^{-3})^2 * \frac{26}{4} = 0.0014 \text{ m}^2$$

$$\text{Density of water, at } 35^\circ\text{C} = 993 \text{ kg/m}^3$$

$$\text{Tube velocity} = \frac{16.86}{993} * \frac{1}{0.0014} = 12.12 \text{ m/s}$$

$$h_i = \frac{4200(1.35 + 0.02t)u_t^{0.8}}{d_i^{0.2}}$$

$$\begin{aligned} h_i &= \frac{4200(1.35 + 0.02 * 35)12.12^{0.8}}{16.8^{0.2}} \\ &= 36036.56 \text{ W/m}^\circ\text{C} \end{aligned}$$

Fouling factors: as neither fluid is heavily fouling, use 6000 W/m²°C for each side.

$$k_w = 50 \text{ W/m}^\circ\text{C}$$

Overall Coefficient

$$\frac{1}{U_o} = \frac{1}{h_o} + \frac{1}{h_{do}} + \frac{d_o \cdot \ln\left(\frac{d_o}{d_i}\right)}{2 \cdot K_W} + \frac{d_o}{d_i \cdot h_{id}} + \frac{d_o}{d_i \cdot h_i}$$
$$\frac{1}{U} = \frac{1}{1392.96} + \frac{1}{6000} + \frac{20 \cdot 10^{-3} \ln\left(\frac{20}{16.8}\right)}{2 \cdot 50} + \frac{20}{16.8} \cdot \left(\frac{1}{6000} + \frac{1}{36036.56}\right)$$
$$\frac{1}{U} = 1.15 \cdot 10^{-3}$$

$$U = 869.56 \text{ W/m}^2\text{°C}$$

Significantly lower than the assumed value of $900 \text{ W/m}^2\text{°C}$.

Repeating calculation using new trial value of $750 \text{ W/m}^2\text{°C}$

$$Area = \frac{704.92 \cdot 10^3}{750 \cdot 99.24} = 9.47 \text{ m}^2$$

$$Number \text{ of tubes} = \frac{9.47}{0.305} = 31$$

Tube bundle diameter

$$D_b = d_o \left(\frac{N_t}{K_1}\right)^{1/n_1}$$

$$D_b = 20 \left(\frac{31}{0.158}\right)^{1/2.263} = 206 \text{ mm}$$

$$\text{Number of tubes in centre row } N_r = \frac{D_b}{P_t} = \frac{206}{25} = 8.24$$

$$\Gamma_h = \frac{W_c}{LN_t} = \frac{704.92}{3600} \cdot \frac{1}{4.88 \cdot 31} = 1.29 \cdot 10^{-3} \text{ kg/m s}$$

$$N_r = \frac{2}{3} \cdot 8.24 = 5.49$$

Using Kern's method, the mean coefficient for a tube bundle is given by:

$$(h_c)_b = 0.95 k_L \left[\frac{\rho_L (\rho_L - \rho_v) g}{\mu_L \Gamma_h} \right]^{1/3} N_r^{-1/6}$$

$$h_c = 0.95 \cdot 0.136 \left[\frac{113.03 \cdot (113.03 - 17.53) \cdot 9.81}{0.0087 \cdot 10^{-3} \cdot 1.29 \cdot 10^{-3}} \right]^{1/3} \cdot 5.49^{-1/6}$$
$$= 2055.50 \text{ W/m}^2\text{°C}$$

$$\text{New Tube velocity} = 12.12 * \frac{26}{31} = 10.16 \text{ m/s}$$

$$h_i = \frac{4200(1.35 + 0.02t)u_t^{0.8}}{d_i^{0.2}}$$

$$h_i = \frac{4200(1.35 + 0.02 * 35)10.16^{0.8}}{16.8^{0.2}}$$

$$= 31293.65 \text{ W/m}^2\text{C}$$

$$\frac{1}{U_o} = \frac{1}{h_o} + \frac{1}{h_{do}} + \frac{d_o \cdot \ln\left(\frac{d_o}{d_i}\right)}{2 \cdot K_W} + \frac{d_o}{d_i \cdot h_{id}} + \frac{d_o}{d_i \cdot h_i}$$

$$\frac{1}{U} = \frac{1}{2055.50} + \frac{1}{6000} + \frac{20 * 10^{-3} \ln\left(\frac{20}{16.8}\right)}{2 * 50} + \frac{20}{16.8} * \left(\frac{1}{6000} + \frac{1}{31293.65}\right)$$

$$U = 1081.67 \text{ W/m}^2\text{C}$$

Close enough to estimate, firm up design.

Shell-Side Pressure Drop

Use pull-through floating head, no need for close clearance.

Select baffle spacing D shell diameter, 45 percent cut.

From Figure 12.10, clearance = 95 mm.

$$\text{Shell i.d.} = 206 + 95 = 301 \text{ mm}$$

Use Kern's method to make an approximate estimate.

$$A_s = \frac{(p_t - d_o)D_s l_B}{p_t}$$

$$\text{Cross-flow area } A_s = \frac{(25 - 20)}{25} * 301 * 301 * 10^{-6}$$

$$= 0.018 \text{ m}^2$$

Mass flow-rate, based on inlet conditions

$$G_s = \frac{1969.09}{3600} * \frac{1}{0.018} = 30.38 \text{ kg/s m}^2$$

For a square pitch arrangement:

$$d_e = \frac{1.27}{d_o} (p_t^2 - 0.785d_o^2)$$

$$\text{Equivalent diameter, } d_e = \frac{1.27}{20} (25^2 - 0.785 * 20^2) \\ = 19.8 \text{ mm}$$

$$\text{Vapor viscosity} = 0.008 \text{ mNs/m}^2$$

$$Re = \frac{30.38 * 19.8 * 10^{-3}}{0.008 * 10^{-3}} = 75190.5$$

$$\text{From Figure 12.30, } j_f = 2.2 * 10^{-2}$$

$$u_s = \frac{G_s}{\rho_v} = \frac{30.38}{17.53} = 1.73 \text{ m/s}$$

Take pressure drop as 50 per cent of that calculated using the inlet flow; neglect viscosity correction.

$$\Delta P_s = 8j_f \left(\frac{D_s}{d_e} \right) * \left(\frac{L}{l_B} \right) * \frac{\rho u_s^2}{2} * \left(\frac{\mu}{\mu_w} \right)^{-0.14} \\ \Delta P_s = \frac{1}{2} [8 * 2.2 * 10^{-2} * \left(\frac{301}{19.8} \right) * \left(\frac{4.88}{0.301} \right) * \frac{17.53(1.73)^2}{2}] \\ = 56896 \text{ N/m}^2 \\ = 56.8 \text{ kPa}$$

Tube-Side Pressure Drop

$$\text{Viscosity of water} = 0.6 \text{ mN s/m}^2$$

$$Re = \frac{u_t \rho d_i}{\mu} = \frac{10.16 * 993 * 16.8 * 10^{-3}}{0.6 * 10^{-3}} = 282488.64$$

$$\text{From Figure 12.24, } j_f = 3.5 * 10^{-3}.$$

Neglect viscosity correction.

$$\Delta P_t = N_p \left[8j_f * \left(\frac{L}{d_i} \right) * \left(\frac{\mu}{\mu_w} \right)^{-m} + 2.5 \right] \frac{\rho u_t^2}{2} \\ \Delta P_t = 4 \left[8 * 3.5 * 10^{-3} * \left(\frac{4.88}{16.8 * 10^{-3}} \right) + 2.5 \right] \frac{993 * 10.16^2}{2} \\ = 2179898 \text{ N/m}^2 \\ = 2180 \text{ kPa (316 psi)},$$

acceptable.

6.2.2 Design of Reboiler B12

Physical Properties:

Temperature = 271.94 °C

Pressure = 0.8 bar

Boiling point = 287 °C

Latent heat = 280.3 kJ/kg

Mean specific heat, liquid = 1.56 kJ/kg°C

Critical pressure, P_c = 47.1 bar

Heat loads:

$$\text{Sensible heat (maximum)} = (287 - 0) * 1.56 = 447.72 \text{ kJ/kg}$$

$$\text{Total heat load} = (447.72 + 280.3) * \frac{2289.22}{3600} = 462.94 \text{ kW},$$

add 5 % for heat losses

$$\text{maximum heat load (duty)} = 1.05 * 462.94$$

$$= 486.09 \text{ kW}$$

From Figure 12.1 assume $U = 1000 \text{ W/m}^2\text{°C}$.

Mean temperature difference; both sides isothermal, steam saturation temperature at 0.8 bar = 346.1°C

$$\Delta T_m = 346.1 - 287 = 59.1^\circ\text{C}$$

$$\text{Area (outside)required} = \frac{486.09 * 10^3}{1000 * 59.1} = 8.22 \text{ m}^2$$

Select 25 mm i.d., 30 mm o.d. plain U-tubes,

Nominal length 4.8 m (one U-tube)

$$\text{Number of U tubes} = \frac{8.22}{(30 * 10^{-3}) * 3.14 * 4.8} = 19$$

Use square pitch arrangement, pitch = 1.5* tube o.d.

$$= 1.5 * 30 = 45 \text{ mm}$$

Draw a tube layout diagram, take minimum bend radius

$$1.5 * \text{tube o.d.} = 45 \text{ mm}$$

Proposed layout gives 20 U-tubes, tube outer limit diameter 420 mm.

Boiling coefficient

Use Mostinski's equation:

heat flux, based on estimated area,

$$q = \frac{486.09}{8.22} = 59.13 \text{ kW/m}^2$$

$$h_{nb} = 0.104 * (P_c)^{0.69} * (q)^{0.7} \left[1.8 * \left(\frac{P}{P_c} \right)^{0.17} + 4 * \left(\frac{P}{P_c} \right)^{1.2} + 10 * \left(\frac{P}{P_c} \right)^{10} \right]$$

$$\begin{aligned} h_{nb} &= 0.104 * (47.1)^{0.69} \\ &\quad * (59.13 * 10^3)^{0.7} \left[1.8 * \left(\frac{0.8}{47.1} \right)^{0.17} + 4 * \left(\frac{0.8}{47.1} \right)^{1.2} + 10 * \left(\frac{0.8}{47.1} \right)^{10} \right] \\ &= 3248.5 * 0.93 \\ &= 3021.10 \text{ W/m}^2\text{°C} \end{aligned}$$

Take steam condensing coefficient as $8000 \text{ W/m}^2\text{°C}$, fouling coefficient $5000 \text{ W/m}^2\text{°C}$; fouling coefficient, essentially clean, $10,000 \text{ W/m}^2\text{°C}$.

Tube material will be plain carbon steel, $k_w = 55 \text{ W/m}^2\text{°C}$

$$\begin{aligned} \frac{1}{U_o} &= \frac{1}{h_o} + \frac{1}{h_{do}} + \frac{d_o \cdot \ln \left(\frac{d_o}{d_i} \right)}{2 \cdot K_w} + \frac{d_o}{d_i \cdot h_{id}} + \frac{d_o}{d_i \cdot h_i} \\ \frac{1}{U_o} &= \frac{1}{3021.10} + \frac{1}{10,000} + \frac{30 * 10^{-3} * \ln \left(\frac{30}{25} \right)}{2 * 55} + \frac{30}{25} \left(\frac{1}{5000} + \frac{1}{8000} \right) \\ U_o &= 1148.46 \text{ W/m}^2\text{°C} \end{aligned}$$

Close enough to original estimate of $1000 \text{ W/m}^2\text{°C}$ for the design to stand.

Myers and Katz (Chem. Eng. Prog. Sym. Ser. 49(5) 107 114) give some data on the boiling on banks of tubes. To compare the value estimate with their values an estimate of the boiling film temperature difference is required:

$$= \frac{1148.46}{3021.10} * 59.1 = 22.46^\circ\text{C} \text{ (72.42}^\circ\text{F)}$$

Myers data, extrapolated, gives a coefficient of around $3000 \text{ Btu/h ft}^2\text{°F}$ at a 72°F temperature difference = $17,100 \text{ W/m}^2\text{°C}$, so the estimated value of 3021.10 is certainly on the safe side.

Check maximum allowable heat flux. Use modified Zuber equation

$$\text{Surface tension (estimated)} = 9.7 * 10^{-3} \text{ N/m}$$

$$\rho_L = 550 \text{ kg/m}^3$$

$$\rho_v = \frac{109.1}{22.4} * \frac{273}{(273 + 287)} * 0.8 = 1.89 \text{ kg/m}^3$$

$$N_t = 52$$

For square arrangement $K_b = 0.44$

$$q_{cb} = K_b * \left(\frac{P_t}{d_o}\right) * \left(\frac{\lambda}{\sqrt{N_t}}\right) [\sigma g * (\rho_L - \rho_v) * \rho_v^2]^{0.25}$$

$$q_{cb} = 0.44 * 1.5 * \frac{280.3 * 10^3}{\sqrt{52}} [9.7 * 10^{-3} * 9.81 * (550 - 1.89) * 1.89^2]^{0.25}$$

$$= 94,781.45 \frac{W}{m^2}$$

$$= 95 \text{ k W/m}^2$$

Applying a factor of 0.7, maximum flux should not exceed $95 * 0.7 = 66.5 \text{ kW/m}^2$.

Actual flux of 59.13 kW/m^2 is well below maximum allowable.

Layout

From tube sheet layout $D_b = 420 \text{ mm}$.

Take shell diameter as twice bundle diameter

$$D_s = 2 * 420 = 840 \text{ mm}.$$

Take liquid level as 500 mm from base,

Freeboard = $840 - 500 = 340 \text{ mm}$, satisfactory.

From sketch, width at liquid level = 0.8 m .

Surface area of liquid = $0.8 * 2.4 = 1.9 \text{ m}^2$.

$$\text{Vapor velocity at surface} = \frac{5000}{3600} * \frac{1}{1.89} * \frac{1}{1.9} = 0.386 \text{ m/s}$$

Maximum allowable velocity

$$v_v < 0.2 * \left[\frac{\rho_L - \rho_v}{\rho_v}\right]^{\frac{1}{2}}$$

$$v_v = 0.2 * \left[\frac{550 - 1.89}{1.89}\right]^{\frac{1}{2}} = 3.40 \text{ m/s}$$

So, actual velocity is well below maximum allowable velocity. A smaller shell diameter could be considered.

6.3 Design of Flash

Data obtained from aspen

Volumetric flow rate of vapour $V_v = 0.2261 \frac{m^3}{sec}$

Density of liquid $\rho_l = 11158.381 \frac{kg}{m^3}$

Density of vapour $\rho_v = 0.118 \frac{kg}{m^3}$

Settling velocity U_t

$$U_t = 0.07 \left[\frac{(\rho_l - \rho_v)}{\rho_v} \right]^{\frac{1}{2}}$$
$$or, U_t = 0.07 \left[\frac{(11158.381 - 0.118088)}{0.118088} \right]^{\frac{1}{2}}$$
$$or, U_t = 6.93 \frac{m}{s}$$

As the separation of condensate from steam is unlikely to be critical, a demister pad will not be specified. So,

$$U_s = 0.15 \times 6.932$$

$$U_s = 1.03 \frac{m}{s}$$

Minimum vessel diameter D_v

$$D_v = \sqrt{\left(\frac{4V_v}{\pi U_s} \right)}$$

$$or, D_v = \sqrt{\left(\frac{4 \times 0.2261}{\pi \times 1.0398} \right)}$$

$$or, D_v = 0.526 m$$

$$or, D_v = 1.72 ft$$

Doing round off

$$or, D_v = 2 ft$$

$$or, D_v = 0.60 m$$

Cross Section Area of the flash, A

$$A = \frac{\pi D_v^2}{4}$$

$$or, A = \frac{\pi \times 0.6096^2}{4}$$

$$or, A = 0.29 \text{ m}^2$$

Considering the L/D ratio to be 4 we get length of the reactor

$$L = 4 \times D_v$$

$$or, L = 4 \times 0.609$$

$$or, L = 2.43 \text{ m}$$

Hold up Volume V_h

$$V_h = A \times L$$

$$or, V_h = 0.2918 \times 2.4384$$

$$or, V_h = 0.71 \text{ m}^3$$

Hold up Time

$$t_h = \frac{V_h}{V_l}$$

$$or, t_h = \frac{0.711525}{0.0015013}$$

$$or, t_h = \frac{0.711525}{0.0015013}$$

$$or, t_h = 473.91 \text{ sec}$$

$$or, t_h = 7.89 \text{ min}$$

As the holdup time is necessary to be more than 10 min we need to modify the diameter of the vessel with given formula,

$$D_{v \text{ new}} = D_{v \text{ old}} \times \left(\frac{10}{D_{v \text{ old}}} \right)^{0.5}$$

$$or, D_{v \text{ new}} = 7.898 \times \left(\frac{10}{7.898} \right)^{0.5}$$

$$or, D_{v \text{ new}} = 0.68 \text{ m}$$

New Cross Sectional Area of the flash, A

$$A = \frac{\pi D_v^2}{4}$$

$$\text{or, } A = \frac{\pi \times 0.6859^2}{4}$$

$$\text{or, } A = 0.369 \text{ m}^2$$

Considering the L/D ratio to be 4 we get length of the reactor

$$L = 4 \times D_v \text{ new}$$

$$\text{or, } L = 4 \times 0.685$$

$$\text{or, } L = 2.74 \text{ m}$$

Hold up Volume, V_h

The volume of liquid held inside the liquid is calculated by

$$V_h = A \times L$$

$$\text{or, } V_h = 0.36954 \times 2.74376$$

$$\text{or, } V_h = 1.013 \text{ m}^3$$

Hold up Time

$$t_h = \frac{V_h}{V_l}$$

$$\text{or, } t_h = \frac{1.01392}{0.0015013}$$

$$\text{or, } t_h = 675.312 \text{ sec}$$

$$\text{or, } t_h = 11.25 \text{ min}$$

This is satisfactory.

Liquid depth required, h_v

$$h_v = \frac{V_h}{A}$$

$$\text{or, } h_v = \frac{1.01392}{0.36954}$$

$$\text{or, } h_v = 2.74 \text{ m}$$

Demister pad thickness

$$\begin{aligned} &= \frac{D_{vnew}}{4} \\ &= \frac{0.6859}{4} \\ &= 0.17 \text{ m} \end{aligned}$$

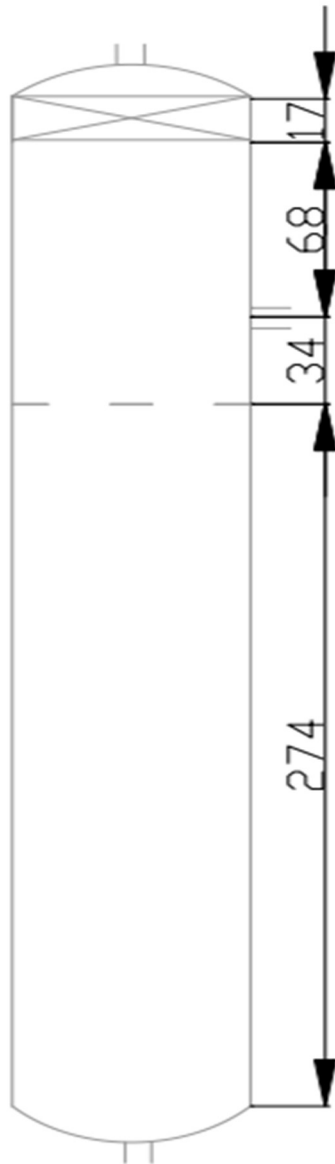


Figure 4. Flash Design Dimension in cm

6.4 Design of Decanter

Data obtained from aspen

$$\text{Density of aniline, } \rho_A = 1020 \frac{\text{kg}}{\text{m}^3}$$

$$\text{Density of water, } \rho_w = 1000 \frac{\text{kg}}{\text{m}^3}$$

$$\text{Viscosity of aniline, } \mu_A = 0.0031 \frac{\text{Nm}}{\text{Sec}}$$

$$\text{Viscosity of water, } \mu_w = 0.00089 \frac{\text{Nm}}{\text{Sec}}$$

$$\text{Mass flow rate of aniline, } m_A = 0.2124 \frac{\text{kg}}{\text{Sec}}$$

$$\text{Mass flow rate of water, } m_w = 0.179444 \frac{\text{kg}}{\text{Sec}}$$

$$\text{Volume flow rate of water, } V_w = 0.00018063 \frac{\text{m}^3}{\text{sec}}$$

$$\text{Volume flow rate of aniline, } V_A = 0.0002254 \frac{\text{m}^3}{\text{sec}}$$

Settling (terminal) velocity of the dispersed phase droplets with diameter d_d which is supposed to be 150×10^{-6} is given by,

$$U_d = \frac{d_d^2 \times g(\rho_A - \rho_w)}{18\mu_w}$$

$$\text{or, } U_d = \frac{(150 \times 10^{-6})^2 \times 9.8(1020 - 1000)}{18 \times 0.00089}$$

$$\text{or, } U_d = \frac{(150 \times 10^{-6})^2 \times 9.8(1020 - 1000)}{18 \times 0.00089}$$

$$\text{or, } U_d = 2.75 \times 10^{-4} \frac{\text{m}}{\text{sec}}$$

$$\text{Interface are} = \frac{\text{flow rate}}{U_d}$$

$$= \frac{0.000180631666}{2.75280899 \times 10^{-4}}$$

$$= 0.656 \text{ m}^2$$

Diameter of Decanter

$$\text{Area} = \frac{\pi \times D^2}{4}$$

$$\text{or, } 0.656 = \frac{\pi \times D^2}{4}$$

$$\text{or, } D = \sqrt{0.83546436}$$

$$\text{or, } D = 0.91 \text{ m}$$

Let's suppose the height of the decanter to be twice the diameter

$$H = 2D$$

$$\text{or, } H = 2 \times 0.914$$

$$\text{or, } H = 1.828 \text{ m}$$

Dispersion Band

$$= 0.1 \times H$$

$$= 0.18 \text{ m}$$

Residence time in dispersion band

$$= \frac{\text{dispersion band}}{U_d}$$

$$= \frac{0.1828}{2.7528089 \times 10^{-4}}$$

$$= 664.07 \text{ sec}$$

$$= 11.06 \text{ m}$$

$$\text{Velocity of aniline} = \frac{\text{mass flow rate of aniline}}{\rho_A \times \text{interface area}}$$

$$= \frac{0.2124}{1020 \times 0.656}$$

$$= 0.00031 \frac{\text{m}}{\text{s}}$$

As aniline is considered to be dispersed phase its settling velocity $U_d = 0.0003174 \frac{\text{m}}{\text{s}}$ using the formula for settling velocity of dispersed phase we get the diameter of dispersed phase droplet.

$$U_d = \frac{d_d^2 \times g(\rho_A - \rho_w)}{18\mu_w}$$

$$\text{or, } d_d = \sqrt{\frac{18\mu_w}{g(\rho_A - \rho_w)}}$$

$$\text{or, } d_d = \sqrt{\frac{18 \times 0.00089}{9.8(1020 - 1000)}}$$

$$\text{or, } d_d = 1.61 \times 10^{-4} \text{ m}$$

Feed pipe

Total volume flow rate, $Q = V_w + V_A$

$$= 0.0001806 + 0.0002254$$

$$= 0.000406 \frac{m^3}{sec}$$

Considering the velocity be $1 \frac{m}{sec}$

$$\text{Cross section area} = \frac{\text{volume flow rate}}{\text{velocity}}$$

$$= 0.000406 \text{ m}^2$$

For finding the diameter of the feed pipe

$$A = \frac{\pi D^2}{4}$$

$$\text{or, } D = \sqrt{\frac{4A}{\pi}}$$

$$\text{or, } D = \sqrt{\frac{4 \times 0.000406031}{\pi}}$$

$$\text{or, } D = 0.022 \text{ m}$$

$$Z_1 = 0.9 H$$

$$\text{or, } Z_1 = 0.9 \times 1.828$$

$$\text{or, } Z_1 = 1.64 \text{ m}$$

$$Z_3 = 0.5 H$$

$$\text{or, } Z_3 = 0.5 \times 1.828$$

$$\text{or, } Z_3 = 0.91 \text{ m}$$

$$Z_2 = \frac{(Z_1 - Z_3)\rho_A}{\rho_w}$$

$$\text{or, } Z_2 = \frac{(1.645 - 0.914)1020}{1000}$$

$$\text{or, } Z_2 = 0.74 \text{ m}$$

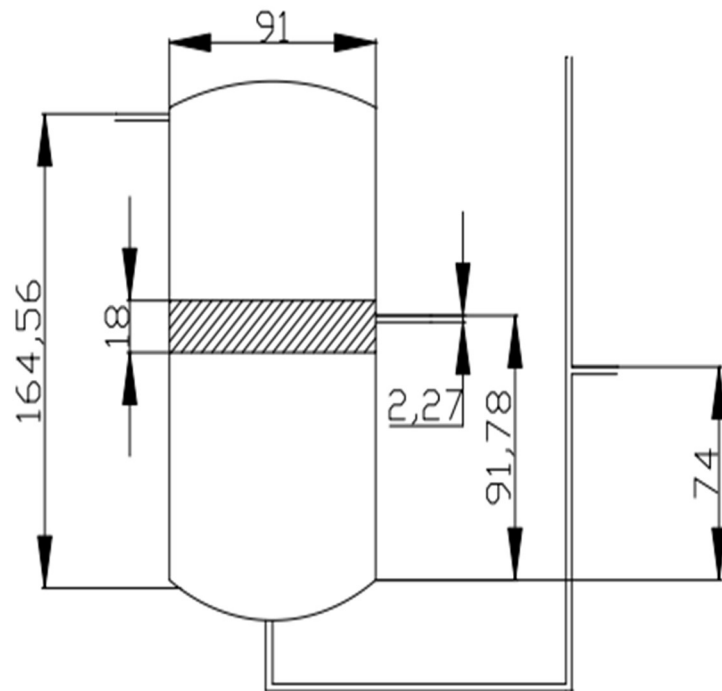


Figure 5. Decanter Design with dimension in cm

6.5 Design of Distillation Column

c - corrosion allowance

D_i - internal diameter of shell/ tray diameter

f - allowable stress

p - design pressure

t_s - shell thickness

ρ_s - density of shell material

h - height of column

t_{ins} - thickness of insulator

ρ_{ins} - density of insulation

D_{ins} - outer diameter of insulation

D_m - mean diameter of shell

μ - Poisson's ratio

E - modulus of elasticity

t_t - thickness of tray

w_d - width of downcomer

h_d - height of downcomer

l_h - loss of head

D_h - Diameter of holes

n_h - number of holes

h_w - height of weir

C_d - Clearance of downcomer

w_w - width of weir

s_t - spacing of trays

D_m - mean diameter of shell

Table 10: Design Parameters for Distillation Column

Values From Aspen	Distillation 1	Distillation 2	Distillation 3
D_i	1.69359	1.553259505	1.378
p	80000	5000	-
h	3.657	3.657	2.4384
t_t	0.0034	0.0034	0.0034
w_d	1.23058324	1.12861583	1.001270302
C_d	0.0381	0.0381	0.0381
h_w	0.0508	0.0508	0.0508
l_h	0.118475	0.10174349	0.1057484328
D_h	0.00127	0.00127	0.00127
n_h	1423	1197	942
s_t	0.6096	0.6096	0.6096
A_t	2.027449975	1.705377116	1.342242007

Table 11: Properties for Distillation Column

Properties	Value for SS	Value for glass wool
ρ	7930	20
E	200000000000	-
μ	0.29	-
C	0.00079375	-

Height of downcomer, h_d

Down comer makes a vertical pathway that allows liquid to flow downward from the higher tray to the lower tray, maintaining proper flow and separation efficiency. Downcomer are present in each tray and in a rectangular shape. They prevent flooding.

$$\begin{aligned}
 h_d &= s_t - l_h \\
 &= 0.6096 - 0.118475 \\
 &= 0.49 \text{ m}
 \end{aligned}$$

External diameter of insulation, D_{ins}

To prevent heat loss 10 cm thick insulation of glass wool covers the distillation. Glass wool has low density and high heat, chemical and fire resistance.

$$\begin{aligned}
 D_{ins} &= (D_i + t_s) + 2 \times t_{ins} \\
 &= (1.69359 + 0.00772576) + 2 \times 0.1 \\
 &= 1.90 \text{ m}
 \end{aligned}$$

Weight of a Tray, Weir and Down Comer, w_t

$$\begin{aligned}w_t &= \rho_s \times (A_t \times (h_d \times w_d) + (h_w \times w_w)) \times t_t \\&= 7930 \times (2.02745 + (0.491125 \times 1.231) + (0.5715 \times 1.231)) \times 0.0034 \\&= 89.93 \text{ kg}\end{aligned}$$

Axial Stress Due to Pressure

Axial stress in a distillation column can occur due to the pressure exerted by the vapour and liquid inside the column. This stress is also caused by the weight of the column and the fluids it contains, which creates a force that acts along the axis of the column. If the axial stress exceeds the material's yield strength, it can cause deformation or even failure of the column.

$$\begin{aligned}f_{ap} &= \frac{pD_i}{4(t_s - c)} \\&= \frac{80000 \times 1.69359}{4(0.00772576 - 0.00079375)} \\&= 4886288.39 \text{ pascal}\end{aligned}$$

Stress Due to Dead Load

Stress due to dead load in a distillation column is caused by the weight of the column and the equipment it supports. This weight creates a force that acts on the column and can cause it to deform or fail if the stress exceeds the material's yield strength. Proper design and material selection are critical to ensure the structural integrity of the column and prevent failure due to dead load stress. Dead load stress is one of several types of stresses that must be considered in the design and operation of distillation columns.

1. Compressive Stress Due to Weight of Shell

$$\begin{aligned}f_{dx} &= \frac{\text{weight of shell}}{\text{crosssection area of shell}} \\&= \rho_s h \\&= 7930 \times 3.6576 \\&= 29004.76 \text{ pascal}\end{aligned}$$

2. Compressive Stress Due to Weight of Insulation

$$f_{dins} = \frac{\pi D_{ins} t_{ins} \rho_{ins}}{\pi D_m (t_s - c)}$$

$$= \frac{3.1429 \times 0.1 \times 20}{\left(3.1429 \times \left(1.69359 + \frac{0.00772576}{2} \right) (0.00772576 - 0.00079375) \right)}$$

$$= 169.97 \text{ pascal}$$

3. Compressive Stress Due to Liquid in Column

$$f_{aliq} = \frac{\text{Total Weight of Liquid}}{\pi D_m (t_s - c)}$$

$$= \frac{7396}{\left(3.1429 \times \left(1.69359 + \frac{0.00772576}{2} \right) (0.00772576 - 0.00079375) \right)}$$

$$= 199990.53 \text{ pascal}$$

Stress Due to Wind Load

Stress due to wind load in a distillation column is caused by the force exerted by the wind on the column and the equipment it supports. Wind load stress can cause the column to bend or sway, which can lead to fatigue failure over time. Wind load stress must be considered during the design phase of the column, and appropriate measures such as the use of wind bracing or other structural reinforcements must be taken to prevent failure. The magnitude of wind load stress depends on factors such as wind speed, wind direction, and the shape and orientation of the column and its supporting equipment.

$$f_w = \frac{1.4 p_w h^2}{\pi D_o (t_s - c)}$$

$$= \frac{1.4 \times 239.4015 \times 3.6576^2}{\pi \times 1.90131576 \times (0.00772576 - 0.00079375)}$$

$$= 129766.03 \text{ pascal}$$

Elastic Stability

Elastic stability in a distillation column refers to the column's ability to resist deformation under external loads without undergoing permanent plastic deformation or failure.

$$f_c = \frac{1}{12} \frac{E}{\sqrt{3(1 - \mu^2)}} \frac{(t_s - c)}{\frac{D_o}{2}}$$

$$= \frac{1}{12} \frac{2000000000000}{\sqrt{3(1 - 0.29^2)}} \times \frac{(0.00772576 - 0.00079375)}{\frac{1.90131576}{2}}$$

$$= 73316043.61$$

Shell Thickness

$$t_s = \frac{pD_i}{2fJ - p} + c$$

$$t_s = 0.007 \text{ m}$$

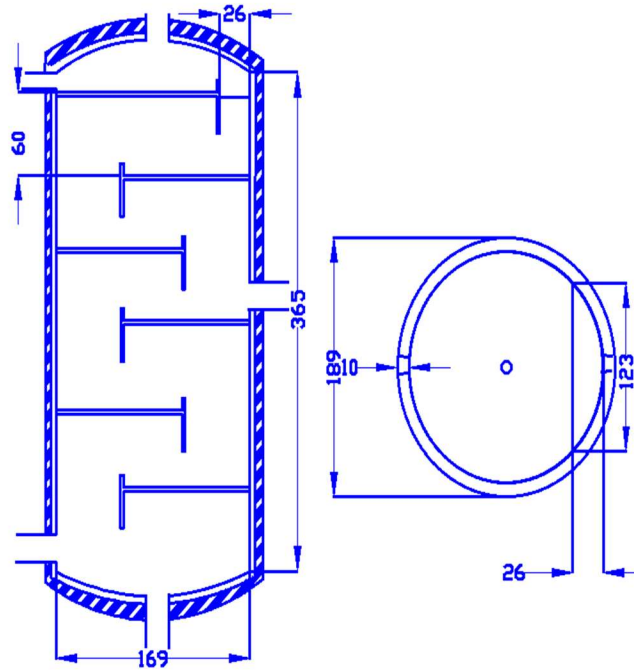


Figure 6. Distillation Column Design K-101 with dimension in cm

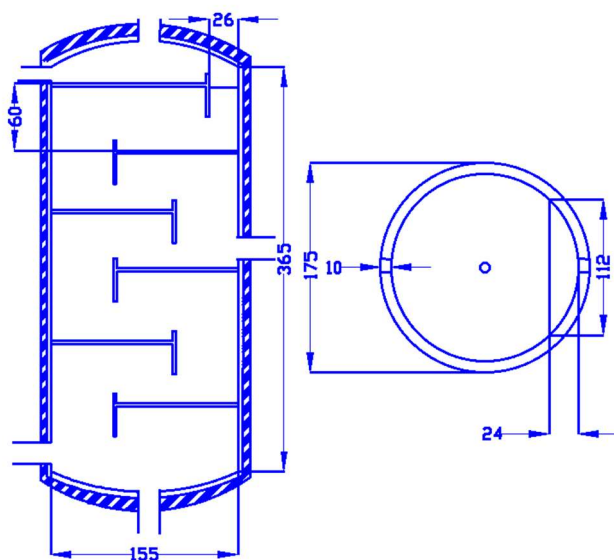


Figure 7. Distillation Column Design K-102 with dimension in cm

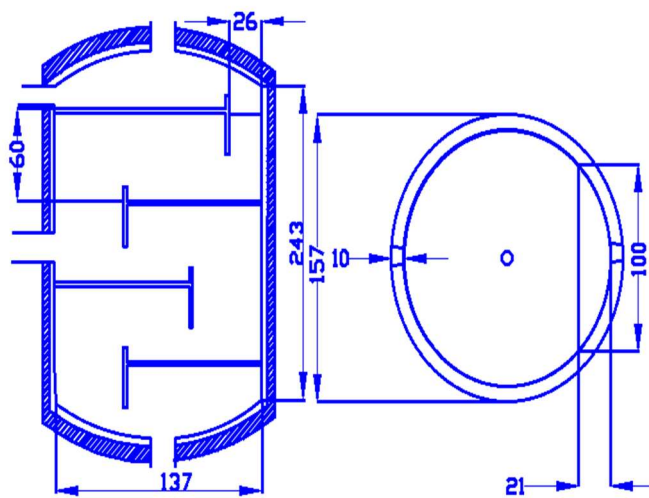


Figure 8. Distillation Column Design K-103 with dimension in cm

6.6 Design of Reactor

Data obtained from Aspen.

Volume of the Reactor, $V = 38.1369 \text{ m}^3$

Operating Pressure, $p = 1 \text{ bar}$

Operating temperature, $T = 108 \text{ }^\circ\text{C}$

The common material used in reactor and pressure vessels is stainless steel 303 because of its availability, strength, corrosion resistance nature and it is accepted material in most of the chemicals.

Data from Selected material

Elasticity, $E = 190 \times 10^9 \text{ Pa}$

Poisson's Ratio, $\mu = 0.3$

Now for the dimension of Reactor we suppose the L/D ratio as 4

$$\begin{aligned} V &= \frac{\pi}{4} D_i^2 \times L \text{ or } 38.1369 \\ &= \frac{\pi}{4} D_i^2 \times 4D_i \text{ or } D_i \\ &= 2.29 \text{ m} \end{aligned}$$

Design Pressure, p

Design pressure is taken 10% more than the operating pressure.

$$\begin{aligned} p &= 1.1 \times 0.1 \times 10^6 \\ &= 1.1 \times 10^5 \text{ Pascal} \end{aligned}$$

Allowable Stress, f

Assuming the ASME Boiler and Pressure Vessel Code (ASME BPVC) Section VIII Division 1 with a factor of safety of 3, the allowable stress for stainless steel 303 at $108 \text{ }^\circ\text{C}$ is approximately $75 \times 10^6 \text{ Pa}$.

Vessel Thickness, t_s

The shell thickness is taken considering corrosion

$$\begin{aligned} t_s &= \frac{pD_i}{2f - p} + C \\ &= \frac{1.1 \times 10^5 \times 2.29825654}{2 \times 75 \times 10^6 - 1.1 \times 10^5} + 0.00079375 \end{aligned}$$

$$= \frac{0.11 \times 2.29825654}{2 \times 75 - 0.11} + 0.00079375$$

$$= 0.0024 \text{ m}$$

External Pressure

As we are using a heat jacket that may itself give an external pressure to the reactor this pressure is necessary while designing the reactor. As the reactor need to be maintained at a temperature of 108 degree Celsius we will be using steam at temperature higher than 108 degree Celsius 1.5 atm pressure.

The length of jacket is considered throughout the length of the reactor so

$$L = 4D_i$$

$$= 9.19 \text{ m}$$

External design pressure is taken 10% more than working Pressure $p_o = 151987.5 \text{ Pa}$

External Diameter of the Vessel, D_o

After adding the thickness of the shell, external diameter is obtained though for very small thickness of vessel and very large diameter of the vessel the external and internal diameter are nearly equal. $D_o = D_i + 2t_s = 2.29825654 + 2 \times 0.002480374988 = 2.30 \text{ m}$

Jacket Length L

We suppose the length of the jacket is the same as the length of the reactor as we need to cover the whole length of the reactor. But when there is an elliptical head it may be slightly more because we need to cover the elliptical head also.

Critical buckling pressure, p_c

$$p_c = \frac{2.42 \times E \times \left(\frac{t_s - C}{D_o}\right)^{\frac{5}{2}}}{(1 - \mu^2)^{\frac{3}{4}} \left(\left(\frac{L}{D_o}\right) - 0.45 \left(\frac{t_s - C}{D_o}\right)^{\frac{1}{2}}\right)}$$

$$= \frac{2.42 \times 190 \times 10^9 \times \left(\frac{0.002480374988 - 0.00079375}{2.30321729}\right)^{\frac{5}{2}}}{(1 - 0.3^2)^{\frac{3}{4}} \left\{ \left(\frac{9.19302616}{2.30321729}\right) - 0.45 \left(\frac{0.002480374988 - 0.00079375}{2.30321729}\right)^{\frac{1}{2}} \right\}}$$

$$= \frac{2.42 \times 190 \times 10^9 \times 1.4511391 \times 10^{-8}}{0.9317106167 \times \{(3.99138466) - 0.45 \times 0.02706087337\}}$$

$$= \frac{6672.337582}{3.707469657}$$

$$= 1799.70 \text{ Pascal}$$

Critical stress, f_c

$$f_c = \frac{p_c D_o}{2t_s}$$

$$= \frac{1799.701198 \times 2.30321729}{2 \times 0.002480374988}$$

$$= 835579.88 \text{ Pascal}$$

Allowable stress, f_a

By considering the factor of safety of 3 we get the allowable stress

$$f_a = \frac{f_c}{3}$$

$$= \frac{835579.8894}{3}$$

$$= 278526.62 \text{ Pascal}$$

This value is far more less than given allowable stress, f_a

Allowable Pressure p_a

Considering the safety factor of 3

$$p_a = \frac{p_c}{3}$$

$$= \frac{1799.701198}{3}$$

$$= 599.90 \text{ Pa}$$

this is far less than design pressure $p \ 1.1 \times 10^5 \text{ Pa}$.

6.7 Design of Pump

Pump types and design parameters are mainly determined by fluid composition and fluid status. Flow from wells has complex composition and several possible phase statuses. In gas status, the flow is natural gas with some oil droplets, water droplets, and vapour. In oil status, except for oil, the flow contains water droplets and gas bubbles. In water status, the water also contains oil droplets, gas bubbles, and dissolved gas. In multiphase status, it is a combination of gas and liquid phases. The major influence factors for pump geometry are the volume flow rate that must be ingested and the fractions of this volume that are liquid and gas.

$$\text{Power} = \text{Volume flow} \times \text{Pressure drop} \quad (7)$$

$$\text{Head} = \frac{10.2 \times P}{S_g} \quad (8)$$

where,

$$\text{Specific gravity } (S_g) = \frac{\text{density of substance}}{\text{density of water } 4^\circ\text{C}} \quad (9)$$

6.8 Design of Compressor

A compressor is a device that increases the pressure of a substance (usually a gas) by reducing the volume of the substance. Compressors are used in many applications, most of which involve increasing the pressure inside a gas storage container, such as: Compression of gases in petroleum refineries and chemical plants.

$$\text{Actual work} = \frac{\text{Enthalpy Difference}}{\text{Mechanical Efficiency}} \quad (10)$$

$$\text{Shaft power} = \frac{\text{Pressure drop} \times \text{Volume flow}}{\text{Efficiency}} \quad (11)$$

$$\text{Brake horsepower (BHP)} = \frac{100 QH}{3960 n} \quad (12)$$

$$\text{Net positive suction head (NPSH)} = \frac{v^2}{2g} + \frac{P}{SW} - \frac{P_v}{SW} \quad (13)$$

The results of the calculations are summarised in the specification sheet of compressor in Chapter 7.

Chapter 7 Equipment and Instrument Specification

7.1 Equipment List

Table 12: Equipment List

S. No.	Item No.	Equipment Name	Qty	Specifications	Material of Construction
1	E105, E111, E119	Condenser	3	Pull-through floating head	Cast Iron, SS-316
2	E106, E110, E112	Reboiler	3	Kettle Type	Cast iron, SS 316
3	E101, E102, E103, E107, E108, E109	Heat Exchanger	6	Shell and Tube	SS 304
4	F105	Crystallizer	1	Isothermal	SS 304
5	C101, C102	Reactor	2	CSTR	SS 304
6	F104	Decanter	1	Isothermal	SS 304
7	F101	Flash	1	Horizontal	SS 304

7.2 Specifications Sheet of Pipes

Table 13: Specification Sheet of Pipe

Pipe number in PFD	Standard Steel Pipe					
	Calculate d inside diameter (in.)	Inside diameter (in.)	Outside Diameter (in.)	Nominal Diameter (in.)	Schedule no.	Wall thickness (in.)
1.	6.66	7.981	8.625	8	40s	0.322
2.	1.22	1.380	1.660	1	40s	0.140
3.	4.29	5.047	5.656	5	40s	0.258
4.	1.24	1.380	1.660	1	40s	0.140
5.	3.62	5.047	5.656	5	40s	0.258
6.	1.25	1.380	1.660	1	40s	0.140

Pipe number in PFD	Standard Steel Pipe					
	Calculate d inside diameter (in.)	Inside diameter (in.)	Outside Diameter (in.)	Nominal Diameter (in.)	Schedule no.	Wall thickness (in.)
7.	3.66	5.047	5.656	5	40s	0.258
8.	9.13	9.981	10.625	10	40s	0.322
9.	3.81	5.047	5.656	5	40s	0.258
10.	1.25	1.380	1.660	1	40s	0.140
11.	15.9	16.380	17.660	17	40s	0.425
12.	29.9	30.981	32.625	32	40s	0.280
13.	0.82	1.049	1.315	1	40s	0.133
14.	18.3	19.065	19.625	19	40s	0.280
15.	29.8	30.049	30.315	30	40s	0.248
16.	0.82	1.049	1.315	1	40s	0.133
17.	0.62	1.049	1.315	1	40s	0.133
18.	0.43	1.049	1.315	1	40s	0.133
19.	0.47	1.049	1.315	1	40s	0.133
20.	1.18	1.380	1.660	1	40s	0.140
21.	1.45	1.59	2.15	2	40s	0.133
22.	1.45	1.680	2.15	2	40s	0.140
23.	1.45	1.80	2.15	2	40s	0.140
24.	24.6	26.049	28.315	28	40s	0.133
25.	1.10	1.380	1.660	1	40s	0.140
26.	0.82	1.049	1.315	1	40s	0.133

7.3 Specifications Sheet of Heat Exchanger

Table 14: Specification Sheet of Heat Exchanger (HE-101)

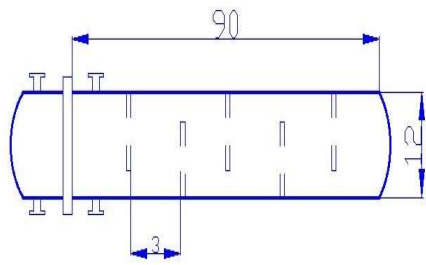
HEAT EXCHANGER SPECIFICATION SHEET					
Company: Kathmandu University					
Tag no.: HE 101			Design Code: TEMA		
P & ID no.: HE 101					
Service of Unit: Cooling					
FLUID PROPERTY TABLE					
Fluid Allocation		Tube Side		Shell Side	
Fluid name		S3		WATER	
Fluid Quantity, Total	kg/s	0.055		0.306	
Vapour(In/out)	kg/s	0.055	0.055	-	-
Liquid (In/Out)	kg/s	-	-	0.306	0.306
Non-Condensable	kg/s				
Temperature	°C	226.09	85	25	95
Density	kg/m^3	0.194	0.270	1000	1000
Viscosity	kg/m/s	3.05e-5	1.84e-5	0.89e-3	0.280e-3
Molecular weight	g/mol	2.01	2.01	18.01	18.01
Specific heat capacity	J/kg/K	8620	14290	4180	4100
Thermal Conductivity	W/m/K	0.172	0.200	0.606	0.644
Mass velocity	kg/s.m2				
Pressure Drop	bar	0	0	0	0
Overall Transfer Coefficient					
CONSTRUCTION DATA					
Heat Transfer area (m^2)					
	Shell Side	Tube Side			
Design Pressure (atm)					
Design Temp. (°C)					
No. of passes per Shell	1				
Material	Stainless Steel				
Tube no: 146 OD: 19.05 mm	Birmingham wire gauge (BWG): 1.651 mm	Length: 1822.8 mm	Pitch: 23.8 mm		
Tube Type: Plain		Tube Pattern: 30-triangular			
Inside shell diameter: 151.87 mm		Shell Thickness:			
Head: Pull through U-Tube	Thickness:		Insulation: none		
Baffle: Vertical	Cut(%d): 25		Spacing: 30.37 mm	Type: Single	

Table 15: Specification Sheet of Heat Exchanger (HE-102)

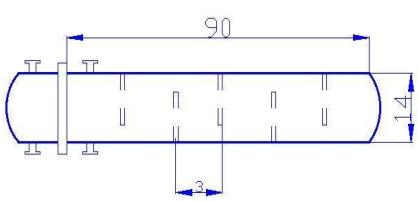
HEAT EXCHANGER SPECIFICATION SHEET					
Company: Kathmandu University					
Tag no.: HE 102			Design Code: TEMA		
P & ID no.: HE 102					
Service of Unit: Heating					
FLUID PROPERTY TABLE					
Fluid Allocation		Tube Side		Shell Side	
Fluid name		S4		Water	
Fluid Quantity, Total	kg/s	1.70		0.542	
Vapor(In/out)	kg/s	-	-	-	-
Liquid (In/Out)	kg/s	1.70	1.70	0.542	0.542
Non-Condensable	kg/s				
Temperature	°C	25.36	85	95	30
Density	kg/m^3	1198.14	1138.59	1000	1000
Viscosity	kg/m/s	0.0428	0.0239	0.00028	0.000798
Molecular weight	g/mol	123.11	123.11	18.01	18.01
Specific heat capacity	J/kg/K	1530	1380	4180	4180
Thermal Conductivity	W/m/K	0.140	0.126	0.652	0.606
Pressure Drop	bar	0	0	0	0
Fouling Resistance (minimum)	W/m^2/°C				
Overall Transfer Coefficient					
CONSTRUCTION DATA					
Heat Transfer area (m^2)					
	Shell Side	Tube Side			
Design Pressure (atm)					
Design Temp. (°C)					
No. of passes per Shell	1				
Material	Stainless Steel				
Tube no: 146 OD: 19.05 mm	Birmingham wire gauge (BWG): 1.651 mm				
Tube Type: Plain			Tube Pattern: 30-triangular		
Inside shell diameter : 164.76 mm			Shell Thickness:		
Head: Pull through U-Tube	Thickness:		Insulation: none		
Baffle: Vertical	Cut(%d): 25		Spacing: 32.95 mm	Type: Single	

Table 16: Specification Sheet of Heat Exchanger (HE-103)

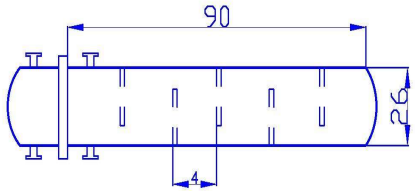
HEAT EXCHANGER SPECIFICATION SHEET					
Company: Kathmandu University					
Tag no.: HE 103			Design Code: TEMA		
P & ID no.: HE 103					
Service of Unit: Heating					
FLUID PROPERTY TABLE					
Fluid Allocation		Tube Side		Shell Side	
Fluid name		S10		WATER	
Fluid Quantity, Total	kg/s	1.73		1.25	
Vapor(In/out)	kg/s	-	-	-	-
Liquid (In/Out)	kg/s	1.73	1.73	1.25	1.25
Non-Condensable	kg/s				
Temperature	°C	25	120	95	30
Density	kg/m^3	1158.38	7.311	1000	1000
Viscosity	kg/m/s	0.0027	0.0016	0.00028	0.000798
Molecular weight	g/mol	71.82	71.82	18.01	18.01
Specific heat capacity	J/kg/K	1933.28	2212.55	4180	4180
Thermal Conductivity	W/m/K	0.192	0.193	0.652	0.606
Pressure Drop	bar	0	0	0	0
Fouling Resistance (minimum)	W/m^2/°C				
Overall Transfer Coefficient					
CONSTRUCTION DATA					
Heat Transfer area (m^2)					
	Shell Side	Tube Side			
Design Pressure (atm)					
Design Temp. (°C)					
No. of passes per Shell	1				
Material	Stainless Steel				
Tube no: 146 OD: 19.05 mm	Birmingham wire gauge (BWG): 1.651 mm	Length: 1822.8 mm	Pitch: 23.8 mm		
Tube Type: Plain		Tube Pattern: 30-triangular			
Inside shell diameter: 224.18 mm		Shell Thickness:			
Head: Pull through U-Tube	Thickness:		Insulation: none		
Baffle: Vertical	Cut(%d): 25	Spacing: 44.83 mm	Type: Single		

Table 17: Specification Sheet of Heat Exchanger (HE-107)

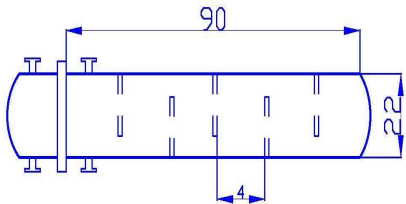
HEAT EXCHANGER SPECIFICATION SHEET						
Company: Kathmandu University						
Tag no.: HE 107			Design Code: TEMA			
P & ID no.: HE 107						
Service of Unit: Cooling						
FLUID PROPERTY TABLE						
Fluid Allocation		Tube Side		Shell Side		
Fluid name		S12		WATER		
Fluid Quantity, Total	kg/s	1.1026		0.66		
Vapor(In/out)	kg/s	-	-	-	-	
Liquid (In/Out)	kg/s	1.1026	1.1026	0.66	0.66	
Non-Condensable	kg/s					
Temperature	°C	167.7	120	25	95	
Density	kg/m^3	1.324	3.524	1000	1000	
Viscosity	kg/m/s	0.0027	0.0016	0.89e-3	0.280e-3	
Molecular weight	g/mol	59.98	59.98	18.01	18.01	
Specific heat capacity	J/kg/K	3834.81	3505.16	4180	4100	
Thermal Conductivity	W/m/K	0.263	0.252	0.606	0.644	
Pressure Drop	bar	-0.2	-0.2	0	0	
Fouling Resistance (minimum)	W/m^2/°C					
Overall Transfer Coefficient						
CONSTRUCTION DATA						
Heat Transfer area (m^2)						
	Shell Side					Tube Side
Design Pressure (atm)						
Design Temp. (°C)						
No. of passes per Shell	1					
Material	Stainless Steel					
Tube no: 146 OD: 19.05 mm	Birmingham wire gauge (BWG): 1.651 mm					Length: 1822.8 mm
Tube Type: Plain		Tube Pattern: 30-triangular				
Inside shell diameter: 204.67 mm		Shell Thickness:				
Head: Pull through U-Tube	Thickness:		Insulation: none			
Baffle: Vertical	Cut(%d): 25		Spacing: 40.93 mm	Type: Single		

Table 18: Specification Sheet of Heat Exchanger (HE-108)

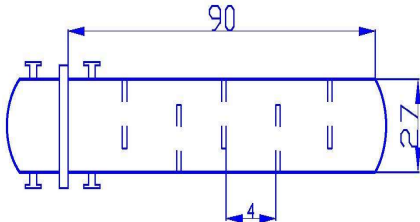
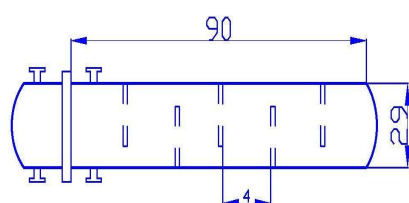
HEAT EXCHANGER SPECIFICATION SHEET					
Company: Kathmandu University					
Tag no.: HE 108			Design Code: TEMA		
P & ID no.: HE 108					
Service of Unit: Cooling					
FLUID PROPERTY TABLE					
Fluid Allocation		Tube Side		Shell Side	
Fluid name		S7		WATER	
Fluid Quantity, Total	kg/s	0.426		0.416	
Vapor(In/out)	kg/s	-	-	-	-
Liquid (In/Out)	kg/s	0.426	0.426	0.416	0.416
Non-Condensable	kg/s				
Temperature	°C	115.88	25	25	95
Density	kg/m^3	0.516	1015.07	1000	1000
Viscosity	kg/m/s	0.255e-03	4.40e-4	0.89e-3	0.280e-3
Molecular weight	g/mol	33.19	33.19	18.01	18.01
Specific heat capacity	J/kg/K	3054.38	3185.72	4180	4100
Thermal Conductivity	W/m/K	0.386	0.408	0.606	0.644
Pressure Drop	bar	-0.5	-0.5	0	0
Fouling Resistance (minimum)	W/m^2/°C				
Overall Transfer Coefficient					
CONSTRUCTION DATA					
Heat Transfer area (m^2)					
	Shell Side	Tube Side			
Design Pressure (atm)					
Design Temp. (°C)					
No. of passes per Shell	1				
Material	Stainless Steel				
Tube no: 146 OD: 19.05 mm	Birmingham wire gauge (BWG): 1.651 mm	Length: 1822.8 mm	Pitch: 23.8 mm		
Tube Type: Plain		Tube Pattern: 30-triangular			
Inside shell diameter: 227.12 mm		Shell Thickness:			
Head: Pull through U-Tube	Thickness:	Insulation: none			
Baffle: Vertical	Cut(%d): 25	Spacing: 45.42 mm	Type: Single		

Table 19: Specification Sheet of Heat Exchanger (HE-109)

HEAT EXCHANGER SPECIFICATION SHEET					
Company: Kathmandu University					
Tag no.: HE 109			Design Code: TEMA		
P & ID no.: HE 109					
Service of Unit: Cooling					
FLUID PROPERTY TABLE					
Fluid Allocation		Tube Side		Shell Side	
Fluid name		S1		WATER	
Fluid Quantity, Total	kg/s	2.054		0.061	
Vapor(In/out)	kg/s	-	-	-	-
Liquid (In/Out)	kg/s	2.054	2.054	0.061	0.061
Non-Condensable	kg/s				
Temperature	°C	111.99	108	25	95
Density	kg/m^3	1007.35	1011.8	1000	1000
Viscosity	kg/m/s	0.000255	0.00044	0.89e-3	0.280e-3
Molecular weight	g/mol	104.17	104.17	18.01	18.01
Specific heat capacity	J/kg/K	2254.51	2094.20	4180	4100
Thermal Conductivity	W/m/K	0.119	0.126	0.606	0.644
Pressure Drop	bar	-0.2	-0.2	0	0
Fouling Resistance (minimum)	W/m^2/°C				
Overall Transfer Coefficient					
CONSTRUCTION DATA					
Heat Transfer area (m^2)					
	Shell Side	Tube Side			
Design Pressure (atm)					
Design Temp. (°C)					
No. of passes per Shell	1				
Material	Stainless Steel				
Tube no: 146 OD: 19.05 mm	Birmingham wire gauge (BWG): 1.651 mm	Length: 1822.8 mm	Pitch: 23.8 mm		
Tube Type: Plain		Tube Pattern: 30-triangular			
Inside shell diameter: 235.56 mm		Shell Thickness:			
Head: Pull through U-Tube	Thickness:		Insulation: none		
Baffle: Vertical	Cut(%d): 25		Spacing: 47.11 mm	Type: Single	

7.4 Specifications Sheet of Condenser

Table 20: Specification Sheet of Condenser (E-105)

PROJECT: BASIC DESIGN OF ACETAMINOPHEN PLANT			
Section:	Specification Sheet Of Condenser	Company:	Kathmandu University
Project Owner:	Ashok, Krishna, Nishant	Checked By:	Dr. Bibek Uprety
CONDENSER SPECIFICATION SHEET			
Tag No		E-105	
Service		Cooling	
PROCESS DATA			
Heat Duty	1969.09 kg/h	U,assume	750 W / m ² °C
Pressure	0.8 bar	ΔT_{lm}	99.24 °C
Vapor Enter	120 °C	<i>Trial area</i>	9.47 m ²
Condensation Complete	168 °C	<i>Surface area of one tube</i>	0.305 m ²
Average Molecular Weight of Vapors	60	<i>Number of tubes</i>	31
Enthalpy of Vapor	1535.79 kJ/kg	Pitch, P_t	25 mm
Enthalpy of condensate	247.0 kJ/kg	Tube bundle diameter D_b	206 mm
Cooling Water Available	30 °C	Number of tubes in center row N_r	5.49
Temperature limited	40 °C	<i>Mean temperature condensate</i>	111 °C
O.D. of tubes	20 mm	Mean coefficient for a tube bundle h_c	2055.50 W /m ² °C
I.D. of tubes	16.8 mm	<i>Tube cross – sectional area</i>	0.0014 m ²
Length	4.88 m (16 ft)	<i>Tube velocity</i>	10.16 m/s
CONSTRUCTION AND MATERIALS			
Head	Pull-through floating head	Mass flow-rate G_s	30.38 kg /s m ²
Baffle spacing D	45% cut	<i>Equivalent diameter d_e</i>	19.8 mm
Clearance	95 mm	Pressure drop	50%
Shell i.d.	301 mm	Shell-side pressure drop ΔP_s	56.8 kPa
<i>Cross – flow area A_s</i>	0.018 m ²	Tube-side pressure drop ΔP_t	2180 kPa
Material of Construction	Ferritic stainless steel		
Remarks : Condenser Should be always operated at specified condition.			

Table 21: Specification Sheet of Condenser (E-106)

PROJECT: BASIC DESIGN OF ACETAMINOPHEN PLANT			
Section:	Specification Sheet Of Condenser	Company:	Kathmandu University
Project Owner:	Ashok, Krishna, Nishant	Checked By:	Dr. Bibek Uprety
CONDENSER SPECIFICATION SHEET			
Tag No		E-106	
Service		Cooling	
PROCESS DATA			
Heat Duty	1372.26 kg/h	U,assume	900 W/m ² °C
Pressure	0.5 bar	ΔT_{lm}	76.32 °C
Vapour Enter	120 °C	<i>Trial area</i>	27.59 m ²
Condensation Complete	116 °C	<i>Surface area of one tube</i>	0.305 m ²
Average Molecular Weight of Vapours	33	<i>Number of tubes</i>	90
Enthalpy of Vapour	5219.67 kJ/kg	Pitch, P_t	25 mm
Enthalpy of condensate	247.0 kJ/kg	Tube bundle diameter D_b	330.14 mm
Cooling Water Available	30 °C	Number of tubes in centre row N_r	13.20
Temperature limited	40 °C	<i>Mean temperature condensate</i>	93 °C
O.D. of tubes	20 mm	Mean coefficient for a tube bundle h_c	1506.15 W/m ² °C
I.D. of tubes	16.8 mm	<i>Tube cross – sectional area</i>	0.0049 m ²
Length	4.88 m (16 ft)	<i>Tube velocity</i>	9.31 m/s
<i>Heat transferred from vapou</i>	1895.5 kW	Tube-side coefficient h_i	29181.05 W/m ² °C
CONSTRUCTION AND MATERIALS			
Head	Pull-through floating head	Mass flow-rate G_s	10.58 kg/s m ²
Baffle spacing D	45% cut	<i>Equivalent diameter d_e</i>	19.8 mm
Clearance	95 mm	Pressure drop	50%
Shell i.d.	425.14 mm	Shell-side pressure drop ΔP_s	0.11 kPa
<i>Cross – flow area A_s</i>	0.036 m ²	Tube-side pressure drop ΔP_t	1830 kPa
Material of Construction	Ferritic stainless steel		
Remarks : Condenser Should be always operated at specified condition.			

Table 22: Specification Sheet of Condenser (E-110)

PROJECT: BASIC DESIGN OF ACETAMINOPHEN PLANT			
Section:	Specification Sheet Of Condenser	Company:	Kathmandu University
Project Owner:	Ashok, Krishna, Nishant	Checked By:	Dr. Bibek Uprety
CONDENSER SPECIFICATION SHEET			
Tag No		E-110	
Service		Cooling	
PROCESS DATA			
Heat Duty	1363.01 kg/h	U,assume	900 W/m ² °C
Pressure	0.8 bar	ΔT_{lm}	74.92 °C
Vapour Enter	108 °C	<i>Trial area</i>	32.17 m ²
Condensation Complete	125 °C	<i>Surface area of one tube</i>	0.305 m ²
Average Molecular Weight of Vapours	85	<i>Number of tubes</i>	105
Enthalpy of Vapour	5977.67 kJ/kg	Pitch, P_t	25 mm
Enthalpy of condensate	247.0 kJ/kg	Tube bundle diameter D_b	353.41 mm
Cooling Water Available	30 °C	Number of tubes in center row N_r	14.13
Temperature limited	40 °C	<i>Mean temperature condensate</i>	92 °C
O.D. of tubes	20 mm	Mean coefficient for a tube bundle h_c	1519.14 W/m ² °C
I.D. of tubes	16.8 mm	<i>Tube cross – sectional area</i>	0.0058 m ²
Length	4.88 m (16 ft)	<i>Tube velocity</i>	9.01 m/s
<i>Heat transferred from vapou</i>	2169.71 kW	Tube-side coefficient h_i	28426.35 W/m ² °C
CONSTRUCTION AND MATERIALS			
Head	Pull-through floating head	Mass flow-rate G_s	9.46 kg/s m ²
Baffle spacing D	45% cut	<i>Equivalent diameter d_e</i>	19.8 mm
Clearance	95 mm	Pressure drop	50%
Shell i.d.	448.41 mm	Shell-side pressure drop ΔP_s	0.035 kPa
<i>Cross – flow area A_s</i>	0.040 m ²	Tube-side pressure drop ΔP_t	1714 kPa
Material of Construction	Ferritic stainless steel		
Remarks : Condenser Should be always operated at specified condition.			

7.5 Specifications Sheet of Reboiler

Table 23: Specification Sheet of Reboiler (E-111)

PROJECT: BASIC DESIGN OF ACETAMINOPHEN PLANT			
Section:	Specification Sheet Of Reboiler	Company:	Kathmandu University
Project Owner:	Ashok, Krishna, Nishant	Checked By:	Dr. Bibek Uprety
REBOILER SPECIFICATION SHEET			
Tag No		E-111	
Service		Boiling	
PROCESS DATA			
Temperature		271.94 °C	
Pressure		0.8 bar	
Boiling Point		287 °C	
Latent heat		280.3 kJ/kg	
Mean specific heat, liquid		1.56 kJ/kg°C	
Critical Pressure, P_c		47.1 bar	
Sensible heat (maximum)		447.72 kJ/kg	
Total heat load		462.94 kW	
Maximum heat load (duty)		486.09 kW	
U , assume		1000 W/m ² °C	
ΔT_m		59.1°C	
Area (outside) required		8.22 m ²	
Internal Diameter		25 mm	
Nominal length (one U-tube)		4.8 m	
Number of U tubes		19	
Pitch		45 mm	
Heat flux, (estimated)		59.13 kW/m ²	
Boiling coefficient		3021.10 W/m ² °C	
Surface tension (estimated)		9.7 * 10 ⁻³ N/m	
Maximum allowable heat flux		66.5 kW/m ²	
CONSTRUCTION AND MATERIALS			
Minimum bend radius		45 mm	
Tube sheet layout D_b		420 mm	
Shell diameter D_s		840 mm	
Surface area of liquid		1.9 m ²	
Vapour velocity at surface		0.386 m/s	
Maximum allowable velocity		3.40 m/s	
Remarks : Reboiler Should be always operated at specified condition.			

Table 24: Specification Sheet of Reboiler (E-112)

PROJECT: BASIC DESIGN OF ACETAMINOPHEN PLANT			
Section:	Specification Sheet Of Reboiler	Company:	Kathmandu University
Project Owner:	Ashok, Krishna, Nishant	Checked By:	Dr. Bibek Uprety
REBOILER SPECIFICATION SHEET			
Tag No		E-112	
Service		Boiling	
PROCESS DATA			
Temperature		183.87 °C	
Pressure		0.5 bar	
Boiling Point		170.2 °C	
Latent heat		254.4 kJ/kg	
Mean specific heat, liquid		0.809 kJ/kg°C	
Critical Pressure, P_c		44.8 bar	
Sensible heat (maximum)		148.75 kJ/kg	
Total heat load		264.33 kW	
Maximum heat load (duty)		277.55 kW	
U , assume		1000 W/m ² °C	
ΔT_m		66.2°C	
Area (outside) required		4.19 m ²	
Internal Diameter		25 mm	
Outside Diameter		30 mm	
Nominal length (one U-tube)		4.8 m	
Number of U tubes		10	
Pitch		45 mm	
Heat flux, (estimated)		46.24 kW/m ²	
Boiling coefficient		2909.98 W/m ² °C	
Surface tension (estimated)		9.7 * 10 ⁻³ N/m	
Maximum allowable heat flux		56.94 kW/m ²	
CONSTRUCTION AND MATERIALS			
Minimum bend radius		45 mm	
Tube sheet layout D_b		420 mm	
Shell diameter D_s		840 mm	
Freeboard		340 mm	
Surface area of liquid		1.9 m ²	
Vapour velocity at surface		0.432 m/s	
Maximum allowable velocity		3.60m/s	
Remarks : Reboiler Should be always operated at specified condition.			

Table 25: Specification Sheet of Reboiler (E-113)

PROJECT: BASIC DESIGN OF ACETAMINOPHEN PLANT			
Section:	Specification Sheet Of Reboiler	Company:	Kathmandu University
Project Owner:	Ashok, Krishna, Nishant	Checked By:	Dr. Bibek Uprety
REBOILER SPECIFICATION SHEET			
Tag No		E-113	
Service		Boiling	
PROCESS DATA			
Temperature		245.72 °C	
Pressure		0.8 bar	
Boiling Point		170 °C	
Latent heat		241.5 kJ/kg	
Mean specific heat, liquid		1.38 kJ/kg°C	
Critical Pressure, P_c		58 bar	
Sensible heat (maximum)		234.6 kJ/kg	
Total heat load		418.60 kW	
Maximum heat load (duty)		436.53 kW	
U , assume		1000 W/m ² °C	
ΔT_m		66.4 °C	
Area (outside)required		6.57 m ²	
Internal Diameter		25 mm	
Outside Diameter		30 mm	
Nominal length (one U-tube)		4.8 m	
Number of U tubes		15	
Pitch		45 mm	
Heat flux, estimated		66.44 kW/m ²	
Boiling coefficient		3631.32 W/m ² °C	
Surface tension (estimated)		9.7 * 10 ⁻³ N/m	
Maximum allowable heat flux		75.6 kW/m ²	
CONSTRUCTION AND MATERIALS			
Minimum bend radius		45 mm	
Tube sheet layout D_b		420 mm	
Shell diameter D_s		840 mm	
Freeboard		340 mm	
Surface area of liquid		1.9 m ²	
Vapour velocity at surface		0.220 m/s	
Maximum allowable velocity		2.56 m/s	
Remarks : Reboiler Should be always operated at specified condition.			

7.6 Equipment Sheet of Pump

Table 26: Specification Sheet of Pump (P-101)

CENTRIFUGAL PUMP SPECIFICATION SHEET			
Client: Kathmandu University		Client's Ref No:1	
Document No: 1			
TAG NO.: P-101 PID-1	DESIGN CODE: P-101		
Service: Pumping	CENTRIFUGAL PUMP		
Fluid Name	Nitrobenzene		
Fluid Density (Kg/m³)	1163.78		
Volumetric Flow Rate (cum/hr)	7.16		
Flow Rate (kg/hr)	6155.56	Electricity (KW)	1.88
Temperature (°C)	Operating		Design
	25.36		28.25
Pressure (bar)	Operating		Design
	4		4.5
Head (m)	25.52		
Pump Efficiency	100 (assumed)		
Material Name (casing, impellers)	Cast iron, SS 316		
Electric Motor	Included		
NPSHA (m)	8.50		
Brake Horsepower (KW)	1.35		
Actual Pump Power (KW)	0.42		
Pump Connection to Pipe (in)	INLET		OUTLET
	1.049		1.049

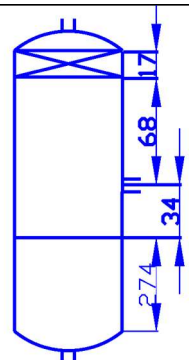
7.7 Equipment Sheet of Compressor

Table 27: Specification Sheet of Compressor (C-101)

COMPRESSOR SPECIFICATION SHEET		DATE: 24 th April. 2023
		REF NO.:
ITEM:		COMPRESSOR
TEG NO:		C-101
SERVICE		FLOW OF AIR
BASIC DESIGN OF ACETAMINOPHEN PLANT		
PROCESS DATA		
Fluid Allocation	Feed air IN	Feed air OUT
Design Temperature,K	298.15	499.24
Pressure, Bar	1.013	4
Enthalpy, MJ/kg	0.00027	2.910
Components (% H2)	1	1
Density, Kg/m3	0.081	0.194
Fluid State	Gas	Gas
Volume Rate cum/sec	0.688	0.288
Molecular Weight	2.01	2.01
Actual Work, Kw	162.95	
Mechanical Efficiency, %	72 (ASSUMED)	
Shaft Power, Kw	162.95	
CONSTRUCTION AND MATERIALS		
Fluid Allocation	Feed IN	Feed OUT
Pipe Line Size, in	7.981	6.065
Casting: Line & Piston Material	Cast Iron, SS-316	
Remarks	Compressor Should be always operated at specified condition.	

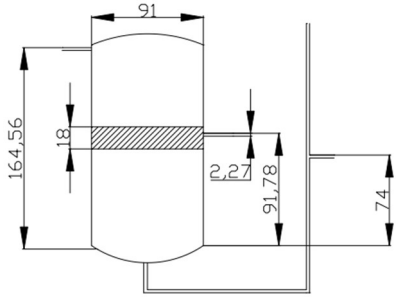
7.8 Equipment Sheet of Flash Drum

Table 28: Specification Sheet of Flash Drum (F-101)

Company : Tirupati Pharmaceuticals	
Location : Janakpur, Nepal	
Service of Unit: Separation of Gas and Liquid	
Tag No: F101	
Date : 25 th APRIL ,2023	
Flash Type: Isothermal	
PERFORMANCE OF UNIT	
Operating Temperature, (°C)	85
Operating Pressure, (bar)	4
Maximum Allowable Temperature, (°C)	94
Maximum Allowable Pressure, (bar)	4.4
DESIGN SPECIFICATION	
Diameter, (m)	0.6
Height, (m)	3.2
Material	Carbon Steel
Design Code	ASME BPVC, Section VIII, Division I
Nozzle Rating	-
Head Type	Ellipsoidal
Shell Type	Cylindrical
	
REMARKS: The demister pad of thickness 0.1 m is used.	

7.9 Equipment Sheet of Decanter

Table 29: Specification Sheet of Decanter (F-104)

Company: Tirupati Pharmaceuticals	
Location: Janakpur, Nepal	
Service of Unit: Separation of Gas and Liquid	
Item No. F104	
Date: 25 th APRIL ,2023	
Type: Isothermal	
PERFORMANCE OF UNIT	
Operating Temperature, (K)	25
Operating Pressure, (bar)	1
Maximum Allowable Temperature, (K)	28
Maximum Allowable Pressure, (bar)	1.1
DESIGN SPECIFICATION	
Diameter, (m)	0.9
Height, (m)	1.6
Material	Carbon Steel
Design Code	ASME BPVC, Section VIII, Division I
Nozzle Rating	
Head Type	Ellipsoidal
Shell Type	Cylindrical
	
REMARKS: The demister pad of thickness 0.2 m is used.	

7.10 Equipment Sheet of Distillation Column

Table 30: Specification Sheet of Distillation Column (K-101)

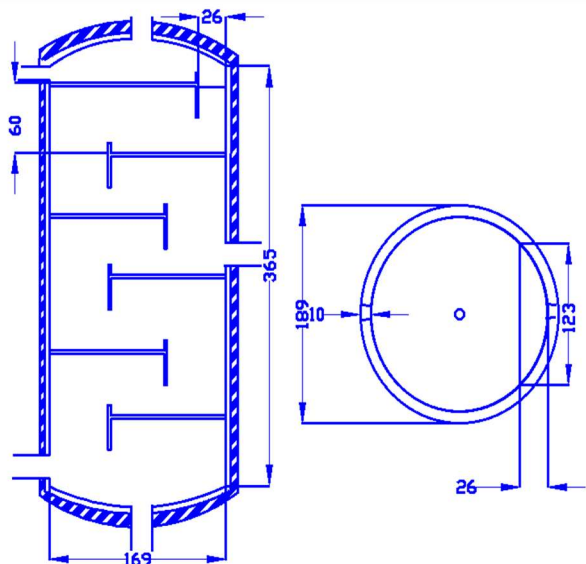
PROJECT: BASIC DESIGN OF ACETAMINOPHEN PLANT			
Section:	Specification Sheet Of Distillation Column	Company:	Kathmandu University
Project Owner:	Ashok, Krishna, Nishant	Checked By:	Dr. Bibek Uprety
DISTILLATION COLUMN SPECIFICATION SHEET			
Tag No		K-101	
Service		Separation	
PROCESS DATA			
Column Internal Summary		Down Comer Geometry	
No. of Trays	6	Downcomer Clearance, (m)	0.03
Total Diameter, (m)	1.8	Downcomer Width Top, (mm)	265
Total height, (m)	3.65	Downcomer Width Bottom, (mm)	265
Tray		Downcomer Area Top, (m ²)	0.225
Tray Type	SIEVE	Downcomer Area Bottom, (m ²)	0.225
Diameter, (m)	1.69		
Tray Spacing, (m)	0.60		
Hole Diameter, (m)	0.01		
Hole Area / Active Area, (m ²)	0.1		
Number of Holes	1423		
Deck Gauge Thickness Value, (mm)	3.4		
Cross-Sectional Area, (m ²)	2.25		
Active Area, (m ²)	1.80		
Net Area, (m ²)	2.02		
Weir Geometry			
Weir Height, (m)	0.05		
Weir Length, (m)	1.23		
Remarks : Distillation Column Should be always operated at specified condition.			

Table 31: Specification Sheet of Distillation Column (K-102)

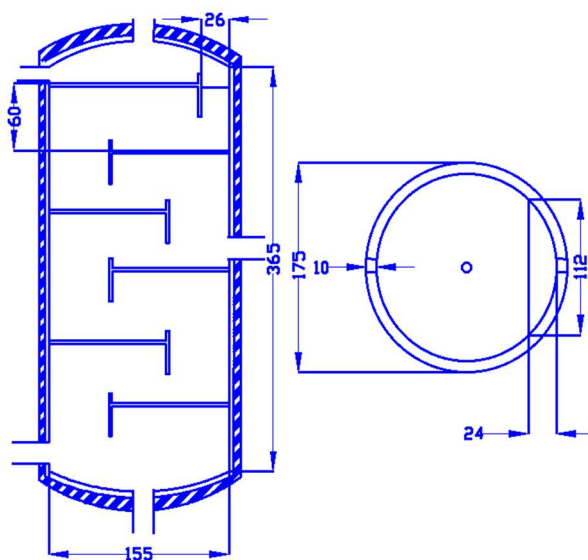
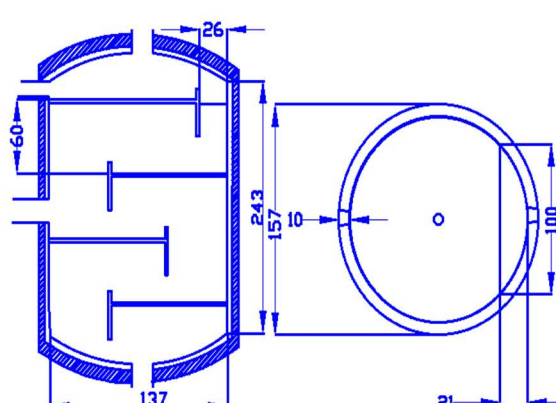
PROJECT: BASIC DESIGN OF ACETAMINOPHEN PLANT			
Section:	Specification Sheet Of Distillation Column	Company:	Kathmandu University
Project Owner:	Ashok, Krishna, Nishant	Checked By:	Dr. Bibek Uprety
DISTILLATION COLUMN SPECIFICATION SHEET			
Tag No		K-102	
Service		Separation	
PROCESS DATA			
Column Internal Summary		Down Comer Geometry	
No. of Trays	6	Downcomer Clearance, (m)	0.03
Total Diameter, (m)	1.75	Downcomer Width Top, (mm)	243.04
Total height, (m)	3.65	Downcomer Width Bottom, (mm)	243.04
Tray		Downcomer Area Top, (m ²)	0.189
Tray Type	SIEVE	Downcomer Area Bottom, (m ²)	0.189
Diameter, (m)	1.55		
Tray Spacing, (m)	0.60		
Number of Passes	1		
Hole Diameter, (m)	0.01		
No of Holes	1197		
Hole Area, (m ²)	0.1		
Deck Thickness, (Gauge)	10		
Deck Thickness, (mm)	3.4		
Cross-Sectional Area, (m ²)	1.89		
Active Area, (m ²)	1.51		
Net Area, (m ²)	1.70		
Weir Geometry			
Weir Height, (m)	0.05		
Weir Length, (m)	1.12		
Remarks : Distillation Column Should be always operated at specified condition.			

Table 32: Specification Sheet of Distillation Column (K-103)

PROJECT: BASIC DESIGN OF ACETAMINOPHEN PLANT			
Section:	Specification Sheet Of Distillation Column	Company:	Kathmandu University
Project Owner:	Ashok, Krishna, Nishant	Checked By:	Dr. Bibek Uprety
DISTILLATION COLUMN SPECIFICATION SHEET			
Tag No		K-103	
Service		Separation	
PROCESS DATA			
Column Internal Summary		Down Comer Geometry	
No. of Trays	4	Downcomer Clearance, (m)	0.03
Total Diameter, (m)	1.8	Downcomer Width Top, (mm)	215.62
Total height, (m)	2.43	Downcomer Width Bottom, (mm)	215.62
Tray		Downcomer Area Top, (m ²)	0.149
Tray Type	SIEVE	Downcomer Area Bottom, (m ²)	0.149
Diameter, (m)	1.37		
Tray Spacing, (m)	0.60		
Number of Passes	1		
Hole Diameter, (m)	0.01		
No of Holes	942		
Hole Area, (m ²)	0.1		
Deck Thickness, (mm)	3.4		
Cross-Sectional Area, (m ²)	1.49		
Active Area, (m ²)	1.19		
Net Area, (m ²)	1.34		
Weir Geometry			
Weir Height, (m)	0.05		
Weir Length, (m)	1.00		
Remarks : Distillation Column Should be always operated at specified condition.			

7.11 Equipment Sheet of Control Valve

Table 33: Specification Sheet of Control Valve (CV-101)

CONTROL VALVE SPECIFICATION SHEET		Date: 25 th April, 2023
Item		Ref No:
TAG No.		Control Valve
Service		CV-101
Flow Of Fluid		
SPECIFICATIONS AND DATA SHEET		
S.No.		
1.	Line Size	8"
2.	Valve Size	9"
3.	End Connection	Reducer
4.	Material	GROUP 2
5.	Body Material	18 Cr-8 Ni
6.	Type Of Body and Bonnet	ASTM A351 CF8 Cast Stainless Steel
7.	Flowing Medium	Gas
8.	Flow Rate, kg/h	20
9.	Inlet Pressure, bar	1
10.	Outlet Pressure, bar	1
11.	Pressure Drop, bar	-
12.	Temperature, °C	25
13.	Design Temperature, °C	28
14.	Opening	(60-80) % (Max Flow Rate) (>20%) (Min Flow Rate)
15.	Class	400
REMARKS: During maintenance operation of control valve, the bypass valve which is around the control valve should be opened first.		

Table 34: Specification Sheet of Control Valve (CV-105)

CONTROL VALVE SPECIFICATION SHEET		Date: 25 th April, 2023
		Ref No:
Item		Control Valve
TAG No.		CV-105
Service		Flow Of Fluid
SPECIFICATIONS AND DATA SHEET		
S.No.		
1.	Line Size	10”
2.	Valve Size	11”
3.	End Connection	Reducer
4.	Material	GROUP 2
5.	Body Material	18 Cr-8 Ni
6.	Type Of Body and Bonnet	ASTM A351 CF8 Cast Stainless Steel
7.	Flowing Medium	Gas
8.	Flow Rate, kg/h	6357
9.	Inlet Pressure, bar	4
10.	Outlet Pressure, bar	4
11.	Pressure Drop, bar	-
12.	Temperature, °C	85
13.	Design Temperature, °C	94
14.	Opening	(60-80) % (Max Flow Rate) (>20%) (Min Flow Rate)
15.	Class	400
REMARKS: During maintenance operation of the control valve, the bypass valve which is around the control valve should be opened first.		

Table 35: Specification Sheet of Control Valve (CV-106)

CONTROL VALVE SPECIFICATION SHEET		Date: 25 th April, 2023
		Ref No:
Item		Control Valve
TAG No.		CV-106
Service		Flow Of Fluid
SPECIFICATIONS AND DATA SHEET		
S.No.		
1.	Line Size	5"
2.	Valve Size	6"
3.	End Connection	Reducer
4.	Material	GROUP 2
5.	Body Material	18 Cr-8 Ni
6.	Type Of Body and Bonnet	ASTM A351 CF8 Cast Stainless Steel
7.	Flowing Medium	Gas
8.	Flow Rate, kg/h	96
9.	Inlet Pressure, bar	1
10.	Outlet Pressure, bar	1
11.	Pressure Drop, bar	-
12.	Temperature, °C	25
13.	Design Temperature, °C	28
14.	Opening	(60-80) % (Max Flow Rate) (>20%) (Min Flow Rate)
15.	Class	400
REMARKS: During maintenance operation of the control valve, the bypass valve which is around the control valve should be opened first.		

Table 36: Specification Sheet of Control Valve (CV-107)

CONTROL VALVE SPECIFICATION SHEET		Date: 25 th April, 2023
Item		Ref No:
TAG No.		Control Valve
Service		CV-107
Service		Flow Of Fluid
SPECIFICATIONS AND DATA SHEET		
S.No.		
1.	Line Size	1"
2.	Valve Size	2"
3.	End Connection	Reducer
4.	Material	GROUP 2
5.	Body Material	18 Cr-8 Ni
6.	Type Of Body and Bonnet	ASTM A351 CF8 Cast Stainless Steel
7.	Flowing Medium	Gas
8.	Flow Rate, kg/h	6261
9.	Inlet Pressure, bar	1
10.	Outlet Pressure, bar	1
11.	Pressure Drop, bar	-
12.	Temperature, °C	25
13.	Design Temperature, °C	28
14.	Opening	(60-80) % (Max Flow Rate) (>20%) (Min Flow Rate)
15.	Class	400
REMARKS: During maintenance operation of control valve, the bypass valve which is around the control valve should be opened first.		

Table 37: Specification Sheet of Control Valve (CV-113)

CONTROL VALVE SPECIFICATION SHEET		Date: 25 th April, 2023
Item		Ref No:
TAG No.		Control Valve
Service		CV-113
Service		Flow Of Fluid
SPECIFICATIONS AND DATA SHEET		
S.No.		
1.	Line Size	32"
2.	Valve Size	33"
3.	End Connection	Reducer
4.	Material	GROUP 2
5.	Body Material	18 Cr-8 Ni
6.	Type Of Body and Bonnet	ASTM A351 CF8 Cast Stainless Steel
7.	Flowing Medium	Gas
8.	Flow Rate, kg/h	3961
9.	Inlet Pressure, bar	0.8
10.	Outlet Pressure, bar	0.8
11.	Pressure Drop, bar	-
12.	Temperature, °C	145
13.	Design Temperature, °C	160
14.	Opening	(60-80) % (Max Flow Rate) (>20%) (Min Flow Rate)
15.	Class	400
REMARKS: During maintenance operation of the control valve, the bypass valve which is around the control valve should be opened first.		

Table 38: Specification Sheet of Control Valve (CV-118)

CONTROL VALVE SPECIFICATION SHEET		Date: 25 th April, 2023
Item		Ref No:
TAG No.		Control Valve
Service		CV-118
Service		Flow Of Fluid
SPECIFICATIONS AND DATA SHEET		
S.No.		
1.	Line Size	30"
2.	Valve Size	31"
3.	End Connection	Reducer
4.	Material	GROUP 2
5.	Body Material	18 Cr-8 Ni
6.	Type Of Body and Bonnet	ASTM A351 CF8 Cast Stainless Steel
7.	Flowing Medium	Gas
8.	Flow Rate, kg/h	1533
9.	Inlet Pressure, bar	0.5
10.	Outlet Pressure, bar	0.5
11.	Pressure Drop, bar	-
12.	Temperature, °C	115
13.	Design Temperature, °C	127
14.	Opening	(60-80) % (Max Flow Rate) (>20%) (Min Flow Rate)
15.	Class	400
REMARKS: During maintenance operation of control valve, the bypass valve which is around the control valve should be opened first.		

Table 39: Specification Sheet of Control Valve (CV-122)

CONTROL VALVE SPECIFICATION SHEET		Date: 25 th April, 2023
Item		Ref No:
TAG No.		Control Valve
Service		CV-122
Service		Flow Of Fluid
SPECIFICATIONS AND DATA SHEET		
S.No.		
1.	Line Size	1"
2.	Valve Size	2"
3.	End Connection	Reducer
4.	Material	GROUP 2
5.	Body Material	18 Cr-8 Ni
6.	Type Of Body and Bonnet	ASTM A351 CF8 Cast Stainless Steel
7.	Flowing Medium	Gas
8.	Flow Rate, kg/h	646
9.	Inlet Pressure, bar	1
10.	Outlet Pressure, bar	1
11.	Pressure Drop, bar	-
12.	Temperature, °C	25
13.	Design Temperature, °C	28
14.	Opening	(60-80) % (Max Flow Rate) (>20%) (Min Flow Rate)
15.	Class	400
REMARKS: During maintenance operation of the control valve, the bypass valve which is around the control valve should be opened first.		

Table 40: Specification Sheet of Control Valve (CV-138)

CONTROL VALVE SPECIFICATION SHEET		Date: 25 th April, 2023
Item		Ref No:
TAG No.		Control Valve
Service		CV-138
Service		Flow Of Fluid
SPECIFICATIONS AND DATA SHEET		
S.No.		
1.	Line Size	1"
2.	Valve Size	2"
3.	End Connection	Reducer
4.	Material	GROUP 2
5.	Body Material	18 Cr-8 Ni
6.	Type Of Body and Bonnet	ASTM A351 CF8 Cast Stainless Steel
7.	Flowing Medium	Gas
8.	Flow Rate, kg/h	-
9.	Inlet Pressure, bar	0.5
10.	Outlet Pressure, bar	0.5
11.	Pressure Drop, bar	-
12.	Temperature, °C	183
13.	Design Temperature, °C	202
14.	Opening	(60-80) % (Max Flow Rate) (>20%) (Min Flow Rate)
15.	Class	400
REMARKS: During maintenance operation of control valve, the bypass valve which is around the control valve should be opened first.		

Table 41: Specification Sheet of Control Valve (CV-124)

CONTROL VALVE SPECIFICATION SHEET		Date: 25 th April, 2023
Item		Ref No:
TAG No.		Control Valve
Service		CV-124
Service		Flow Of Fluid
SPECIFICATIONS AND DATA SHEET		
S.No.		
1.	Line Size	1"
2.	Valve Size	2"
3.	End Connection	Reducer
4.	Material	GROUP 2
5.	Body Material	18 Cr-8 Ni
6.	Type Of Body and Bonnet	ASTM A351 CF8 Cast Stainless Steel
7.	Flowing Medium	Gas
8.	Flow Rate, kg/h	2291
9.	Inlet Pressure, bar	0.8
10.	Outlet Pressure, bar	0.8
11.	Pressure Drop, bar	-
12.	Temperature, °C	271
13.	Design Temperature, °C	298
14.	Opening	(60-80) % (Max Flow Rate) (>20%) (Min Flow Rate)
15.	Class	400
REMARKS: During maintenance operation of the control valve, the bypass valve which is around the control valve should be opened first.		

Table 42: Specification Sheet of Control Valve (CV-131)

CONTROL VALVE SPECIFICATION SHEET		Date: 25 th April, 2023
Item		Ref No:
TAG No.		Control Valve
Service		CV-131
Service		Flow Of Fluid
SPECIFICATIONS AND DATA SHEET		
S.No.		
1.	Line Size	2"
2.	Valve Size	3"
3.	End Connection	Reducer
4.	Material	GROUP 2
5.	Body Material	18 Cr-8 Ni
6.	Type Of Body and Bonnet	ASTM A351 CF8 Cast Stainless Steel
7.	Flowing Medium	Gas
8.	Flow Rate, kg/h	7396
9.	Inlet Pressure, bar	1
10.	Outlet Pressure, bar	1
11.	Pressure Drop, bar	-
12.	Temperature, °C	108
13.	Design Temperature, °C	119
14.	Opening	(60-80) % (Max Flow Rate) (>20%) (Min Flow Rate)
15.	Class	400
REMARKS: During maintenance operation of the control valve, the bypass valve which is around the control valve should be opened first.		

Table 43: Specification Sheet of Control Valve (CV-133)

CONTROL VALVE SPECIFICATION SHEET		Date: 25 th April, 2023
Item		Ref No:
TAG No.		Control Valve
Service		CV-133
Service		Flow Of Fluid
SPECIFICATIONS AND DATA SHEET		
S.No.		
1.	Line Size	28"
2.	Valve Size	29"
3.	End Connection	Reducer
4.	Material	GROUP 2
5.	Body Material	18 Cr-8 Ni
6.	Type Of Body and Bonnet	ASTM A351 CF8 Cast Stainless Steel
7.	Flowing Medium	Gas
8.	Flow Rate, kg/h	-
9.	Inlet Pressure, bar	0.8
10.	Outlet Pressure, bar	0.8
11.	Pressure Drop, bar	-
12.	Temperature, °C	125
13.	Design Temperature, °C	138
14.	Opening	(60-80) % (Max Flow Rate) (>20%) (Min Flow Rate)
15.	Class	400
REMARKS: During maintenance operation of control valve, the bypass valve which is around the control valve should be opened first.		

Chapter 8 Economic Analysis

Stakeholders involved in the establishment of a chemical plant must be assured that the project is worthwhile, based on an economic assessment and projected performance. Any investment is made to generate a profit, and in the case of chemical plants, an estimate of the total investment required and the cost of production are required before the project's profitability and other economic performance can be evaluated. This design project evaluates Total Capital Investment (TCI), Total Product Cost (TPC), and Total Revenue (TR), and the economic performance of the project is then assessed by calculating Gross Profit (GP), Net Profit (NP), Payback Period (PP), and Return on Investment (ROI) (ROI). Several factors have been considered before calculating every parameter related to all TCI, TPC, and TR calculations, which are briefly discussed in the sections below.

8.1 Total Capital Investment

Total capital investment covers the price of planning, building, and installing a facility as well as any associated site preparation costs. Under the fixed-capital program, all necessary production and industrial facilities are provided. The total capital investment (TCI) is the sum of the fixed-capital costs (FCC) and the working capital (WC) that are required for its operation.

8.1.1 Fixed Capital Cost and working Capital cost

Fixed capital cost (FCC) is defined as capital cost which is financially immobile. This is simply analogous to the fact that money, once spent, cannot be quickly converted back into cash or another asset. This includes both direct and indirect manufacturing costs. The fixed capital investment consists of:

1. The Inside Battery Limits (ISBL) investment – the cost of the plant itself.
2. The modifications and improvements that must be made to the site infrastructure known as off-site or OSBL investment.
3. Engineering and construction costs.
4. Contingency charges.

Working capital is the additional investment needed, over and above the fixed capital, to start the plant up and operate it to the point when income is earned.

It includes the cost of:

1. Start-up.
2. Initial catalyst charges.

3. Raw materials and intermediates in the process.
4. Finished product inventories.
5. Funds to cover outstanding accounts from customers.

Most of the working capital is recovered at the end of the project. The total investment needed for a project is the sum of the fixed and working capital.

Estimation of purchased equipment cost

$$C_e = C \cdot S^n$$

C_e = purchased equipment cost

C = Cost constant

S = Characteristics size parameter

N = index of that type equipment

8.2 Equipment Cost

Distillation cost

Distillation 1

No of tray = 6

Diameter in ft = 6.2ft

Purchased bare cost equipment = 50000

Distillation 2

No of tray = 6

Diameter in ft = 6.2

Purchased bare cost equipment = 50000

Distillation 3

No of tray = 4

Diameter in ft = 5.18

Purchased bare cost equipment = 25000

Total bare cost of equipment = 125000

For stainless steel = $1.7 \cdot 125000 = 212500$

Reactor Cost

Cost of vessel:

Volume of vessel: 38 m^3

$$C_e = C \cdot S^n$$

C_e = purchased equipment cost

$$C = 15000$$

$$S = 38$$

$$N = 0.4$$

$$\text{Cost of Reactor} = \$64269$$

Cost of propellor:

$$\text{Power of propellor} = 15\text{KW}$$

$$C_e = C * S^n$$

C_e = purchased equipment cost

$$C = 1900$$

$$S = 15$$

$$N = 0.5$$

$$\text{Cost of Reactor} = \$7358$$

$$\text{Cost of 1 reactor} = \text{cost of propellor} + \text{vessel} + \text{jacketed}$$

$$= \$64269 + \$7358$$

$$= \$71627$$

$$\text{Total cost of Reactor} = 2 * \$71627$$

$$= \$143254$$

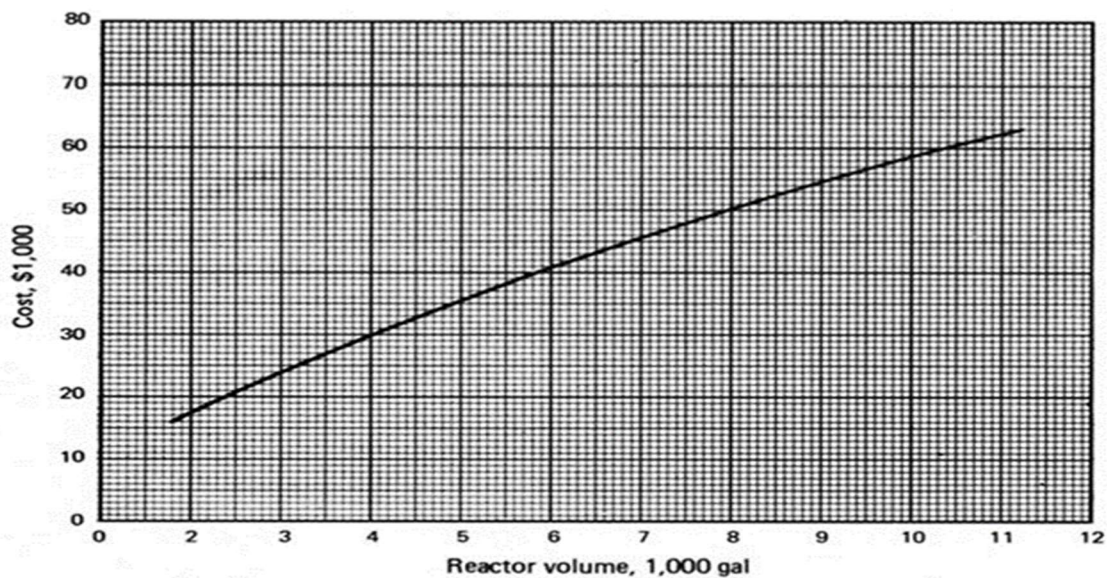


Figure 9. Purchase Cost of Reactor with respect to Volume (Adapted from Richardson, J.F. et al. Chemical Engineering, 3rd ed)

Shell and tube heat exchanger cost

Table 44: Shell and Tube Heat Exchanger Cost

Tag	Type	MOC	U- Tube type factor	Pressure (Pressure Factor)	Heat transfer Area (m ²)	Bare cost of Equipment in 2004 in \$	Bare cost *type factor*pressure factor in \$
E-101	Shell and Tube	Stainless steel for both shell and tube	0.85	4 bar (1)	0.359	10000	8500
E-102	Shell and Tube	Stainless steel for both shell and tube	0.85	4 bar (1)	0.652	10000	8500
E-103	Shell and Tube	Stainless steel for both shell and tube	0.85	1 bar(1)	2.267	10000	8500
E-107	Shell and Tube	Stainless steel for both shell and tube	0.85	1 bar(1)	2.64	10000	8500
E-108	Shell and Tube	Stainless steel for both shell and tube	0.85	1 bar(1)	1.588	10000	8500
E-109	Shell and Tube	Stainless steel for both shell and tube	0.85	1 bar(1)	2.362	10000	8500

*Purchase cost = Bare cost * pressure factor* type factor*

Total cost of heat exchanger in 2004 =\$ 51000

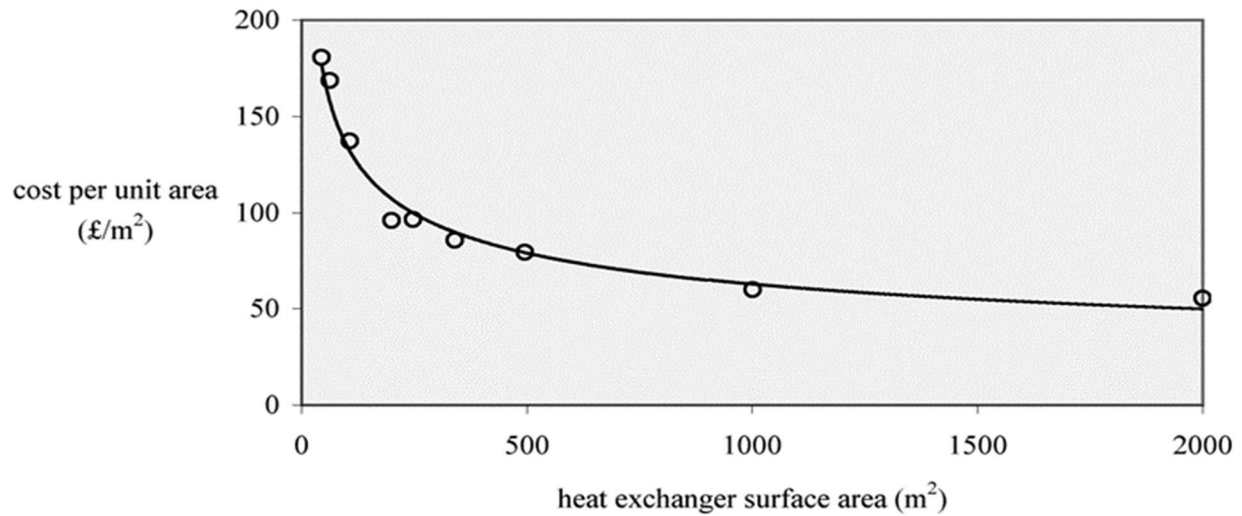


Figure 10. Cost per Unit Area with respect to Heat Exchanger Surface Area (Adapted from Richardson, J.F. et al. Chemical Engineering, 3rd ed)

Reboiler and Condenser Cost

Table 45: Reboiler and Condenser Cost

Tag	Type	MOC	U- Tube type factor	Pressure (Pressur e Factor)	Heat transfer Area (m ²)	Bare cost of Equipme nt in 2004 in \$	Bare cost *type factor*pr essure factor in \$
E-105	Shell and Tube	Stainless steel for both shell and tube	0.85	1 bar (1)	8.22	11000	935
E-106	Shell and Tube	Stainless steel for both shell and tube	0.85	1 bar (1)	7.89	10500	8925
E-110	Shell and Tube	Stainless steel for both shell and tube	0.85	0.5 bar(1)	4.19	10300	8755

E-111	Shell and Tube	Stainless steel for both shell and tube	0.85	0.5 bar(1)	5.88	10400	8840
E-112	Shell and Tube	Stainless steel for both shell and tube	0.85	0.8 bar(1)	2.759	10000	8500
E-113	Shell and Tube	Stainless steel for both shell and tube	0.85	0.8 bar(1)	6.57	10600	9010

$$\text{Purchase cost} = \text{Bare cost} \times \text{pressure factor} \times \text{type factor}$$

Total cost of heat exchanger in 2004 = \$ 44965

Crystallizer cost:

Volume = 5 m³

$$C_e = C \cdot S^n$$

C = Purchased equipment cost

$$C = 2400$$

$$S = 5$$

$$N = 0.6$$

Cost of crystallizer = \$6303

Cost of stainless steel crystalizer = 1.7*6303 = \$10716

Flash Cost

Height = 1.645 m

Diameter = 0.914 m

Material Factor = Stainless steel *2.0

Pressure Factor = 1 bar

Bare cost of equipment = \$11000

$$\begin{aligned} \text{Cost in 2004} &= \text{bare cost} \times \text{pressure factor} \times \text{material factor} \\ &= 11000 \times 1 \times 2.0 \\ &= \$22000 \end{aligned}$$

Pump Cost:

Centrifugal pump

Flow rate = 22.6 GPM

Design Pressure = 4.4 bar

Design Temperature = 85 °C

Efficiency = 82%

Driver Type = Standard Motor

Seal type = Single mechanical seal

Pump cost in 2004 = \$5000

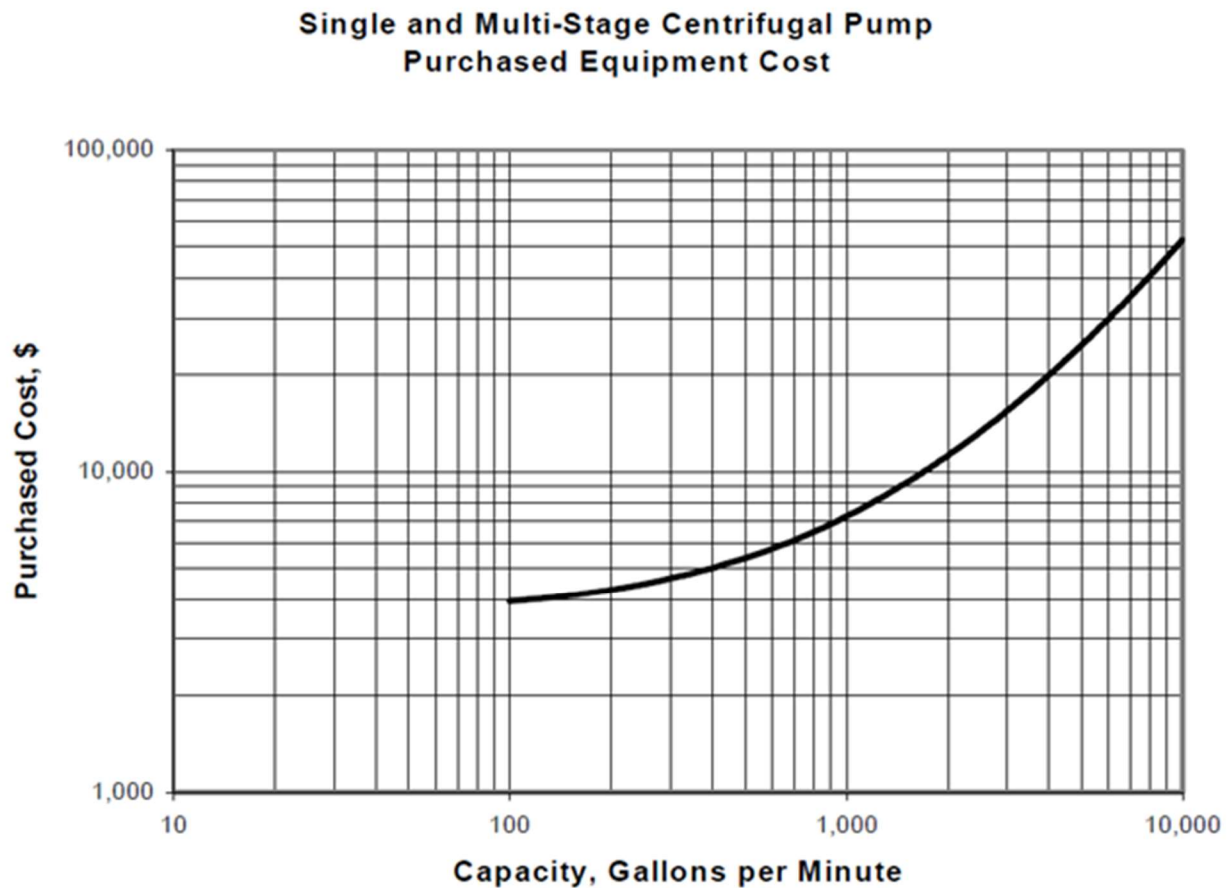


Figure 11. Purchase Cost with respect to Capacity (Adapted from Richardson, J.F. et al. *Chemical Engineering*, 3rd ed)

Compressor Cost:

Material: Carbon Steel

Inlet Temperature: 77°F

Inlet Pressures: 1 bar

Pressure Ratios: 4:1

Molecular Weight: 2

Specific Heat Ratio: 1.405

Volumetric flow rate (ft³/min) = 1459

Cost of compressor in 2004 = \$110000

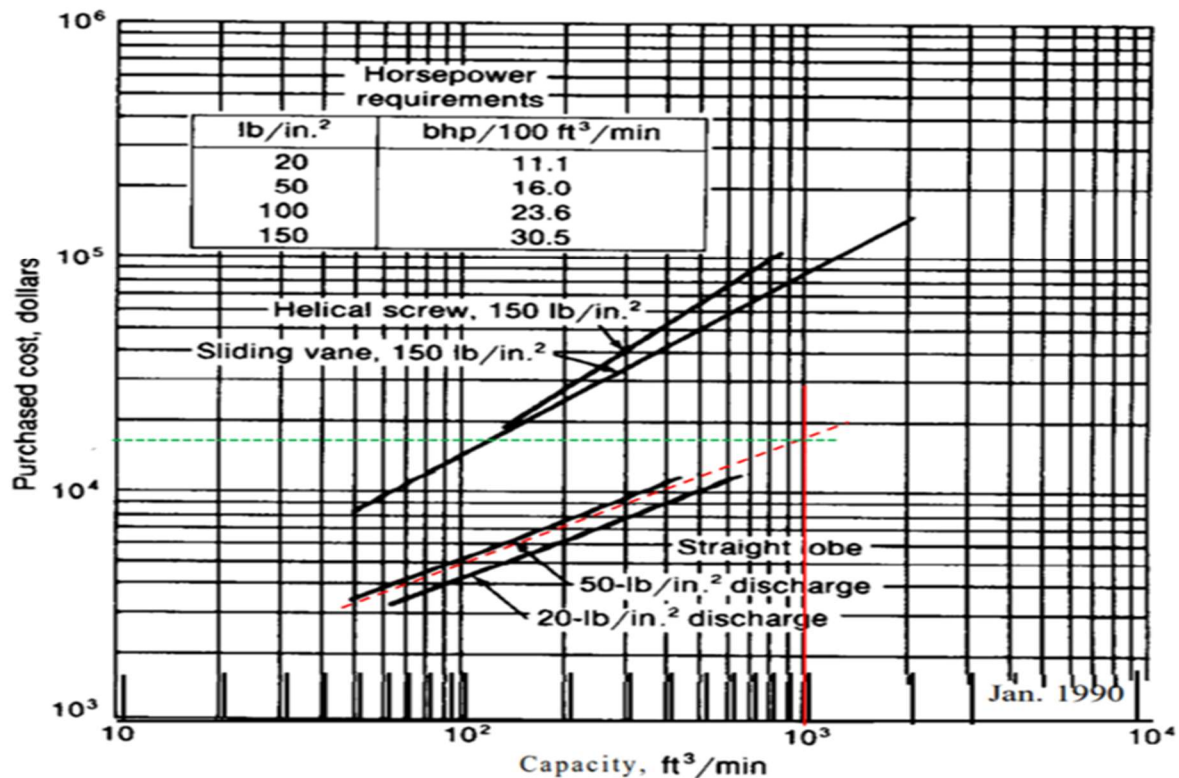


Figure 12. Purchased Cost with Respect to Capacity of Compressor (Adapted from Richardson, J.F. et al. Chemical Engineering, 3rd ed)

Decanter Cost

Area of Decanter = 0.656 m²

$$C_e = C \cdot S^N$$

C = purchased equipment cost

C = 5800

S = 0.656

N = 1.3

Cost of crystallizer = \$3352

Table 46: Total Cost of all equipment In 2004

Equipment Name	Quantity	Total Cost in Dollar
Heat Exchangers	6	51000
Reboiler and Condenser	6	44965
Distillation Column	3	212500
Crystallizer	1	10716
Flash	1	22000
Pump	1	5000
Compressor	1	110000
Decanter	1	3352
Reactor	2	143254

Total cost of equipment = \$602787

Cost Escalation (Inflation)

Cost in year A = Cost in year B * *Cost index in year A / Cost index in year B*

Cost index of 2004 = 113

Cost index of 2023 = 348

Cost in year 2023 = \$602787 * 348 / 113
= \$1856370

8.3 Total Investment cost

Lang Factor (Indirect Cost)

Total purchase cost of major equipment (PCE) = \$1856370

Table 47: Major Equipment and Total Purchase Cost

Major Equipment and Total Purchase Cost	Fluids - Solids
F1 Equipment Erection	0.45
F2 Piping	0.45
F3 Instrumentation	0.15
F4 Electrical	0.10
F5 Buildings, Process	0.10
F6 Utilities	0.45
F7 Storage	0.20
F8 Site Development	0.05
F9 Ancillary Buildings	0.20
F10 Design and Engineering	0.25
F11 Contractors Fee	0.05
F12 Contingency	0.1

$$\begin{aligned} \text{PPC} &= \text{PCE} (1 + f_1 + f_2 + \dots + f_9) \\ &= \$5847565.5 \end{aligned}$$

$$\begin{aligned} \text{Fixed Cost} &= \text{PPC}(1 + f_{10} + f_{11} + f_{12}) \\ &= \$8186591.7 \end{aligned}$$

Working Capital

$$\begin{aligned} \text{Working capital} &= 10 \% \text{ of fixed capital} \\ &= \$818659.17 \end{aligned}$$

$$\begin{aligned} \text{Total investment required} &= \text{Fixed capital} + \text{working capital} \\ &= \$9005250.87 \end{aligned}$$

8.4 Operating Cost

8.4.1 Variable cost

Operating time in year = 4320 hr

Raw material cost

Table 48: Raw Material Cost

Item	Quantity	Rate	Amount
Nitro Benzene	6647400 kg	0.85\$/kg	90404640
Hydrogen	113810.4 kg	1.15 \$ /kg	2094111.36
Acetic Anhydride	2313889.2 kg	0.79\$/kg	29247559.36
Total			123840422.1

Utilities Cost

Power = \$ 1.122 M/year

Total Variable Operating Cost = \$124962422

8.4.2 Fixed Cost

Maintenance cost = \$409329.585 (5% of fixed capital)

Operating labour cost = \$613994.3775 (0.075 of capital cost)

Plant overheads = \$306997.1887 (50% of operating labour)

Laboratory = \$184198.3 (30% of operating labour)

Capital charges = \$491195.502 (6% of fixed capital)

Insurance, = \$81865.917 (1% of fixed capital)

Local taxes = neglect

Royalties = neglect

R&D = neglect

* Other minor cost are neglected and kept as miscellaneous cost

Total Fixed Cost = \$2087580.865

Total Operating cost = Fixed operating cost + variable operating cost
= 2087580.865 + 124962422

= \$127050002.9

Total cost of the project = Fixed cost + working capital + operating cost
= 9005250.87 + 127050002.9
= \$136055253.8

8.5 Economics Evaluation of the Projects

8.5.1 Total Revenue Generated by plant by selling product and by-product

Table 49: Total Revenue Generated by plant by selling product and by-product

S.N.	Item	Type	Quantity	Rate	Total amount(\$)
1	Acetaminophen (Paracetamol)	Main product	3418416 kg/year	0.5\$/5g (USA)	341841600
				0.2\$/5g (Nepal)	136736640
2	Aniline	By product	3620160kg/year	0.3\$/kg	1086048
3	Acetic Acid	By product	5447520kg/year	0.1\$/kg	544752

Total Revenue = \$69999120

Gross profit = Total Revenue - production cost
= 138367440 - 127050002.9
= \$11317437.1

Net profit = Gross profit - 25% of gross profit
= \$8488077.825

Levelized Cost = (CAPEX + OPEX) / Annual production capacity

= (9005250.87 + 127050002.9) / 3418416

= \$39.80067194

8.5.2 Return on Investment

Return on Investment (ROI) is a performance metric used to assess the efficiency of an investment or to compare the efficiency of several investments. It is a simple index of the performance of the money invested. It can also be used to compare different investments within a portfolio. Typically, the investment with the highest ROI is Prioritised

$$\begin{aligned}
 \text{ROI} &= (\text{Net Profit} / \text{Total Investment}) \times 100 \\
 &= (8488077.825 / 136055253.8) \times 100 \\
 &= 6.238699049 \%
 \end{aligned}$$

8.5.3 Cash Flow Diagram and Payback Period

Cash flow diagrams are graphical representations of revenue and expenditure over time. The diagram is built around a horizontal line with markers at various time intervals. Expenses and expenditures are shown at the appropriate times. Cash flow refers to the movement of money into and out of a business. The point in time when the cash flow moves from negative to positive is known as payback period. The term payback period refers to the amount of time it takes to recover the cost of an investment. The cash flow for our industry is shown in the figure below.

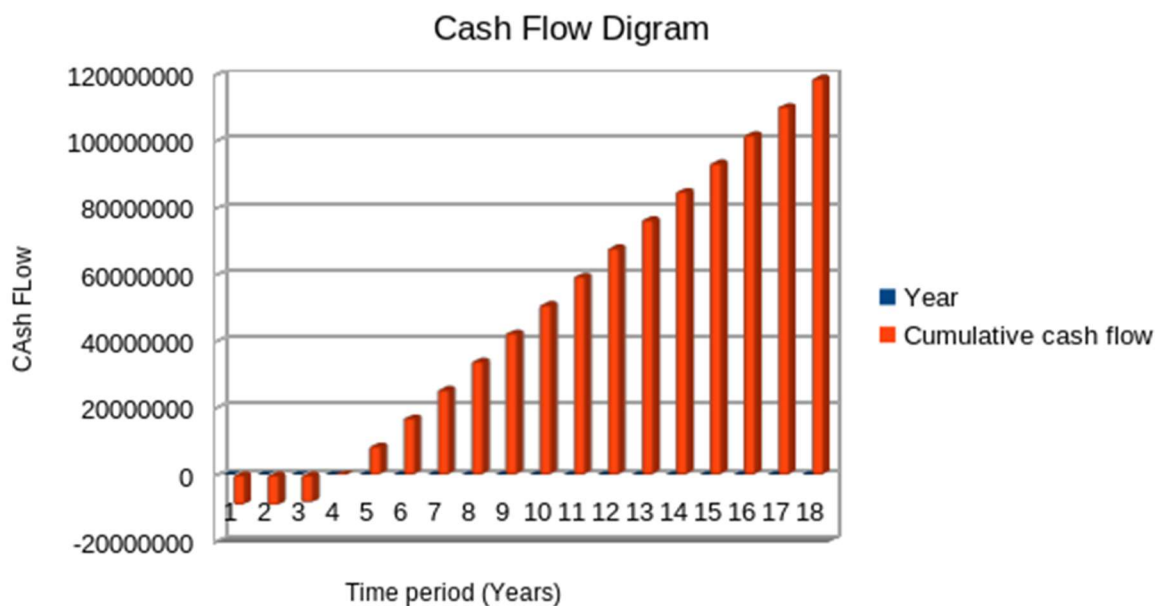


Figure 13. Cash Flow Diagram

Payback period of the project is 4 years.

8.5.4 Net Present Value (NPV)

Net Present Value (NPV) is the present value of cash inflows and outflows added together. In other words, it is the difference between the present values of cash inflows and cash outflows over a given time period. The net present value (NPV) is a good way to determine whether or not a project is profitable. It considers the interest rate, which

is generally equivalent to the inflation rate. As a result, the real value of money now is considered at each year of operation.

As fixed amount is gained every year we can find the Present Value with the following formula

$$\begin{aligned}
 NPV &= A \times \frac{1 - \left(\frac{1}{(1+r)^t}\right)}{r} - \text{initial investment} \\
 &= 8488077.825 \times \frac{1 - \left(\frac{1}{(1+0.1)^{15}}\right)}{0.1} - 136055253.8 \\
 &= \$72,603,112.60
 \end{aligned}$$

8.5.5 Net Future Value (NFV)

$$\begin{aligned}
 NFV &= NPV(1+r)^t \\
 &= 64560994.79(1+0.1)^{15} \\
 &= \$269,687,297.3
 \end{aligned}$$

8.5.6 Discounted Cash Flow rate of Return (DCFRR)

$$\begin{aligned}
 DCFRR &= \frac{NFV}{(1+r)^t} \\
 &= \frac{269687297.3}{(1+0.1)^{15}} \\
 &= \$645,609,94.79
 \end{aligned}$$

8.5.7 Rate of Return (ROR)

$$\begin{aligned}
 ROR &= \frac{F - \text{Initial Investment}}{t \times \text{Initial Investment}} \times 100\% = \frac{269687297.3 - 136055253.8}{15 \times 136055253.8} \times 100\% \\
 &= 6.5 \%
 \end{aligned}$$

SUMMARY OF ACTIVATED ENERGY ANALYSIS RESULTS

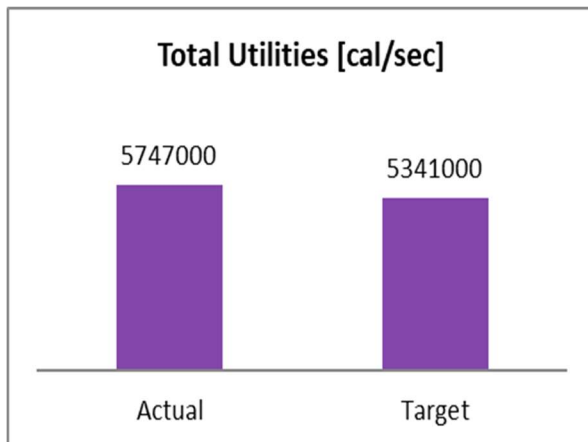


Figure 14: Total Utilities (cal/sec)

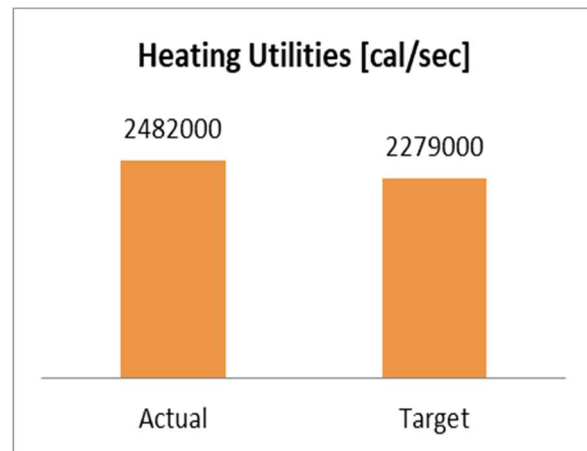


Figure 15: Heating Utilities (cal/sec)

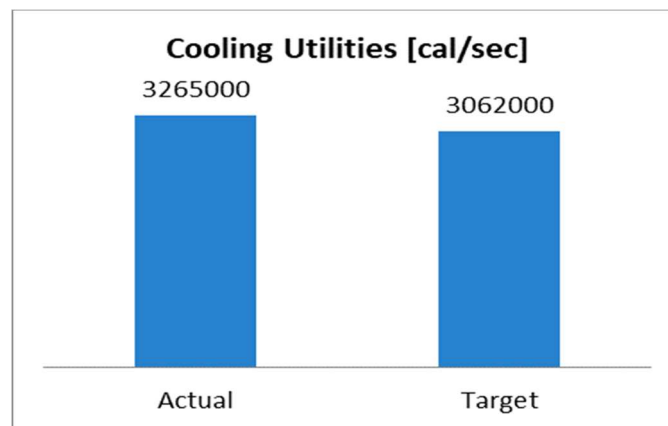


Figure 16: Cooling Utilities (cal/sec)

Summary Table

Table 50: Summary of Utilities

Property	Actual	Target	Available Savings	% of Actual
Total Utilities [cal/sec]	5747000	5341000	406000	7.09
Heating Utilities [cal/sec]	2482000	2279000	203000	8.21
Cooling Utilities [cal/sec]	3265000	3062000	203000	6.24
Carbon Emissions [kg/hr]	0	0	0	0

Chapter 9 Environmental and Safety Hazards

SAFETY ANALYSIS

Every manufacturing process including chemical processes are to some extent hazardous because of the use of different chemicals, flammable objects etc. Safety and loss prevention can be considered by identifying the hazards, assessment of hazards and its control and prevention of the possible hazardous deviations in process variables by provision of automatic control systems, alarms and interlocks. The risks can always be minimised with good operating practices and management.

The main concern is mainly with precautions and protocols that are to be followed while handling materials in the plant. Safety equipment includes: splash goggles, protective coats, gloves and safety shoes are all required in dealing with these materials regardless of their reactivity and stability. These sections in this chapter are the documentations that will include the target materials and compounds encountered and utilised in the plant.

9.1 Material Safety Data Sheet

A material safety data sheet is a technical document which provides detailed and comprehensive information on a controlled product related to:

- Health effects of exposure to the product
- Hazard evaluation related to the product's handling, storage or use
- Measure to protect workers at risk of exposure
- Emergency procedures.

Given Table is the material safety and data sheet (MSDS) for the Acetaminophen.

Table 51: Material Safety and Data Sheet (MSDS) for the Acetaminophen

Section 1: Product Identification			
Product Name:	Acetaminophen		
Chemical formula:	C ₈ H ₉ NO ₂		
Synonyms:	Paracetamol, N-(4-Hydroxyphenyl) acetamide, 4'-Hydroxyacetanilide, 4-Acetamidophenol, APAP		
Section 2: Hazardous Ingredients			
Name	CAS #	%	TLV
1. Acetaminophen	103-90-2	100	Exposure limits: 0.3 ppm (0.37mg/m3)
2. Acetic Anhydride	108-24-7	5 – 15	Exposure limits: 200 ppm

3. Water	7732-18-5	Balance	(262mg/m3) N/A
Section 3: Physical Data			
Physical state	Solid, White, Odorless, Crystalline Powder, Odorless.		
pH	5.5 - 6.5 (20 degrees Celsius) (10% solution)		
Auto-Ignition Temperature	540 °C, 1004 °F		
Solubility	Freely soluble in alcohol; soluble in boiling water and in 1 N sodium hydroxide.		
Freezing point	(168 - 172)°C, (334.4-341.6)°F		
Boiling point	> 500 °C, 932 °F		
Specific gravity	1.293 (21 °C)		
Vapor pressure	0.0000003 kPa (25 °C)		
Vapor density	5.2 (Air = 1)		
Relative density	5.2 (Water = 1)		
Decomposition temperature	> 180 °C		
Section 4: Fire and Explosion Data			
Flash point	168-172 °C		
Flammability	May be combustible at high temperature		
Fire extinguishing procedures	i. Small fire: dry chemical, CO₂ or water spray. Large fire: dry chemical, CO₂, alcohol resistant foam or water spray. Do not get water inside containers. ii. Suppress (knock down) gases/vapors/mists with a water spray jet. Prevent fire extinguishing water from contaminating surface water or the ground water system.		
Section 5: Reactivity Data			
Stability	i. Stable under recommended storage conditions ii. Dry, pure paracetamol is stable to 45°C iii. Contamination and humid conditions must be avoided		
Incompatibility	Acetaminophen is physically incompatible with dexketoprofen, diazepam, fentanyl, ketorolac, metamizole, phenytoin, phenobarbital and propofol.		
Section 6: Toxicological Properties			
Routes of entry	i. Oral: Rat: LD50: (mg/kg): 1944 ii. Dermal: Rabbit LD50: (mg/kg): Not available iii. Inhalation: Rat: LD50: (mg/L/4hr): Not available		

Effects of acute exposure	i. Inhalation: May be harmful if inhaled. May cause respiratory tract irritation. ii. Ingestion: May be harmful if swallowed. iii. Skin: May be harmful if absorbed through skin. May cause skin irritation. May cause allergic reaction. iv. Eyes May cause serious eye irritation.
Effects of chronic exposure	i. Irregular heartbeat ii. Cardiovascular collapse iii. Reproductive disorders iv. Blood disorders
Section 7: Preventative Measures	
Protective clothing and PPE	Wear respiratory protection. Avoid dust formation. Avoid breathing vapours, mist or gas. Ensure adequate ventilation. Evacuate personnel to safe areas. Avoid breathing dust. Safety glasses for eye protection Handle with gloves
Storage Conditions	Store in original container, tightly sealed container, protected from direct sunlight, in a dry, well-ventilated area, away from incompatible materials. Store in accordance with local regulations. Eliminate all ignition sources. Separate from oxidizing materials. Containers that have been opened must be carefully resealed and kept upright to prevent leakage. Do not store in unlabelled containers. Use appropriate containment to avoid environmental contamination. Preserve in tight, light-resistant containers, and store at room temperature. Protect from moisture and heat.
Cleanup Procedure	Do not touch damaged containers or spilled material unless wearing appropriate protective clothing. Wear a respirator, chemical safety goggles, rubber boots and heavy rubber gloves. Stop leaking if you can do it without risk. Prevent entry into waterways, sewers, basements or confined areas. Shut off all sources of ignition. Evacuate the area. If necessary, employ water fog to disperse the vapours. Absorb the matter with compatible vermiculite or other absorbing material. Place in a suitable container and retain for disposal. Ventilate and clean the affected area. Do not flush into the sewerage system or to drains.
Section 8: First Aid Measures	
Eye contact	Flush with copious amounts of water for 15 minutes, separating eyelids with fingers. If irritation persists, seek medical aid.
Skin contact	Wash with soap & water for 15 minutes. If irritation persists, seek medical aid.

Inhalation	Remove to fresh air. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. Call a physician.
Ingestion	Call a physician. Wash out your mouth with water. Do not induce vomiting without medical advice.

9.2 HAZOP Analysis

HAZOP study is a procedure for the systematic, critical examination of the operability of a process. It determines whether deviations from the design or operational intent can lead to undesirable consequences. The following are the objectives of the HAZOP study done.

- To check a design
- To decide whether and where to build
- To decide whether to buy a piece of equipment
- To obtain a list of questions to put to a supplier
- To check running instructions
- To improve the safety of existing facilities.

Generally, the HAZOP team comprises the following personnel: Project Manager, Production Manager, Chemical Engineer, Instrument Engineer, Electrical Engineer, and Quality Control Engineer.

9.2.1 HAZOP Study of Heat Exchanger

Unit: Heat exchanger, Node: Inlet and outlet streams, Parameter: Pressure drop

Table 52: Summary of HAZOP Analysis of Heat Exchanger

Guide word	Deviation	Cause	Consequence	Action
Very high	Pressure drop	Pump control fail	Outlet stream has unwanted properties	Spare pump for emergency
		Fault in pressure transmitter	Accumulation in the reactor	Instrument must be inspected on a regular basis
Very low	Pressure drop	Pump tripping	Increasing try flooding	Regular maintenance required
		No inlet flow	Damaging of pump	Inspect the Inlet flow

9.2.2 HAZOP Study of Condenser

Unit: Condenser, Parameter: Pressure

Table 53: Summary of HAZOP Analysis of Condenser

Guide word	Deviation	Cause	Consequence	Action
High	Pressure	Open and failed relief valve	Leak in Condenser	Backup relief valve
		Blockage of stream	Increase in temperature	Line inspected regularly
Low	Pressure	Closed and failed relief valve	Condensation of gas is low	Pressure sensor installation
		Blockage in pipeline of product	Condensation does not take place	Installation of flow metre before Condenser

9.2.3 HAZOP Study of Reboiler

Unit: Reboiler, Parameter: Flow rate

Table 54: Summary of HAZOP Analysis of Reboiler

Guide word	Deviation	Cause	Consequence	Action
No	Inlet flow	Pump problem	Decrease in product quality	Installation of micrometre
		Blocked pipe	Pressure drop	Maintenance of transferring lines
More	Pressure drop	Open and failed feed valve	Large amount of product not used	Use of flow metre
		Heat exchanger leakage	Decrease in product quality	Use ratio sensor
Less	Inlet flow	Feed value failure and close	Decrease in product quality	Regular inspection of transferring lines
		Pipe plugging	Pump damage	Use of controller for the valves

9.3 Environmental Impact Analysis

An environmental impact study establishes a baseline for the state of the environment and offers an estimate of the environmental impact of planned operations. An evaluation of the short- and long-term effects on the physical environment, such as noise, air, and water pollution, as well as the implications on local services, way of life, and health standards, may be included in an EIA. Our primary focus will be on reducing the negative effects on the environment as much as we can.

9.3.1 Solid and Liquid Waste

There is no liquid waste involved. In case of solid waste, the two categories of degradable and non-degradable garbage are initially applied to all solid waste produced. Each portion must include closed bins for both biodegradable and non-biodegradable waste. The biodegradable waste container is sent to a solid waste management facility that has been approved. The non-biodegradable trash container's contents are submitted to a recycler for recycling.

9.3.2 Process Waste

According to the current plan, no processed waste will be produced; but, if any do, they must be stored and disposed of in accordance with the regulations endorsed by the Pollution Control Board.

9.3.3 Hazardous Waste

Used machine oil will be securely stored in HDPE drums in a secure, covered facility. The spent oil will be sold to fund additional care. The necessary precautions will be taken to prevent spills and leaks of spent oil from storage.

9.3.4 Air Emission and Control

Throughout the entire process, all exhaust gas emissions will be channelled and repurposed for things like heating and remaining chemical use. The residual gas will be expelled by a chimney at the conclusion. Exhaust gases from manufacturing processes will be released as emissions, and they must only be composed of nitrogen, hydrogen, carbon dioxide, carbon monoxide, and traces of formaldehyde and methanol.

Public areas around the plant site may be directly impacted by the noise pollution brought on by the operation of machinery. Consequently, the plant will be constructed far from a residential neighbourhood.

Chapter 10 Plant Layout

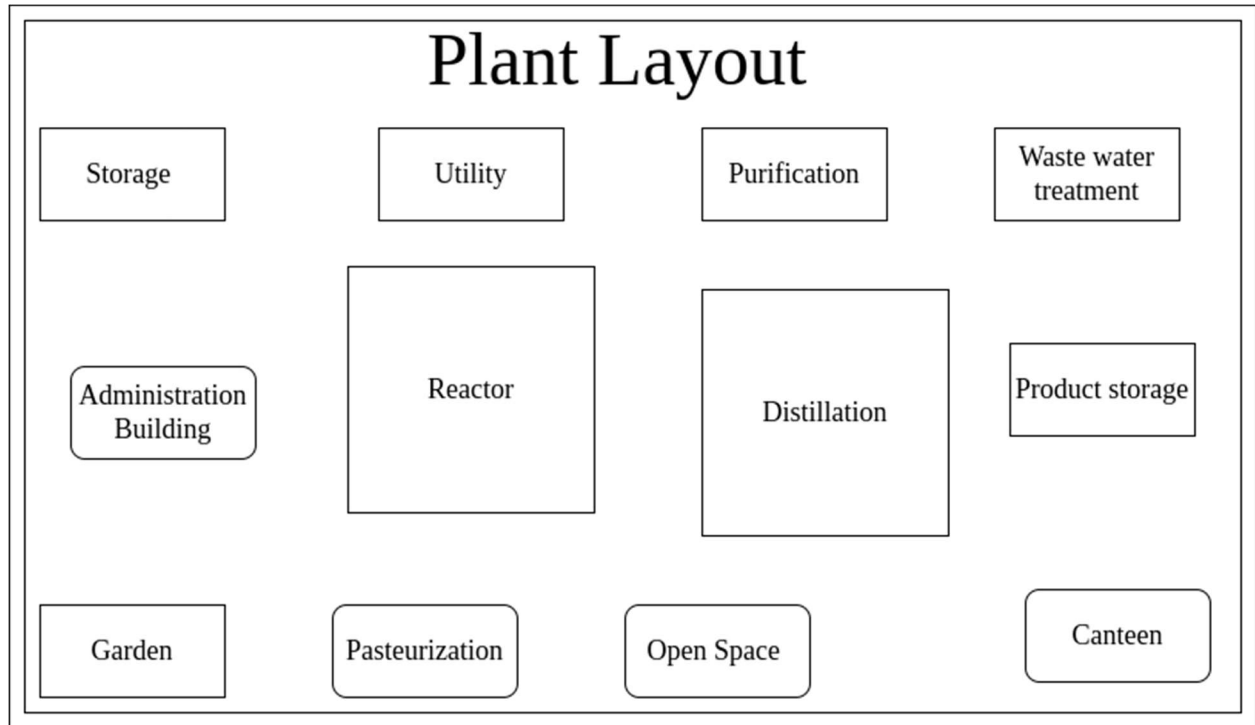


Figure 17: Plant Layout

Chapter 11 Conclusion

In conclusion, the design of a Para-Aminophenol plant from the hydrogenation of nitrobenzene and its use in producing Acetaminophen in Nepal has been presented. The study involves a comprehensive process of material and energy balance calculations, process flow diagrams, piping and instrumentation diagrams, equipment design, economic analysis, environmental impact assessment, and safety and HAZOP studies. The plant is designed to produce 40.97 kmol/hr of Para-Aminophenol and 20.97 kmol/hr of Acetaminophen with a purity of 99.9%, which surpasses the quality of Paracetamol currently found in Nepal. The successful implementation of this plant will reduce Nepal's dependence on imports from India for its Paracetamol requirements, ensuring a stable supply and price of the key starting material, Para-Aminophenol, and the final product, Paracetamol. This will contribute to the development of the local pharmaceutical industry, create employment opportunities, and ultimately benefit the economy of Nepal.

The CAPEX and OPEX of the design are calculated 9 million USD and 127 million USD respectively. The levelized cost of product was calculated 40 USD and ROR and ROI was calculated 6.5 % and 6.2 % respectively taking 4 years period of study. Therefore, the plant is economically attractive and technically and economically feasible.

Furthermore, the environmental impact assessment shows that the plant is designed to comply with local and international environmental regulations. The carbon footprint, wastewater management, and waste disposal methods are evaluated, ensuring that the plant operates sustainably. Safety and HAZOP studies are also conducted to identify potential hazards and mitigate risks associated with the plant's operation.

In conclusion, the design of the Para-Aminophenol plant and its use in producing Acetaminophen in Nepal presents an opportunity to reduce the country's dependence on imports from India and contribute to the development of the local pharmaceutical industry. The study is comprehensive and covers all aspects of plant design, economic analysis, environmental impact assessment, and safety and HAZOP studies. The success of this project will ultimately benefit Nepal's economy, and the quality of the produced Paracetamol will meet international standards.

Chapter 12 References

1. Ansari, M., Neupane, D., & Sohani, S. (2014). Health care and pharmacy practice in Nepal. *Canadian Journal of Hospital Pharmacy*, 67(1), 74-80.
2. Bhattarai, M. (2017). Factors affecting the growth of the pharmaceutical industry in Nepal. *South Asian Journal of Business Studies*, 6(1), 76-91.
3. Bollyky, T. J., Gostin, L. O., & Hamburg, M. A. (2021). The equitable distribution of COVID-19 therapeutics and vaccines. *JAMA*, 323(24), 2462-2463.
4. Chen, H., Engkvist, O., Wang, Y., Olivecrona, M., & Blaschke, T. (2021). The rise of deep learning in drug discovery. *Drug Discovery Today*, 23(6), 1241-1250.
5. Jackson, N. A. C., Kester, K. E., Casimiro, D., Gurunathan, S., & DeRosa, F. (2020). The promise of mRNA vaccines: a biotech and industrial perspective. *NPJ Vaccines*, 5(1), 11.
6. KC, B., Acharya, R. P., & Sharma, A. K. (2020). Prospects and challenges of the pharmaceutical industry in Nepal. *Kathmandu University Medical Journal*, 18(70), 9-16.
7. Kupferschmidt, K., & Cohen, J. (2020). Race to find COVID-19 treatments accelerates. *Science*, 367(6485), 1412-1413.
8. Lurie, N., Saville, M., Hatchett, R., & Halton, J. (2020). Developing Covid-19 vaccines at pandemic speed. *New England Journal of Medicine*, 382(21), 1969-1973.
9. Mullard, A. (2020). How COVID vaccines are being divided up around the world. *Nature*, 588(7837), 208-209.
10. Poudel, A., & Ishii, K. (2016). Pharmaceutical industry in least developed countries of Asia with special reference to Nepal. *Journal of Pharmaceutical Health Services Research*, 7(4), 231-238
11. Poudel, A., & Ishii, K. (2016). Pharmaceutical industry in least developed countries of Asia with special reference to Nepal. *Journal of Pharmaceutical Health Services Research*, 7(4), 231-238.
12. Shankar, P. R., Subish, P., Mishra, P., & Lalitha, K. (2015). Ambiguous pricing of Nepalese medicines. *Journal of Generic Medicines*, 5(2), 60-65.
13. Shrestha, R., Ghale, A., & Lohani, S. (2017). National drug policy in Nepal: Implementation challenges in a resource-limited setting. *Journal of Pharmaceutical Policy and Practice*, 10(1), 36.
14. Singh, G., Bhandari, G. P., & Kafle, R. (2018). Current scenario of pharmaceutical production in Nepal. *International Journal of Health & Allied Sciences*, 7(3), 130-134.
15. Yamey, G., Schäferhoff, M., Hatchett, R., Pate, M., & Zhao, F. (2020). Ensuring global access to COVID-19 vaccines.

16. Shrestha, S., Dhimal, M., & Gautam, I. (2018). Dengue in Nepal: Epidemiology, Diagnosis, and Preparedness. *Journal of Nepal Health Research Council*, 16(2), 123-128.
17. Shrestha, R., Ghale, A., & Lohani, S. (2017). National drug policy in Nepal: Implementation challenges in a resource-limited setting. *Journal of Pharmaceutical Policy and Practice*, 10(1), 36.
18. Shankar, P., Subish, P., Mishra, P., & Lalit, M. (2007). Ambiguous pricing of Nepalese medicines. *Journal of Institute of Medicine Nepal*, 28(3), 35–38. Retrieved from <https://www.nepjol.info/index.php/JIOM/article/view/627>
19. Poudel, A., & Ishii, K. (2016). Pharmaceutical industry in least developed countries of Asia with special reference to Nepal. *Journal of Pharmaceutical Health Services Research*, 7(4), 231-238.
20. Bhattarai, M. (2017). Factors affecting the growth of the pharmaceutical industry in Nepal. *South Asian Journal of Business Studies*, 6(1), 76-91.
21. KC, B., Acharya, R. P., & Sharma, A. K. (2020). Prospects and challenges of the pharmaceutical industry in Nepal. *Kathmandu University Medical Journal*, 18(70), 9-16.
22. Nepal Ministry of Health. (2016). National List of Essential Medicines Nepal (Fifth Revision). Kathmandu: Government of Nepal, Ministry of Health.
23. Chen, X., Li, Y., Li, X., Wang, H., & Ma, Y. (2021). Highly efficient and selective synthesis of para-aminophenol via hydrogenation of nitrophenol over a Pt/SiO₂ catalyst. *Applied Catalysis A: General*, 626, 118171.
24. Guo, S., Ma, J., Zhang, L., & Han, H. (2013). Highly efficient synthesis of para-aminophenol by catalytic hydrogenation of nitrobenzene over reduced graphene oxide-supported platinum catalyst. *Applied Catalysis A: General*, 466, 125-130.
25. Ma, J., Guo, S., Zhang, L., & Han, H. (2018). Synthesis of para-aminophenol by the reduction of 4-nitroaniline with H₂ over Pt/WO₃ catalyst. *Applied Catalysis A: General*, 562, 187-193.
26. Ma, J., Guo, S., Zhang, L., & Han, H. (2018). Synthesis of para-aminophenol by the reduction of 4-nitroaniline with H₂ over Pt/WO₃ catalyst. *Applied Catalysis A: General*, 562, 187-193.
27. Rajput, S. S., Nadaf, A. H., Chakraborty, S., Kadam, P. S., & Ghadge, S. R. (2019). Production of para-amino phenol from nitrobenzene using iron as a reducing agent. *Materials Today: Proceedings*, 16, 1046-1050.
28. Suriya, P., Baskar, R., Ramesh, P. S., & Sivanesan, S. (2014). Production of para-amino phenol from nitrobenzene via azobenzene by *Pseudomonas stutzeri*. *Journal of hazardous materials*, 280, 527-532.
29. Wang, H., Jia, J., Wu, X., & Chen, W. (2014). A green and efficient synthesis of para-aminophenol by catalytic oxidation of 1,4-phenylenediamine with hydrogen peroxide. *Chemical Engineering Journal*, 239, 309-314

30. Wang, H., Jia, J., Wu, X., & Chen, W. (2014). A green and efficient synthesis of para-aminophenol by catalytic oxidation of 1,4-phenylenediamine with hydrogen peroxide. *Chemical Engineering Journal*, 239, 309-314.
31. Noyes, A. A., & Bray, W. C. (1902). The catalysis of the reduction of nitrobenzene by hydrogen. *Journal of the American Chemical Society*, 24(6), 521-535.
32. Riggs, W. W. (1955). The Raney nickel catalysed hydrogenation of nitrobenzene to aniline. *Journal of the American Chemical Society*, 77(17), 4647-4648.
33. Xu, Y., Zhu, X., Chen, Y., Chen, C., & Chen, Y. (2016). Hydrogenation of nitrobenzene to aniline over biochar-supported Pd catalysts. *Chemical Engineering Journal*, 286, 145-152.
34. Feng, Q., Hou, J., Chen, Q., Yang, H., & Liu, S. (2019). Highly efficient hydrogenation of nitrobenzene to aniline over chitosan modified Pd/C catalysts. *Applied Catalysis A: General*, 570, 338-347.
35. Chen, X., Chen, J., Huang, R., Zhang, M., & Yan, N. (2016). Hydrogenation of nitrobenzene to aniline in water using microwave-assisted Pd/C catalysts. *Catalysis Communications*, 83, 43-47.
36. Chen, Y., Chen, C., & Zhang, Y. (2017). Highly efficient hydrogenation of nitrobenzene to aniline under ultrasonic irradiation over a Pd/C catalyst. *Ultrasonics Sonochemistry*, 39, 679-684.
37. Feng, Y., Wang, Q., Wang, L., Wang, M., Jiang, X., & Han, Y. (2020). Biocatalytic synthesis of acetaminophen using recombinant *Escherichia coli* expressing a thermostable aminotransferase. *Bioresource Technology*, 307, 123251.
38. Feng, Y., Wang, Q., Wang, Y., & Li, Z. (2020). A critical review of the biocatalytic synthesis of acetaminophen. *Critical Reviews in Biotechnology*, 40(1), 75-89.
39. Gao, H., Wang, Y., Cao, Y., & Cao, Y. (2019). Reductive amination of p-nitrophenol to acetaminophen over Mn-doped TiO₂. *Catalysis Letters*, 149(5), 1285-1292.
40. García-Sancho, C., Belver, C., & Marcilla, A. (2019). Green synthesis of acetaminophen: An introductory experiment in a laboratory of organic chemistry. *Journal of Chemical Education*, 96(4), 751-754.
41. García-Sancho, C., Lafuente, P., López-Sanz, J., & García-García, P. (2019). Acetaminophen production in water by catalytic hydrogenation of p-nitrophenol: Sustainable processes for a sustainable world. *Green Chemistry*, 21(8), 1951-1961.
42. Geng, Y., Zhao, Y., Chen, Z., Zhang, Y., & Ma, W. (2019). Electrocatalytic reduction of p-nitrophenol to acetaminophen by Au nanoparticles modified carbon electrode. *Journal of Electroanalytical Chemistry*, 840, 77-82.
43. Gholami, A., Rahmani-Sani, A., & Jalilian, R. (2019). A review on recent advances in synthesis of acetaminophen. *Journal of Molecular Liquids*, 290, 111232.
44. Kilic, E., & Balköse, D. (2014). Production of acetaminophen via reductive amination of p-nitrophenol using copper oxide catalyst. *Chemical Engineering Communications*, 201(5), 584-593.

45. Kilic, F. S., & Balköse, D. (2014). Synthesis of acetaminophen by reductive amination of p-nitrophenol using a copper oxide catalyst. *Industrial & Engineering Chemistry Research*, 53(46), 18103-18109.
46. Kumar, A., Chandra, A., & Yadav, S. K. (2016). An overview on mechanochemical synthesis of acetaminophen. *Journal of Materials Science: Materials in Electronics*, 27(9), 9415-9422.
47. Kumar, P., Chopra, A., & Khurana, J. M. (2019). Hydrogenation of nitrobenzene using graphene-supported Pd-Cu and Pd-Au bimetallic catalysts. *RSC Advances*, 9(11), 6340-6349.
48. Liang, G., Huang, H., & Lin, X. (2018). Synthesis of acetaminophen via oxidative coupling of phenols with CuFe₂O₄ as catalyst. *Chinese Journal of Chemical Engineering*, 26(3), 466-471.
49. Liu, X., Ma, L., Yang, Y., Huang, W., & Zhao, C. (2019). Synthesis of acetaminophen from nitrobenzene using nanoscale zero-valent iron as the reducing agent. *Journal of Chemical Technology & Biotechnology*, 94(8), 2615-2623.
50. Morse, G. K., & Fraenkel, G. (1949). Acetylation of p-aminophenol. *Journal of the American Chemical Society*, 71(1), 226-227.
51. Morse, S. V., & Fraenkel, G. (1949). Acetylation of aminophenol. *Journal of the American Chemical Society*, 71(2), 499-501.
52. Sehgal, D., Datta, A., & Kumar, D. (2018). Acetaminophen synthesis via hydrogenation in PEG-based ionic liquids. *Separation and Purification Technology*, 201, 85-90.
53. Sehgal, S., Saini, S., Gupta, M. K., & Singh, V. K. (2018). Synthesis of acetaminophen via nitrobenzene reduction in ionic liquid media. *Journal of Molecular Liquids*, 261, 135-142.
54. Song, X., Gao, J., & Jiang, L. (2020). Catalytic reduction of p-nitrophenol to acetaminophen over CoFe₂O₄. *Materials Science and Engineering: B*, 252, 114380.
55. Weiss, J., Horvath, C., Van Zyl, G. (1966). The hydrogenation of p-nitrophenol to p-aminophenol over Raney nickel. *Journal of Catalysis*, 5(3), 297-302.
56. Weiss, S., Nudelman, A., & Popper, R. (1966). The catalytic hydrogenation of p-nitrophenol to p-aminophenol. *Journal of the American Chemical Society*, 88(11), 2533-2537.
57. Xie, X., Wu, L., Li, G., Li, J., Li, G., & Li, C. (2018). Hydrophobic Pd/C: An efficient and selective catalyst for the hydrogenation of nitrobenzene to aniline. *Catalysis Communications*, 103, 61-65.
58. Zhang, H., Feng, X., Wang, M., Cao, Z., & Lu, X. (2020). Synthesis of acetaminophen via acid-catalysed coupling of 4-aminophenol with acetic anhydride. *Journal of Chemical Research*, 44(4), 193-198.
59. Zhang, J., Zhang, Y., & Wang, G. (2019). Production of acetaminophen by reduction of p-nitrophenol with activated carbon supported iron catalyst. *Bulletin of Materials Science*, 42(5), 157.

Appendix

Calculation of Pipe Sizes

Table 55: Design Parameters for Pipe 2

S.N.	Design Parameters	Values
1	Temperature (°C)	25
2	Pressure (Bar)	1
3	Density (kg/m ³)	1198.49
4	Mass Flow Rate (Kg/s)	1.7098
5	Volume Flow (m ³ /s)	0.001426
6	Linear velocity(m/s)	1.828

$$\text{Volumetric flow rate } (\dot{v}) = \frac{\text{Mass Flow Rate}}{\text{Density}} = 0.00142 \text{ m}^3/\text{s}$$

$$\text{Cross Sectional Area } (A_c) = \frac{\text{Volumetric flow rate}}{\text{velocity}} = 0.00078 \text{ m}^2$$

$$\text{Internal diameter of pipe}(di) = \sqrt{\frac{A_c * 4}{\pi}} = 0.031 \text{ m} = 1.22 \text{ in}$$

Corresponding standard stainless steel pipe;

Inside diameter = 1.380 in

Wall thickness = 0.140 in

Schedule number = 40s

Outside diameter =1.660 in

Nominal diameter = 1 in

Table 56: Design Parameters for Pipe 3

S.N.	Design Parameters	Values
1	Temperature (°C)	226.097
2	Pressure (Bar)	4
3	Density (kg/m ³)	0.1940
4	Mass Flow Rate (Kg/s)	0.0559
5	Volume Flow (m ³ /s)	0.28863
6	Linear velocity(m/s)	30.48

$$\text{Volumetric flow rate } (\dot{v}) = \frac{\text{Mass Flow Rate}}{\text{Density}} = 0.2881 \text{ m}^3/\text{s}$$

$$\text{Cross Sectional Area } (A_c) = \frac{\text{Volumetric flow rate}}{\text{velocity}} = 0.00945 \text{ m}^2$$

$$\text{Internal diameter of pipe}(di) = \sqrt{\frac{A_c * 4}{\pi}} = 0.109 \text{ m} = 4.29 \text{ in}$$

Corresponding standard stainless steel pipe;

Inside diameter = 5.047 in
 Wall thickness = 0.258 in
 Schedule number = 40s
 Outside diameter = 5.656 in
 Nominal diameter = 5 in

Table 57: Design Parameters for Pipe 4

S.N.	Design Parameters	Values
1	Temperature (°C)	25.36
2	Pressure (Bar)	4
3	Density (kg/m ³)	1198.14
4	Mass Flow Rate (Kg/s)	1.7098
5	Volume Flow (m ³ /s)	0.00142
6	Linear velocity(m/s)	1.828

$$\text{Volumetric flow rate } (\dot{v}) = \frac{\text{Mass Flow Rate}}{\text{Density}} = 0.00142 \text{ m}^3/\text{s}$$

$$\text{Cross Sectional Area } (A_c) = \frac{\text{Volumetric flow rate}}{\text{velocity}} = 0.00078 \text{ m}^2$$

$$\text{Internal diameter of pipe}(d_i) = \sqrt{\frac{A_c \times 4}{\pi}} = 0.0315 \text{ m} = 1.24 \text{ in}$$

Corresponding standard stainless steel pipe;

Inside diameter = 1.380 in
 Wall thickness = 0.140 in
 Schedule number = 40s
 Outside diameter = 1.660 in
 Nominal diameter = 1 in

Table 58: Design Parameters for Pipe 5

S.N.	Design Parameters	Values
1	Temperature (°C)	85
2	Pressure (Bar)	4
3	Density (kg/m ³)	0.2703
4	Mass Flow Rate (Kg/s)	0.0559
5	Volume Flow (m ³ /s)	0.2071
6	Linear velocity(m/s)	30.48

$$\text{Volumetric flow rate } (\dot{v}) = \frac{\text{Mass Flow Rate}}{\text{Density}} = 0.2068 \text{ m}^3/\text{s}$$

$$\text{Cross Sectional Area } (A_c) = \frac{\text{Volumetric flow rate}}{\text{velocity}} = 0.00678 \text{ m}^2$$

$$\text{Internal diameter of pipe}(di) = \sqrt{\frac{Ac*4}{\pi}} = 0.092 \text{ m} = 3.62 \text{ in}$$

Corresponding standard stainless steel pipe;

Inside diameter = 5.047 in

Wall thickness = 0.258 in

Schedule number = 40s

Outside diameter = 5.656 in

Nominal diameter = 5 in

Table 59: Design Parameters for Pipe 6

S.N.	Design Parameters	Values
1	Temperature (°C)	85
2	Pressure (Bar)	4
3	Density (kg/m ³)	1138.59
4	Mass Flow Rate (Kg/s)	1.7098
5	Volume Flow (m ³ /s)	0.00150
6	Linear velocity(m/s)	1.828

$$\text{Volumetric flow rate } (\dot{v}) = \frac{\text{Mass Flow Rate}}{\text{Density}} = 0.0015 \text{ m}^3/\text{s}$$

$$\text{Cross Sectional Area } (A_c) = \frac{\text{Volumetric flow rate}}{\text{velocity}} = 0.00082 \text{ m}^2$$

$$\text{Internal diameter of pipe}(di) = \sqrt{\frac{Ac*4}{\pi}} = 0.032 \text{ m} = 1.25 \text{ in}$$

Corresponding standard stainless steel pipe;

Inside diameter = 1.380 in

Wall thickness = 0.140 in

Schedule number = 40s

Outside diameter = 1.660 in

Nominal diameter = 1 in

Table 60: Design Parameters for Pipe 7

S.N.	Design Parameters	Values
1	Temperature (°C)	83.91
2	Pressure (Bar)	4
3	Density (kg/m ³)	8.4711
4	Mass Flow Rate (Kg/s)	1.7658
5	Volume Flow (m ³ /s)	0.2084
6	Linear velocity(m/s)	30.48

$$\text{Volumetric flowrate } (\dot{v}) = \frac{\text{Mass Flowrate}}{\text{Density}} = 0.2084 \text{ m}^3/\text{s}$$

$$\text{Cross Sectional Area } (A_c) = \frac{\text{Volumetric flow rate}}{\text{velocity}} = 0.0068 \text{ m}^2$$

$$\text{Internal diameter of pipe}(di) = \sqrt{\frac{A_c * 4}{\pi}} = 0.093 \text{ m} = 3.66 \text{ in}$$

Corresponding standard stainless steel pipe;

Inside diameter = 5.047 in

Wall thickness = 0.258 in

Schedule number = 40s

Outside diameter = 5.656 in

Nominal diameter = 5 in

Table 61: Design Parameters for Pipe 8

S.N.	Design Parameters	Values
1	Temperature (°C)	85
2	Pressure (Bar)	4
3	Density (kg/m ³)	22.678
4	Mass Flow Rate (Kg/s)	1.7658
5	Volume Flow (m ³ /s)	0.07786
6	Linear velocity(m/s)	1.828

$$\text{Volumetric flowrate } (\dot{v}) = \frac{\text{Mass Flowrate}}{\text{Density}} = 0.0778 \text{ m}^3/\text{s}$$

$$\text{Cross Sectional Area } (A_c) = \frac{\text{Volumetric flowrate}}{\text{velocity}} = 0.0425 \text{ m}^2$$

$$\text{Internal diameter of pipe}(di) = \sqrt{\frac{A_c * 4}{\pi}} = 0.232 \text{ m} = 9.13 \text{ in}$$

Corresponding standard stainless steel pipe;

Inside diameter = 9.981 in

Wall thickness = 0.322 in

Schedule number = 40s

Outside diameter = 10.625 in

Nominal diameter = 10 in

Table 62: Design Parameters for Pipe 9

S.N.	Design Parameters	Values
1	Temperature (°C)	25
2	Pressure (Bar)	1
3	Density (kg/m ³)	0.1180

4	Mass Flow Rate (Kg/s)	0.0267
5	Volume Flow (m^3/s)	0.2261
6	Linear velocity(m/s)	30.48

$$\text{Volumetric flowrate } (\dot{v}) = \frac{\text{Mass Flowrate}}{\text{Density}} = 0.2262 \text{ m}^3/\text{s}$$

$$\text{Cross Sectional Area } (A_c) = \frac{\text{Volumetric flowrate}}{\text{velocity}} = 0.0074 \text{ m}^2$$

$$\text{Internal diameter of pipe}(di) = \sqrt{\frac{Ac*4}{\pi}} = 0.097 \text{ m} = 3.81 \text{ in}$$

Corresponding standard stainless steel pipe;

Inside diameter = 5.047 in

Wall thickness = 0.258 in

Schedule number = 40s

Outside diameter = 5.656 in

Nominal diameter = 5 in

Table 63: Design Parameters for Pipe 10

S.N.	Design Parameters	Values
1	Temperature (°C)	25
2	Pressure (Bar)	1
3	Density (kg/m ³)	1158.38
4	Mass Flow Rate (Kg/s)	1.73917
5	Volume Flow (m^3/s)	0.00150
6	Linear velocity(m/s)	1.828

$$\text{Volumetric flowrate } (\dot{v}) = \frac{\text{Mass Flowrate}}{\text{Density}} = 0.0015 \text{ m}^3/\text{s}$$

$$\text{Cross Sectional Area } (A_c) = \frac{\text{Volumetric flowrate}}{\text{velocity}} = 0.00082 \text{ m}^2$$

$$\text{Internal diameter of pipe}(di) = \sqrt{\frac{Ac*4}{\pi}} = 0.032 \text{ m} = 1.25 \text{ in}$$

Corresponding standard stainless steel pipe;

Inside diameter = 1.380 in

Wall thickness = 0.140 in

Schedule number = 40s

Outside diameter = 1.660 in

Nominal diameter = 1 in

Table 64: Design Parameters for Pipe 11

S.N.	Design Parameters	Values
1	Temperature (°C)	120
2	Pressure (Bar)	1
3	Density (kg/m ³)	7.3119
4	Mass Flow Rate (Kg/s)	1.7391
5	Volume Flow (m ³ /s)	0.2378
6	Linear velocity(m/s)	1.828

$$\text{Volumetric flowrate } (\dot{v}) = \frac{\text{Mass Flowrate}}{\text{Density}} = 0.2378 \text{ m}^3/\text{s}$$

$$\text{Cross Sectional Area } (A_c) = \frac{\text{Volumetric flowrate}}{\text{velocity}} = 0.130 \text{ m}^2$$

$$\text{Internal diameter of pipe}(d_i) = \sqrt{\frac{A_c * 4}{\pi}} = 0.406 \text{ m} = 15.9 \text{ in}$$

Corresponding standard stainless steel pipe;

Inside diameter = 16.380 in

Wall thickness = 0.425 in

Schedule number = 40s

Outside diameter = 17.660 in

Nominal diameter = 17 in

Table 65: Design Parameters for Pipe S12

S.N.	Design Parameters	Values
1	Temperature (°C)	167.78
2	Pressure (Bar)	0.8
3	Density (kg/m ³)	1.3248
4	Mass Flow Rate (Kg/s)	1.1026
5	Volume Flow (m ³ /s)	0.8322
6	Linear velocity(m/s)	1.828

$$\text{Volumetric flowrate } (\dot{v}) = \frac{\text{Mass Flowrate}}{\text{Density}} = 0.8322 \text{ m}^3/\text{s}$$

$$\text{Cross Sectional Area } (A_c) = \frac{\text{Volumetric flowrate}}{\text{velocity}} = 0.4552 \text{ m}^2$$

$$\text{Internal diameter of pipe}(d_i) = \sqrt{\frac{A_c * 4}{\pi}} = 0.761 \text{ m} = 29.9 \text{ in}$$

Corresponding standard steel pipe;

Inside diameter = 30.981 in

Wall thickness = 0.280 in

Schedule number = 40s

Outside diameter = 32.625 in

Nominal diameter = 32 in

Table 66: Design Parameters for Pipe S6

S.N.	Design Parameters	Values
1	Temperature (°C)	271.94
2	Pressure (Bar)	0.8
3	Density (kg/m ³)	932.66
4	Mass Flow Rate (Kg/s)	0.6365
5	Volume Flow (m ³ /s)	0.000682
6	Linear velocity(m/s)	1.828

$$\text{Volumetric flowrate } (\dot{v}) = \frac{\text{Mass Flowrate}}{\text{Density}} = 0.00068 \text{ m}^3/\text{s}$$

$$\text{Cross Sectional Area } (A_c) = \frac{\text{Volumetric flowrate}}{\text{velocity}} = 0.00037 \text{ m}^2$$

$$\text{Internal diameter of pipe}(di) = \sqrt{\frac{Ac*4}{\pi}} = 0.021 \text{ m} = 0.82 \text{ in}$$

Corresponding standard stainless steel pipe;

Inside diameter = 1.049 in

Wall thickness = 0.133 in

Schedule number = 40s

Outside diameter =1.315 in

Nominal diameter = 1 in

Table 67: Design Parameters for Pipe S13

S.N.	Design Parameters	Values
1	Temperature (°C)	120
2	Pressure (Bar)	1
3	Density (kg/m ³)	3.5246
4	Mass Flow Rate (Kg/s)	1.1026
5	Volume Flow (m ³ /s)	0.3128
6	Linear velocity(m/s)	1.828

$$\text{Volumetric flowrate } (\dot{v}) = \frac{\text{Mass Flowrate}}{\text{Density}} = 0.3128 \text{ m}^3/\text{s}$$

$$\text{Cross Sectional Area } (A_c) = \frac{\text{Volumetric flowrate}}{\text{velocity}} = 0.1711 \text{ m}^2$$

$$\text{Internal diameter of pipe}(di) = \sqrt{\frac{Ac*4}{\pi}} = 0.466 \text{ m} = 18.3 \text{ in}$$

Corresponding standard stainless steel pipe;

Inside diameter = 19.065 in

Wall thickness = 0.280 in

Schedule number = 40s

Outside diameter = 19.625 in

Nominal diameter = 19 in

Table 68: Design Parameters for Pipe S7

S.N.	Design Parameters	Values
1	Temperature (°C)	115.88
2	Pressure (Bar)	0.5
3	Density (kg/m ³)	0.5162
4	Mass Flow Rate (Kg/s)	0.4260
5	Volume Flow (m ³ /s)	0.8252
6	Linear velocity(m/s)	1.828

$$\text{Volumetric flowrate } (\dot{v}) = \frac{\text{Mass Flowrate}}{\text{Density}} = 0.8252 \text{ m}^3/\text{s}$$

$$\text{Cross Sectional Area } (Ac) = \frac{\text{Volumetric flowrate of stream 3}}{\text{velocity of stream 3}} = 0.4514 \text{ m}^2$$

$$\text{Internal diameter of pipe}(di) = \sqrt{\frac{Ac*4}{\pi}} = 0.758 \text{ m} = 29.8 \text{ in}$$

Corresponding standard stainless steel pipe;

Inside diameter = 30.049 in

Wall thickness = 0.248 in

Schedule number = 40s

Outside diameter = 30.315 in

Nominal diameter = 30 in

Table 69: Design Parameters for Pipe S9

S.N.	Design Parameters	Values
1	Temperature (°C)	183.87
2	Pressure (Bar)	0.5
3	Density (kg/m ³)	1026.09
4	Mass Flow Rate (Kg/s)	0.6765
5	Volume Flow (m ³ /s)	0.000659
6	Linear velocity(m/s)	1.828

$$\text{Volumetric flowrate } (\dot{v}) = \frac{\text{Mass Flowrate}}{\text{Density}} = 0.00065 \text{ m}^3/\text{s}$$

$$\text{Cross Sectional Area } (A_c) = \frac{\text{Volumetric flowrate}}{\text{velocity}} = 0.00036 \text{ m}^2$$

$$\text{Internal diameter of pipe}(di) = \sqrt{\frac{A_c \cdot 4}{\pi}} = 0.021 \text{ m} = 0.82 \text{ in}$$

Corresponding standard stainless steel pipe;

Inside diameter = 1.049 in

Wall thickness = 0.133 in

Schedule number = 40s

Outside diameter = 1.315 in

Nominal diameter = 1 in

Table 70: Design Parameters for Pipe S16

S.N.	Design Parameters	Values
1	Temperature (°C)	25
2	Pressure (Bar)	1
3	Density (kg/m ³)	1015.07
4	Mass Flow Rate (Kg/s)	0.4260
5	Volume Flow (m ³ /s)	0.00041
6	Linear velocity(m/s)	1.828

$$\text{Volumetric flowrate } (\dot{v}) = \frac{\text{Mass Flowrate}}{\text{Density}} = 0.00041 \text{ m}^3/\text{s}$$

$$\text{Cross Sectional Area } (A_c) = \frac{\text{Volumetric flowrate}}{\text{velocity}} = 0.00022 \text{ m}^2$$

$$\text{Internal diameter of pipe}(di) = \sqrt{\frac{A_c \cdot 4}{\pi}} = 0.016 \text{ m} = 0.62 \text{ in}$$

Corresponding standard stainless steel pipe;

Inside diameter = 1.049 in

Wall thickness = 0.133 in

Schedule number = 40s

Outside diameter = 1.315 in

Nominal diameter = 1 in

Table 71: Design Parameters for Pipe S17

S.N.	Design Parameters	Values
1	Temperature (°C)	25
2	Pressure (Bar)	1
3	Density (kg/m ³)	993.974

4	Mass Flow Rate (Kg/s)	0.1795
5	Volume Flow (m^3/s)	0.00018
6	Linear velocity(m/s)	1.828

$$\text{Volumetric flowrate } (\dot{v}) = \frac{\text{Mass Flowrate}}{\text{Density}} = 0.00018 \text{ m}^3/\text{s}$$

$$\text{Cross Sectional Area } (A_c) = \frac{\text{Volumetric flowrate}}{\text{velocity}} = 0.000098 \text{ m}^2$$

$$\text{Internal diameter of pipe}(di) = \sqrt{\frac{A_c \times 4}{\pi}} = 0.011 \text{ m} = 0.43 \text{ in}$$

Corresponding standard stainless steel pipe;

Inside diameter = 1.049 in

Wall thickness = 0.133 in

Schedule number = 40s

Outside diameter = 1.315 in

Nominal diameter = 1 in

Table 72: Design Parameters for Pipe S18

S.N.	Design Parameters	Values
1	Temperature (°C)	25
2	Pressure (Bar)	1
3	Density (kg/m ³)	1093.46
4	Mass Flow Rate (Kg/s)	0.2464
5	Volume Flow (m^3/s)	0.0002254
6	Linear velocity(m/s)	1.828

$$\text{Volumetric flow rate } (\dot{v}) = \frac{\text{Mass Flow Rate}}{\text{Density}} = 0.00022 \text{ m}^3/\text{s}$$

$$\text{Cross Sectional Area } (A_c) = \frac{\text{Volumetric flow rate}}{\text{velocity}} = 0.00012 \text{ m}^2$$

$$\text{Internal diameter of pipe}(di) = \sqrt{\frac{A_c \times 4}{\pi}} = 0.012 \text{ m} = 0.47 \text{ in}$$

Corresponding standard stainless steel pipe;

Inside diameter = 1.049 in

Wall thickness = 0.133 in

Schedule number = 40s

Outside diameter = 1.315 in

Nominal diameter = 1 in

Table 73: Design Parameters for Pipe S19

S.N.	Design Parameters	Values
1	Temperature (°C)	25
2	Pressure (Bar)	1
3	Density (kg/m ³)	1074.08
4	Mass Flow Rate (Kg/s)	1.4179
5	Volume Flow (m ³ /s)	0.00132
6	Linear velocity(m/s)	1.828

$$\text{Volumetric flowrate } (\dot{v}) = \frac{\text{Mass Flowrate}}{\text{Density}} = 0.00132 \text{ m}^3/\text{s}$$

$$\text{Cross Sectional Area } (A_c) = \frac{\text{Volumetric flowrate}}{\text{velocity}} = 0.00072 \text{ m}^2$$

$$\text{Internal diameter of pipe}(di) = \sqrt{\frac{Ac*4}{\pi}} = 0.030 \text{ m} = 1.18 \text{ in}$$

Corresponding standard stainless steel pipe;

Inside diameter = 1.380 in

Wall thickness = 0.140 in

Schedule number = 40s

Outside diameter =1.660 in

Nominal diameter = 1 in

Table 74: Design Parameters for Pipe S1

S.N.	Design Parameters	Values
1	Temperature (°C)	111.99
2	Pressure (Bar)	0.8
3	Density (kg/m ³)	1007.35
4	Mass Flow Rate (Kg/s)	2.0544
5	Volume Flow (m ³ /s)	0.00203
6	Linear velocity(m/s)	1.828

$$\text{Volumetric flow rate } (\dot{v}) = \frac{\text{Mass Flow Rate}}{\text{Density}} = 0.0020 \text{ m}^3/\text{s}$$

$$\text{Cross Sectional Area } (A_c) = \frac{\text{Volumetric flow rate}}{\text{velocity}} = 0.0011 \text{ m}^2$$

$$\text{Internal diameter of pipe}(di) = \sqrt{\frac{Ac*4}{\pi}} = 0.037 \text{ m} = 1.45 \text{ in}$$

Corresponding standard stainless steel pipe;

Inside diameter = 1.59 in

Wall thickness = 0.133 in

Schedule number = 40s

Outside diameter =2.15 in

Nominal diameter = 2 in

Table 75: Design Parameters for Pipe S4

S.N.	Design Parameters	Values
1	Temperature (°C)	108
2	Pressure (Bar)	1
3	Density (kg/m ³)	1011.8
4	Mass Flow Rate (Kg/s)	2.0544
5	Volume Flow (m ³ /s)	0.00203
6	Linear velocity(m/s)	1.828

$$\text{Volumetric flowrate } (\dot{v}) = \frac{\text{Mass Flowrate}}{\text{Density}} = 0.0020 \text{ m}^3/\text{s}$$

$$\text{Cross Sectional Area } (A_c) = \frac{\text{Volumetric flowrate}}{\text{velocity}} = 0.0011 \text{ m}^2$$

$$\text{Internal diameter of pipe}(di) = \sqrt{\frac{Ac*4}{\pi}} = 0.037 \text{ m} = 1.45 \text{ in}$$

Corresponding standard stainless steel pipe;

Inside diameter = 1.680 in

Wall thickness = 0.140 in

Schedule number = 40s

Outside diameter = 2.15 in

Nominal diameter = 2 in

Table 76: Design Parameters for Pipe S5

S.N.	Design Parameters	Values
1	Temperature (°C)	108
2	Pressure (Bar)	1
3	Density (kg/m ³)	969.731
4	Mass Flow Rate (Kg/s)	2.0544
5	Volume Flow (m ³ /s)	0.00211
6	Linear velocity(m/s)	1.828

$$\text{Volumetric flowrate } (\dot{v}) = \frac{\text{Mass Flowrate}}{\text{Density}} = 0.0021 \text{ m}^3/\text{s}$$

$$\text{Cross Sectional Area } (A_c) = \frac{\text{Volumetric flowrate}}{\text{velocity}} = 0.0011 \text{ m}^2$$

$$\text{Internal diameter of pipe}(di) = \sqrt{\frac{Ac*4}{\pi}} = 0.037 \text{ m} = 1.45 \text{ in}$$

Corresponding standard stainless steel pipe;

Inside diameter = 1.80 in

Wall thickness = 0.140 in
Schedule number = 40s
Outside diameter = 2.15 in
Nominal diameter = 2 in

Table 77: Design Parameters for Pipe S14

S.N.	Design Parameters	Values
1	Temperature (°C)	125.65
2	Pressure (Bar)	0.8
3	Density (kg/m ³)	2.0888
4	Mass Flow Rate (Kg/s)	1.175
5	Volume Flow (m ³ /s)	0.5625
6	Linear velocity(m/s)	1.828

$$\text{Volumetric flowrate } (\dot{v}) = \frac{\text{Mass Flowrate}}{\text{Density}} = 0.5625 \text{ m}^3/\text{s}$$

$$\text{Cross Sectional Area } (A_c) = \frac{\text{Volumetric flowrate}}{\text{velocity}} = 0.3077 \text{ m}^2$$

$$\text{Internal diameter of pipe}(di) = \sqrt{\frac{A_c \cdot 4}{\pi}} = 0.625 \text{ m} = 24.6 \text{ in}$$

Corresponding standard stainless steel pipe;

Inside diameter = 26.049 in

Wall thickness = 0.133 in

Schedule number = 40s

Outside diameter = 28.315 in

Nominal diameter = 28 in

Table 78: Design Parameters for Pipe S15

S.N.	Design Parameters	Values
1	Temperature (°C)	245.72
2	Pressure (Bar)	0.8
3	Density (kg/m ³)	754.958
4	Mass Flow Rate (Kg/s)	0.8794
5	Volume Flow (m ³ /s)	0.00116
6	Linear velocity(m/s)	1.828

$$\text{Volumetric flowrate } (\dot{v}) = \frac{\text{Mass Flowrate}}{\text{Density}} = 0.0011 \text{ m}^3/\text{s}$$

$$\text{Cross Sectional Area } (A_c) = \frac{\text{Volumetric flowrate}}{\text{velocity}} = 0.00063 \text{ m}^2$$

$$\text{Internal diameter of pipe}(d_i) = \sqrt{\frac{Ac*4}{\pi}} = 0.028 \text{ m} = 1.10 \text{ in}$$

Corresponding standard stainless steel pipe;

Inside diameter = 1.380 in

Wall thickness = 0.140 in

Schedule number = 40s

Outside diameter = 1.660 in

Nominal diameter = 1 in

Table 79: Design Parameters for Pipe S2

S.N.	Design Parameters	Values
1	Temperature (°C)	0
2	Pressure (Bar)	1
3	Density (kg/m ³)	1292.99
4	Mass Flow Rate (Kg/s)	0.8794
5	Volume Flow (m ³ /s)	0.00068
6	Linear velocity(m/s)	1.828

$$\text{Volumetric flowrate } (\dot{v}) = \frac{\text{Mass Flow Rate}}{\text{Density}} = 0.00068 \text{ m}^3/\text{s}$$

$$\text{Cross Sectional Area } (A_c) = \frac{\text{Volumetric flowrate}}{\text{velocity}} = 0.00037 \text{ m}^2$$

$$\text{Internal diameter of pipe}(d_i) = \sqrt{\frac{Ac*4}{\pi}} = 0.021 \text{ m} = 0.82 \text{ in}$$

Corresponding standard stainless steel pipe;

Inside diameter = 1.049 in

Wall thickness = 0.133 in

Schedule number = 40s

Outside diameter = 1.315 in

Nominal diameter = 1 in

Heat Exchanger B4

Step 1: Find mean temperature difference (ΔT_m)

From Aspen,

$$\text{Log mean temperature difference (LMTD)} = 327.42 \text{ K}$$

$$\text{Correction factor } (F_T) = 1 \text{ (As countercurrent)}$$

$$\text{So, } \Delta T_m = \text{LMTD} \cdot F_T = 327.42 \text{ K}$$

Step 2: Find heat duty (Q)

From Aspen,

$$Q = 160138 \text{ Watt}$$

Step 3: Assume a reasonable value of overall heat transfer coefficient ($U_{o,assume}$)

Overall heat transfer coefficient is assumed as,

$$U_{o,assume} = 750 \frac{\text{Watt}}{\text{m}^2 \cdot \text{K}}$$

Step 4: Decide tentative number of shell and tube passes (n_p)

$$n_p = 2$$

Step 5: Calculate heat transfer area (A) required

$$\text{Heat transfer area (A)} = \frac{Q}{U_o \cdot \text{LMTD}} = 0.652 \text{ m}^2$$

Step 6: Select tube material, decide the tube diameter (ID = d_i , OD = d_o), its wall thickness (in terms of BWG or SWG) and tube length (L). Calculate the number of tubes (n_T).

Tube side material is a Nitrobenzene.

Since a significant amount of Nitrobenzene is present. As Nitrobenzene emits toxic fumes of nitrogen oxides upon combustion. Thus, Stainless steel 304 (as it is cheaper than 316) is used as the tube material.

Let,

$$\text{Tube length (L)} = 1.8228 \text{ m (6 ft is a standard length)}$$

$$\text{Outside diameter } (d_o) = 19.05 \text{ mm (For the process industry, 19.05 is most common)}$$

$$\text{Let, Birmingham Wire Gauge (BWG)} = 16 = 1.651 \text{ mm}$$

$$\text{Internal diameter } (d_i) = d_o - BWG = 17.399 \text{ mm}$$

$$\text{So, Area of 1 tube} = \pi \cdot d_o \cdot L = \pi \cdot 19.05 \times 10^{-3} \cdot 1.8228 = 0.1091 \text{ m}^2$$

$$\text{Thus, Number of tubes } (N_T) = \frac{A}{\text{Area of 1 tube}} = \frac{0.652}{0.1091} = 5.97 \approx 6$$

Step 7: Decide type of shell and tube exchanger (fixed tube sheet, U-tube etc.). Select the tube pitch (P_T), determine inside shell diameter (D_s) that can accommodate the calculated number of tubes (n_T)

As the heat exchanger is subjected to a temperature of (328 K), U-tube heat exchanger.

Triangular pitch is better for heat transfer and surface area per unit length (greatest tube density). Thus, triangular pitch is used.

$$\text{Tube pitch } (P_T) = 1.25 \cdot d_o = 0.0238 \text{ m}$$

Now,

$$\text{Bundle diameter } (BD) = d_o \cdot \left(\frac{N_T}{K_1} \right)^{\frac{1}{n_1}}$$

For single pass and triangular pitch,

$$n_1 = 2.142 \text{ and } K_1 = 0.319$$

So,

$$BD = 0.0190 \cdot \left(\frac{6}{0.319} \right)^{\frac{1}{2.142}} = 0.0747 \text{ m} = 74.76 \text{ mm}$$

Bundle diameter clearance (BDC) = 90 mm for pull through U – Tube.

$$\text{Inside shell diameter } (D_s) = (74.76 + 90) \text{ mm} = 164.76 \text{ mm}$$

Step 8: Assign fluid to shell side or tube side. Select the type of baffle, its size (i.e. percentage cut), spacing and number.

Tube side material is Nitrobenzene Meanwhile, the shell side material is water.

Baffle type is cut-segmental baffle.

$$\text{Baffle spacing } (B_s) = 0.2 \cdot 164.76 = 32.95 \text{ mm}$$

Percentage cut of baffles = 25% (The most widely used)

$$\text{Number of baffles} = \frac{\text{Inside shell diameter}}{\text{Baffle spacing}} = \frac{164.76}{32.95} = 5$$

Step 9: Comparison of calculated and assumed overall heat transfer coefficient (U)

A. Shell side coefficient calculation

$$h_o = \frac{j_n}{D_e} \cdot k \cdot Re \cdot Pr^{\frac{1}{3}}$$

Now,

$$Re = \frac{G_s \cdot D_e}{\mu}$$

Where,

$$\text{Mass velocity } (G_s) = \frac{\text{Mass flow rate}}{\text{Area of Shell}} = \frac{0.542}{\frac{(P_T - d_o) \cdot D_s \cdot B_s}{P_T}} = \frac{0.542}{0.0010} = 499.33 \frac{\text{kg}}{\text{m}^2 \cdot \text{s}}$$

$$D_e = \frac{1.1}{d_o} (P_T^2 - 0.917 \cdot d_o^2) = 0.0136 \text{ m}$$

$$\mu = 5.39 \times 10^{-4} \text{ Pa} \cdot \text{s}$$

So,

$$Re = 12493.03$$

And,

$$Pr = \frac{\mu \cdot C_p}{k} = 3.58$$

Here,

$$C_p = 4180 \frac{\text{J}}{\text{kg} \cdot \text{K}}$$

$$k = 0.629 \frac{\text{W}}{\text{m} \cdot \text{K}}$$

Again,

$$h_o = 124769.60 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

B. Tube side calculation

$$h_i = \frac{j_n}{D_e} \cdot k \cdot Re \cdot Pr^{\frac{1}{3}}$$

$$T_{mean} = 328.33 \text{ K}$$

$$Re = \frac{4 \cdot \text{mass flow rate} \cdot \frac{n_p}{N_T}}{\pi \cdot d_i \cdot \mu} = 1251.17$$

Here,

$$\mu = 33.35 \times 10^{-3} \text{ Pa} \cdot \text{s}$$

$$C_p = 1455 \frac{\text{J}}{\text{kg} \cdot \text{K}}$$

$$k = 0.133 \frac{\text{W}}{\text{m} \cdot \text{K}}$$

$$Pr = \frac{\mu \cdot C_p}{k} = \frac{33.35 \times 10^{-3} \cdot 1455}{0.133} = 364.84$$

So,

$$h_i = \frac{0.15}{0.0136} \cdot 0.133 \cdot 1250.53 \cdot (364.84)^{\frac{1}{3}} = 13220.63 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

Thus,

$$U_{o,calc} = \frac{1}{\frac{1}{h_o} + \frac{1}{h_{do}} + \frac{d_o \cdot \ln\left(\frac{d_o}{d_i}\right)}{2 \cdot K_w} + \frac{d_o}{d_i \cdot h_i} + \frac{d_o}{d_i \cdot h_{di}}}$$

From literature,

$$h_{do} = 4500 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

$$h_{di} = 3500 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

So,

$$U_{o,calc} = \frac{1}{7.38 \times 10^{-6} + 2.22 \times 10^{-4} + 5.59 \times 10^{-4} + 8.37 \times 10^{-5} + 3.13 \times 10^{-4}}$$

$$\text{or, } U_{o,calc} = 842.36 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

$$\text{Since } \frac{U_{o,calc} - U_{o,assume}}{U_{o,assume}} \cdot 100\%$$

= 13%, which is less than 30%, our assumptions for design are valid.

Step 10: Over design calculation

Over design percentage is zero according from Aspen

Heat Exchanger B15

Step 1: Find mean temperature difference (ΔT_m)

From Aspen,

$$\text{Log mean temperature difference (LMTD)} = 343.46 \text{ K}$$

$$\text{Correction factor } (F_T) = 1 \text{ (As countercurrent)}$$

$$\text{So, } \Delta T_m = \text{LMTD} \cdot F_T = 343.46 \text{ K}$$

Step 2: Find heat duty (Q)

From Aspen,

$$Q = 583971 \text{ Watt}$$

Step 3: Assume a reasonable value of overall heat transfer coefficient ($U_{o,assume}$)

Overall heat transfer coefficient is assumed as,

$$U_{o,assume} = 750 \frac{\text{Watt}}{\text{m}^2 \cdot \text{K}}$$

Step 4: Decide tentative number of shell and tube passes (n_p)

$$n_p = 2$$

Step 5: Calculate heat transfer area (A) required

$$\text{Heat transfer area (A)} = \frac{Q}{U_o \cdot \text{LMTD}} = 2.267 \text{ m}^2$$

Step 6: Select tube material, decide the tube diameter (ID = d_i , OD = d_o), its wall thickness (in terms of BWG or SWG) and tube length (L). Calculate the number of tubes (n_T).

Tube side material is a mixture of Hydrogen, Nitrobenzene, Para Amino Phenol, Aniline, Water .

Since a significant amount of Nitrobenzene and Para Amino Phenol are present. As Nitrobenzene emits toxic fumes of nitrogen oxides upon combustion. Thus, Stainless steel 304 (as it is cheaper than 316) is used as the tube material.

Let,

$$\text{Tube length (L)} = 1.8228 \text{ m (6 ft is a standard length)}$$

$$\text{Outside diameter } (d_o) = 19.05 \text{ mm (For the process industry, 19.05 is most common)}$$

$$\text{Let, Birmingham Wire Gauge (BWG)} = 16 = 1.651 \text{ mm}$$

$$\text{Internal diameter } (d_i) = d_o - BWG = 17.399 \text{ mm}$$

$$\text{So, Area of 1 tube} = \pi \cdot d_o \cdot L = \pi \cdot 19.05 \times 10^{-3} \cdot 1.8228 = 0.1091 \text{ m}^2$$

$$\text{Thus, Number of tubes } (N_T) = \frac{A}{\text{Area of 1 tube}} = \frac{2.267}{0.1091} = 20.77 \approx 21$$

Step 7: Decide type of shell and tube exchanger (fixed tube sheet, U-tube etc.). Select the tube pitch (P_T), determine inside shell diameter (D_s) that can accommodate the calculated number of tubes (n_T)

As the heat exchanger is subjected to a temperature of (344 K), U-tube heat exchanger.

Triangular pitch is better for heat transfer and surface area per unit length (greatest tube density). Thus, triangular pitch is used.

$$\text{Tube pitch } (P_T) = 1.25 \cdot d_o = 0.0238 \text{ m}$$

Now,

$$\text{Bundle diameter } (BD) = d_o \cdot \left(\frac{N_T}{K_1} \right)^{\frac{1}{n_1}}$$

For single pass and triangular pitch,

$$n_1 = 2.142 \text{ and } K_1 = 0.319$$

So,

$$BD = 0.0190 \cdot \left(\frac{21}{0.319} \right)^{\frac{1}{2.142}} = 0.1341 \text{ m} = 134.18 \text{ mm}$$

And,

$$\text{Bundle diameter clearance } (BDC) = 90 \text{ mm for pull through U - Tube.}$$

$$\text{Inside shell diameter } (D_s) = (134.18 + 90) \text{ mm} = 224.18 \text{ mm}$$

Step 8: Assign fluid to shell side or tube side. Select the type of baffle, its size (i.e. percentage cut), spacing and number.

Tube side material is a mixture of Hydrogen, Nitrobenzene, Para Amino Phenol, Aniline, Water. Meanwhile, the shell side material is water.

Baffle type is cut-segmental baffle.

$$\text{Baffle spacing } (B_s) = 0.2 \cdot 224.18 = 44.83 \text{ mm}$$

$$\text{Percentage cut of baffles} = 25\% \text{ (The most widely used)}$$

$$\text{Number of baffles} = \frac{\text{Inside shell diameter}}{\text{Baffle spacing}} = \frac{224.18}{44.83} = 5$$

Step 9: Comparison of calculated and assumed overall heat transfer coefficient (U)

A. Shell side coefficient calculation

$$h_o = \frac{j_n}{D_e} \cdot k \cdot Re \cdot Pr^{\frac{1}{3}}$$

Now,

$$Re = \frac{G_s \cdot D_e}{\mu}$$

Where,

$$\text{Mass velocity } (G_s) = \frac{\text{Mass flow rate}}{\text{Area of Shell}} = \frac{1.25}{\frac{(P_T - d_o) \cdot D_s \cdot B_s}{P_T}} = \frac{1.25}{0.0019} = 621.80 \frac{\text{kg}}{\text{m}^2 \cdot \text{s}}$$

$$D_e = \frac{1.1}{d_o} (P_T^2 - 0.917 \cdot d_o^2) = 0.0136 \text{ m}$$

$$\mu = 5.39 \times 10^{-4} \text{ Pa} \cdot \text{s}$$

So,

$$Re = 15563.37$$

And,

$$Pr = \frac{\mu \cdot C_p}{k} = 3.58$$

Here,

$$C_p = 4180 \frac{\text{J}}{\text{kg} \cdot \text{K}}$$

$$k = 0.629 \frac{\text{W}}{\text{m} \cdot \text{K}}$$

Again,

$$h_o = 164441.43 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

B. Tube side calculation

$$h_i = \frac{j_n}{D_e} \cdot k \cdot Re \cdot Pr^{\frac{1}{3}}$$

$$T_{mean} = 345.65 \text{ K}$$

Here,

$$\mu = 2.15 \times 10^{-3} \text{ Pa} \cdot \text{s}$$

$$k = 0.192 \frac{\text{W}}{\text{m} \cdot \text{K}}$$

$$Re = \frac{4 \cdot \text{mass flow rate} \cdot \frac{n_p}{N_T}}{\pi \cdot d_i \cdot \mu} = 5642.90$$

$$Pr = \frac{\mu \cdot C_p}{k} = \frac{2.15 \times 10^{-3} \cdot 2072.91}{0.192} = 23.15$$

So,

$$h_i = \frac{0.15}{0.0136} \cdot 0.192 \cdot 5640.04 \cdot (23.21)^{\frac{1}{3}} = 34422.71 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

Thus,

$$U_{o,calc} = \frac{1}{\frac{1}{h_o} + \frac{1}{h_{do}} + \frac{d_o \cdot \ln\left(\frac{d_o}{d_i}\right)}{2 \cdot K_w} + \frac{d_o}{d_i \cdot h_i} + \frac{d_o}{d_i \cdot h_{di}}}$$

From literature,

$$h_{do} = 4500 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

$$h_{di} = 3500 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

So,

$$U_{o,calc} = \frac{1}{6.08 \times 10^{-6} + 1.66 \times 10^{-4} + 5.60 \times 10^{-4} + 3.22 \times 10^{-4} + 2.19 \times 10^{-4}}$$

$$\text{or, } U_{o,calc} = 881.53 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

$$\text{Since } \frac{U_{o,calc} - U_{o,assume}}{U_{o,assume}} \cdot 100\%$$

= 18 %, which is less than 30%, our assumptions for design are valid.

Step 10: Over design calculation

Over design percentage is zero according from Aspen

Heat Exchanger B16

Step 1: Find mean temperature difference (ΔT_m)

From Aspen,

$$\text{Log mean temperature difference (LMTD)} = 416.77 \text{ K}$$

$$\text{Correction factor } (F_T) = 1 \text{ (As countercurrent)}$$

$$\text{So, } \Delta T_m = \text{LMTD} \cdot F_T = 416.5 \text{ K}$$

Step 2: Find heat duty (Q)

From Aspen,

$$Q = 495979 \text{ Watt}$$

Step 3: Assume a reasonable value of overall heat transfer coefficient ($U_{o,assume}$)

Overall heat transfer coefficient is assumed as,

$$U_{o,assume} = 750 \frac{\text{Watt}}{\text{m}^2 \cdot \text{K}}$$

Step 4: Decide tentative number of shell and tube passes (n_p)

$$n_p = 2$$

Step 5: Calculate heat transfer area (A) required

$$\text{Heat transfer area (A)} = \frac{Q}{U_o \cdot \text{LMTD}} = 1.588 \text{ m}^2$$

Step 6: Select tube material, decide the tube diameter (ID = d_i , OD = d_o), its wall thickness (in terms of BWG or SWG) and tube length (L). Calculate the number of tubes (n_T).

Tube side material is a mixture of Hydrogen, Nitrobenzene, Para Amino Phenol, Aniline, Water.

Since a significant amount of Nitrobenzene and Para Amino Phenol are present. As Nitrobenzene emits toxic fumes of nitrogen oxides upon combustion. Thus, Stainless steel 304 (as it is cheaper than 316) is used as the tube material.

Let,

$$\text{Tube length (L)} = 1.8228 \text{ m (6 ft is a standard length)}$$

$$\text{Outside diameter } (d_o) = 19.05 \text{ mm (For the process industry, 19.05 is most common)}$$

$$\text{Let, Birmingham Wire Gauge (BWG)} = 16 = 1.651 \text{ mm}$$

$$\text{Internal diameter } (d_i) = d_o - BWG = 17.399 \text{ mm}$$

$$\text{So, Area of 1 tube} = \pi \cdot d_o \cdot L = \pi \cdot 19.05 \times 10^{-3} \cdot 1.8228 = 0.1091 \text{ m}^2$$

$$\text{Thus, Number of tubes } (N_T) = \frac{A}{\text{Area of 1 tube}} = \frac{1.586}{0.1091} = 14.53 \approx 15$$

Step 7: Decide type of shell and tube exchanger (fixed tube sheet, U-tube etc.). Select the tube pitch (P_T), determine inside shell diameter (D_s) that can accommodate the calculated number of tubes (n_T)

As the heat exchanger is subjected to a temperature of (417 K), U-tube heat exchanger.

Triangular pitch is better for heat transfer and surface area per unit length (greatest tube density). Thus, triangular pitch is used.

$$\text{Tube pitch } (P_T) = 1.25 \cdot d_o = 0.0238 \text{ m}$$

Now,

$$\text{Bundle diameter } (BD) = d_o \cdot \left(\frac{N_T}{K_1} \right)^{\frac{1}{n_1}}$$

For single pass and triangular pitch,

$$n_1 = 2.142 \text{ and } K_1 = 0.319$$

So,

$$BD = 0.0190 \cdot \left(\frac{15}{0.319} \right)^{\frac{1}{2.142}} = 0.1146 \text{ m} = 114.67 \text{ mm}$$

And,

$$\text{Bundle diameter clearance } (BDC) = 90 \text{ mm for pull through U - Tube.}$$

$$\text{Inside shell diameter } (D_s) = (114.67 + 90) \text{ mm} = 204.67 \text{ mm}$$

Step 8: Assign fluid to shell side or tube side. Select the type of baffle, its size (i.e. percentage cut), spacing and number.

Tube side material is a mixture of Hydrogen, Nitrobenzene, Para Amino Phenol, Aniline, Water. Meanwhile, the shell side material is water.

Baffle type is cut-segmental baffle.

$$\text{Baffle spacing } (B_s) = 0.2 \cdot 204.67 = 40.934 \text{ mm}$$

$$\text{Percentage cut of baffles} = 25\% \text{ (The most widely used)}$$

$$\text{Number of baffles} = \frac{\text{Inside shell diameter}}{\text{Baffle spacing}} = \frac{204.67}{40.934} = 5$$

Step 9: Comparison of calculated and assumed overall heat transfer coefficient (U)

A. Shell side coefficient calculation

$$h_o = \frac{j_n}{D_e} \cdot k \cdot Re \cdot Pr^{\frac{1}{3}}$$

Now,

$$Re = \frac{G_s \cdot D_e}{\mu}$$

Where,

$$\text{Mass velocity } (G_s) = \frac{\text{Mass flow rate}}{\text{Area of Shell}} = \frac{0.66}{\frac{(P_T - d_o) \cdot D_s \cdot B_s}{P_T}} = \frac{0.66}{0.00164} = 393.87 \frac{\text{kg}}{\text{m}^2 \cdot \text{s}}$$

$$D_e = \frac{1.1}{d_o} (P_T^2 - 0.917 \cdot d_o^2) = 0.0136 \text{ m}$$

$$\mu = 5.85 \times 10^{-4} \text{ Pa} \cdot \text{s}$$

So,

$$Re = 9083.15$$

And,

$$Pr = \frac{\mu \cdot C_p}{k} = 3.88$$

Here,

$$C_p = 4140 \frac{\text{J}}{\text{kg} \cdot \text{K}}$$

$$k = 0.625 \frac{\text{W}}{\text{m} \cdot \text{K}}$$

Again,

$$h_o = 92532.44 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

B. Tube side calculation

$$h_i = \frac{j_n}{D_e} \cdot k \cdot Re \cdot Pr^{\frac{1}{3}}$$

$$T_{mean} = 417.04 \text{ K}$$

Here,

$$Re = \frac{4 \cdot \text{mass flow rate} \cdot \frac{n_p}{N_T}}{\pi \cdot d_i \cdot \mu} = 5035.04$$

Here,

$$k = 0.257 \frac{W}{m \cdot K}$$

$$Pr = \frac{\mu \cdot C_p}{k} = \frac{2.15 \times 10^{-3} \cdot 3669.98}{0.257} = 30.64$$

So,

$$h_i = \frac{0.15}{0.0136} \cdot 0.257 \cdot 5032.48 \cdot (30.70)^{\frac{1}{3}} = 45109.74 \frac{W}{m^2 \cdot K}$$

Thus,

$$U_{o,calc} = \frac{1}{\frac{1}{h_o} + \frac{1}{h_{do}} + \frac{d_o \cdot \ln \ln \left(\frac{d_o}{d_i} \right)}{2 \cdot K_w} + \frac{d_o}{d_i \cdot h_i} + \frac{d_o}{d_i \cdot h_{di}}}$$

From literature,

$$h_{do} = 4500 \frac{W}{m^2 \cdot K}$$

$$h_{di} = 3500 \frac{W}{m^2 \cdot K}$$

So,

$$U_{o,calc} = \frac{1}{1.05 \times 10^{-5} + 2.22 \times 10^{-4} + 5.60 \times 10^{-4} + 2.45 \times 10^{-5} + 3.13 \times 10^{-4}}$$

$$\text{or, } U_{o,calc} = 884.01 \frac{W}{m^2 \cdot K}$$

$$\text{Since } \frac{U_{o,calc} - U_{o,assume}}{U_{o,assume}} \cdot 100\%$$

= 18 %, which is less than 30%, our assumptions for design are valid.

Step 10: Over design calculation

Over design percentage is zero according from Aspen

Heat Exchanger B18

Step 1: Find mean temperature difference (ΔT_m)

From Aspen,

$$\text{Log mean temperature difference (LMTD)} = 341.58 \text{ K}$$

$$\text{Correction factor } (F_T) = 1 \text{ (As countercurrent)}$$

$$\text{So, } \Delta T_m = \text{LMTD} \cdot F_T = 341.58 \text{ K}$$

Step 2: Find heat duty (Q)

From Aspen,

$$Q = 605178 \text{ Watt}$$

Step 3: Assume a reasonable value of overall heat transfer coefficient ($U_{o,assume}$)

Overall heat transfer coefficient is assumed as,

$$U_{o,assume} = 750 \frac{\text{Watt}}{\text{m}^2 \cdot \text{K}}$$

Step 4: Decide tentative number of shell and tube passes (n_p)

$$n_p = 2$$

Step 5: Calculate heat transfer area (A) required

$$\text{Heat transfer area (A)} = \frac{Q}{U_o \cdot \text{LMTD}} = 2.362 \text{ m}^2$$

Step 6: Select tube material, decide the tube diameter (ID = d_i , OD = d_o), its wall thickness (in terms of BWG or SWG) and tube length (L). Calculate the number of tubes (n_T).

Tube side material is a mixture of Hydrogen, Nitrobenzene, Para Amino Phenol, Aniline, Water.

Since a significant amount of Nitrobenzene and Para Amino Phenol are present. As Nitrobenzene emits toxic fumes of nitrogen oxides upon combustion. Thus, Stainless steel 304 (as it is cheaper than 316) is used as the tube material.

Let,

$$\text{Tube length (L)} = 1.8228 \text{ m (6 ft is a standard length)}$$

$$\text{Outside diameter } (d_o) = 19.05 \text{ mm (For the process industry, 19.05 is most common)}$$

$$\text{Let, Birmingham Wire Gauge (BWG)} = 16 = 1.651 \text{ mm}$$

$$\text{Internal diameter } (d_i) = d_o - BWG = 17.399 \text{ mm}$$

$$\text{So, Area of 1 tube} = \pi \cdot d_o \cdot L = \pi \cdot 19.05 \times 10^{-3} \cdot 1.8228 = 0.1091 \text{ m}^2$$

$$\text{Thus, Number of tubes } (N_T) = \frac{A}{\text{Area of 1 tube}} = \frac{2.362}{0.1091} = 21.64 \approx 22$$

Step 7: Decide type of shell and tube exchanger (fixed tube sheet, U-tube etc.). Select the tube pitch (P_T), determine inside shell diameter (D_s) that can accommodate the calculated number of tubes (n_T)

As the heat exchanger is subjected to a temperature of (342 K), U-tube heat exchanger.

Triangular pitch is better for heat transfer and surface area per unit length (greatest tube density). Thus, triangular pitch is used.

$$\text{Tube pitch } (P_T) = 1.25 \cdot d_o = 0.0238 \text{ m}$$

Now,

$$\text{Bundle diameter } (BD) = d_o \cdot \left(\frac{N_T}{K_1} \right)^{\frac{1}{n_1}}$$

For single pass and triangular pitch,

$$n_1 = 2.142 \text{ and } K_1 = 0.319$$

So,

$$BD = 0.0190 \cdot \left(\frac{22}{0.319} \right)^{\frac{1}{2.142}} = 0.1371 \text{ m} = 137.12 \text{ mm}$$

And,

$$\text{Bundle diameter clearance } (BDC) = 90 \text{ mm for pull through U - Tube.}$$

$$\text{Inside shell diameter } (D_s) = (137.12 + 90) \text{ mm} = 227.12 \text{ mm}$$

Step 8: Assign fluid to shell side or tube side. Select the type of baffle, its size (i.e. percentage cut), spacing and number.

Tube side material is a mixture of Hydrogen, Nitrobenzene, Para Amino Phenol, Aniline, Water. Meanwhile, the shell side material is water.

Baffle type is cut-segmental baffle.

$$\text{Baffle spacing } (B_s) = 0.2 \cdot 227.12 = 45.424 \text{ mm}$$

$$\text{Percentage cut of baffles} = 25\% \text{ (The most widely used)}$$

$$\text{Number of baffles} = \frac{\text{Inside shell diameter}}{\text{Baffle spacing}} = \frac{227.12}{45.424} = 5$$

Step 9: Comparison of calculated and assumed overall heat transfer coefficient (U)

A. Shell side coefficient calculation

$$h_o = \frac{j_n}{D_e} \cdot k \cdot Re \cdot Pr^{\frac{1}{3}}$$

Now,

$$Re = \frac{G_s \cdot D_e}{\mu}$$

Where,

$$\text{Mass velocity } (G_s) = \frac{\text{Mass flow rate}}{\text{Area of Shell}} = \frac{0.416}{\frac{(P_T - d_o) \cdot D_s \cdot B_s}{P_T}} = \frac{0.416}{0.0020} = 201.60 \frac{\text{kg}}{\text{m}^2 \cdot \text{s}}$$

$$D_e = \frac{1.1}{d_o} (P_T^2 - 0.917 \cdot d_o^2) = 0.0136 \text{ m}$$

$$\mu = 5.85 \times 10^{-4} \text{ Pa} \cdot \text{s}$$

So,

$$Re = 4649.22$$

And,

$$Pr = \frac{\mu \cdot C_p}{k} = 3.88$$

Here,

$$k = 0.625 \frac{\text{W}}{\text{m} \cdot \text{K}}$$

Again,

$$h_o = 47362.80 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

B. Tube side calculation

$$h_i = \frac{j_n}{D_e} \cdot k \cdot Re \cdot Pr^{\frac{1}{3}}$$

$$T_{mean} = 343.59 \text{ K}$$

Here,

$$Re = \frac{4 \cdot \text{mass flow rate} \cdot \frac{n_p}{N_T}}{\pi \cdot d_i \cdot \mu} = 8206.28$$

Here,

$$\mu = 3.475 \times 10^{-4} \text{ Pa} \cdot \text{s}$$

$$Pr = \frac{\mu \cdot C_P}{k} = \frac{3.475 \times 10^{-4} \cdot 3120.05}{0.397} = 2.73$$

So,

$$h_i = \frac{0.15}{0.0136} \cdot 0.397 \cdot 8202.12 \cdot (2.73)^{\frac{1}{3}} = 50632.29 \frac{W}{m^2 \cdot K}$$

Thus,

$$U_{o,calc} = \frac{1}{\frac{1}{h_o} + \frac{1}{h_{do}} + \frac{d_o \cdot \ln \ln \left(\frac{d_o}{d_i} \right)}{2 \cdot K_W} + \frac{d_o}{d_i \cdot h_i} + \frac{d_o}{d_i \cdot h_{di}}}$$

From literature,

$$h_{do} = 4500 \frac{W}{m^2 \cdot K}$$

$$h_{di} = 3500 \frac{W}{m^2 \cdot K}$$

So,

$$U_{o,calc} = \frac{1}{2.04 \times 10^{-5} + 2.22 \times 10^{-4} + 5.60 \times 10^{-4} + 2.18 \times 10^{-5} + 3.13 \times 10^{-4}}$$

$$\text{or, } U_{o,calc} = 878.08 \frac{W}{m^2 \cdot K}$$

$$\text{Since } \frac{U_{o,calc} - U_{o,assume}}{U_{o,assume}} \cdot 100\% = 17 \%, \text{ which is less than } 30\%, \text{ our assumptions for design are valid.}$$

Step 10: Over design calculation

Over design percentage is zero according from Aspen

Step 11: Pressure drop

Pressure drop in our design is -0.5 bar.

Heat Exchanger B7

Step 1: Find mean temperature difference (ΔT_m)

From Aspen,

Log mean temperature difference (LMTD) = 383.14 K

Correction factor (F_T) = 1 (As countercurrent)

$$\text{So, } \Delta T_m = LMTD \cdot F_T = 383.14 \text{ K}$$

Step 2: Find heat duty (Q)

From Aspen,

$$Q = 758879.3 \text{ Watt}$$

Step 3: Assume a reasonable value of overall heat transfer coefficient ($U_{o,assume}$)

Aspen estimates that overall heat transfer coefficient

$$U_{o,assume} = 750 \frac{\text{Watt}}{\text{m}^2 \cdot \text{K}}$$

Step 4: Decide tentative number of shell and tube passes (n_p)

$$n_p = 2$$

Step 5: Calculate heat transfer area (A) required

From Aspen,

$$\text{Heat transfer area (A)} = 2.64 \text{ m}^2$$

Step 6: Select tube material, decide the tube diameter (ID = d_i , OD = d_o), its wall thickness (in terms of BWG or SWG) and tube length (L). Calculate the number of tubes (n_T).

Tube side material is a mixture of Hydrogen, Nitrobenzene, Para Amino Phenol, Aniline, Acetic Anhydride, Water.

Since a significant amount of Para Amino Phenol and Acetic Anhydride are present. As Acetic Anhydride is Corrosive to metals and tissue. Thus, Stainless steel 304 (as it is cheaper than 316) is used as the tube material.

Let,

$$\text{Tube length (L)} = 1.8228 \text{ m (6 ft is a standard length)}$$

$$\text{Outside diameter (} d_o \text{)} = 19.05 \text{ mm (For the process industry, 19.05 is most common)}$$

Let, Birmingham Wire Gauge (BWG) = 16 = 1.651 mm

Internal diameter (d_i) = $d_o - BWG = 17.399 \text{ mm}$

So, Area of 1 tube = $\pi \cdot d_o \cdot L = \pi \cdot 19.05 \times 10^{-3} \cdot 1.8228 = 0.1091 \text{ m}^2$

Thus, Number of tubes (N_T) = $\frac{A}{\text{Area of 1 tube}} = \frac{2.64}{0.1091} = 24.19 \approx 25$

Step 7: Decide type of shell and tube exchanger (fixed tube sheet, U-tube etc.). Select the tube pitch (P_T), determine inside shell diameter (D_s) that can accommodate the calculated number of tubes (n_T)

As the heat exchanger is subjected to a temperature of (384 K), U-tube heat exchanger.

Triangular pitch is better for heat transfer and surface area per unit length (greatest tube density). Thus, triangular pitch is used.

Tube pitch (P_T) = $1.25 \cdot d_o = 0.0238 \text{ m}$

Now,

Bundle diameter (BD) = $d_o \cdot \left(\frac{N_T}{K_1}\right)^{\frac{1}{n_1}}$

For single pass and triangular pitch,

$n_1 = 2.142$ and $K_1 = 0.319$

$BD = 0.0190 \cdot \left(\frac{25}{0.319}\right)^{\frac{1}{2.142}} = 0.1455 \text{ m} = 145.56 \text{ mm}$

And,

Bundle diameter clearance (BDC) = 90 mm for pull through U – Tube.

Inside shell diameter (D_s) = $(145.56 + 90) \text{ mm} = 235.56 \text{ mm}$

Step 8: Assign fluid to shell side or tube side. Select the type of baffle, its size (i.e. percentage cut), spacing and number.

Tube side material is a mixture of Methanol, Formaldehyde, Water, Hydrogen, Nitrogen, Oxygen and Carbon monoxide. Meanwhile, the shell side material is cooling water.

Baffle type is cut-segmental baffle.

Baffle spacing (B_s) = $0.2 \cdot 235.56 = 47.11 \text{ mm}$

Percentage cut of baffles = 25% (The most widely used)

$$\text{Number of baffles} = \frac{\text{Inside shell diameter}}{\text{Baffle spacing}} = \frac{235.56}{47.11} = 5$$

Step 9: Comparison of calculated and assumed overall heat transfer coefficient (U)

A. Shell side coefficient calculation

$$h_o = \frac{j_n}{D_e} \cdot k \cdot Re \cdot Pr^{\frac{1}{3}}$$

Now,

$$Re = \frac{G_s \cdot D_e}{\mu}$$

Where,

$$\text{Mass velocity } (G_s) = \frac{\text{Mass flow rate}}{\text{Area of Shell}} = \frac{0.061}{\frac{(P_T - d_o) \cdot D_s \cdot B_s}{P_T}} = \frac{0.061}{0.0022} = 27.48 \frac{\text{kg}}{\text{m}^2 \cdot \text{s}}$$

$$D_e = \frac{1.1}{d_o} (P_T^2 - 0.917 \cdot d_o^2) = 0.0136 \text{ m}$$

$$\mu = 5.85 \times 10^{-4} \text{ Pa} \cdot \text{s}$$

So,

$$Re = 633.80$$

And,

$$Pr = \frac{\mu \cdot C_p}{k} = 3.88$$

Here,

$$k = 0.625 \frac{\text{W}}{\text{m} \cdot \text{K}}$$

Again,

$$h_o = 6456.68 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

B. Tube side calculation

$$h_i = \frac{j_n}{D_e} \cdot k \cdot Re \cdot Pr^{\frac{1}{3}}$$

$$T_{mean} = 383.14 \text{ K}$$

Here,

$$Re = \frac{4 \cdot \text{mass flow rate} \cdot \frac{n_P}{N_T}}{\pi \cdot d_i \cdot \mu} = 34819.29$$

Here,

$$k = 0.122 \frac{W}{m \cdot K}$$

$$Pr = \frac{\mu \cdot C_P}{k} = \frac{86.55 \times 10^{-3} \cdot 2174.35}{0.122} = 6.17$$

So,

$$h_i = \frac{0.15}{0.0136} \cdot 0.122 \cdot 139.72 \cdot (1542.54)^{\frac{1}{3}} = 86973.65 \frac{W}{m^2 \cdot K}$$

Thus,

$$U_{o,calc} = \frac{1}{\frac{1}{h_o} + \frac{1}{h_{do}} + \frac{d_o \cdot \ln \ln \left(\frac{d_o}{d_i} \right)}{2 \cdot K_W} + \frac{d_o}{d_i \cdot h_i} + \frac{d_o}{d_i \cdot h_{di}}}$$

From literature,

$$h_{do} = 4500 \frac{W}{m^2 \cdot K}$$

$$h_{di} = 3500 \frac{W}{m^2 \cdot K}$$

So,

$$U_{o,calc} = \frac{1}{1.53 \times 10^{-4} + 2.22 \times 10^{-4} + 5.60 \times 10^{-4} + 5.05 \times 10^{-4} + 3.13 \times 10^{-4}}$$

$$\text{or, } U_{o,calc} = 791.42 \frac{W}{m^2 \cdot K}$$

$$\text{Since } \frac{U_{o,calc} - U_{o,assume}}{U_{o,assume}} \cdot 100\% = 6\%, \text{ which is less than } 30\%, \text{ our assumptions for design are valid.}$$

Step 10: Over design calculation

Over design percentage is zero according from Aspen

Step 11: Pressure drop

Pressure drop in our design is -0.2 bar.

Condenser B14

Physical Properties

Heat Duty = 1372.26 kg/h

Pressure = 0.5 bar

Vapour Enter = 120 °C

Condensation Complete = 116 °C

Average Molecular Weight of Vapours = 33

Enthalpy of Vapour = 5219.67 kJ/kg

Enthalpy of condensate = 247.0 kJ/kg

Cooling Water Available = 30 °C

Temperature limited = 40 °C

O.D. of tubes = 20 mm

I.D. of tubes = 16.8 mm

Length = 4.88 m (16 ft)

The physical properties of the mixture will be taken as the mean, at the average temperature.

$$\text{Heat transferred from vapour} = \frac{1372.26}{3600} (5219.67 - 247.0) = 1895.5 \text{ kW}$$

$$\text{Cooling water flow} = \frac{1895.5}{(40 - 30) * 4.18} = 45.34 \text{ kg/s}$$

$$\text{Assumed overall coefficient (Table 12.1)} = 900 \text{ W/m}^2\text{°C}$$

Mean temperature difference: the condensation range is small and the change in saturation temperature will be linear, so the corrected logarithmic mean temperature difference can be used.

$$R = \frac{(T_1 - T_2)}{(t_2 - t_1)}$$
$$R = \frac{(120 - 116)}{(40 - 30)} = 0.4$$

$$S = \frac{(t_2 - t_1)}{(T_1 - t_1)}$$

$$S = \frac{(40 - 30)}{(120 - 30)} = 0.11$$

Try a horizontal exchanger, condensation in the shell, four tube passes. For one shell pass, four tube passes, from Figure 12.19, $F_t = 0.92$

$$\Delta T_{lm} = \frac{(120 - 40) - (116 - 30)}{\ln \frac{(120 - 40)}{(116 - 30)}} = 82.96 \text{ } ^\circ\text{C}$$

$$\Delta T_m = 0.92 * 82.96 = 76.32 \text{ } ^\circ\text{C}$$

$$\text{Trial area} = \frac{1895.5 * 10^3}{900 * 76.32} = 27.59 \text{ m}^2$$

$$\begin{aligned} \text{Surface area of one tube} &= 20 * 10^{-3} * 3.14 * 4.88 \\ &= 0.305 \text{ m}^2 \text{ (ignore tube sheet thickness)} \end{aligned}$$

$$\text{Number of tubes} = \frac{27.59}{0.305} = 90$$

Use square pitch, $P_t = 1.25 * 20 \text{ mm} = 25 \text{ mm}$.

Tube bundle diameter

$$D_b = d_o \left(\frac{N_t}{K_1} \right)^{1/n_1}$$

$$D_b = 20 \left(\frac{90}{0.158} \right)^{1/2.263} = 330.14 \text{ mm}$$

$$\text{Number of tubes in center row } N_r = \frac{D_b}{P_t} = \frac{330.14}{25} = 13.20$$

Shell-Side Coefficient

Estimate tube wall temperature, T_w ; assume condensing coefficient of $1500 \text{ W/m}^2\text{ } ^\circ\text{C}$,

Mean temperature

$$\text{Shell} - \text{Side} = \frac{120 + 116}{2} = 118 \text{ } ^\circ\text{C}$$

$$\text{Tube} - \text{Side} = \frac{40 + 30}{2} = 35 \text{ } ^\circ\text{C}$$

$$(118 - T_w)1500 = (118 - 35)900$$

$$T_w = 68 \text{ } ^\circ\text{C}$$

$$\text{Mean temperature condensate} = \frac{118 + 68}{2} = 93^\circ\text{C}$$

Physical properties at 93 °C

$$\mu_L = 0.861 \text{ mNs/m}^2$$

$$\rho_L = 554.2 \text{ kg/m}^3$$

$$k_L = 0.16 \text{ W/m}^\circ\text{C}$$

vapor density at mean vapor temperature

$$\rho_v = \frac{33}{22.4} * \frac{273}{(273 + 118)} * \frac{10}{1} = 10.28 \text{ kg/m}^3$$

$$\Gamma_h = \frac{W_c}{LN_t} = \frac{1895.5}{3600} * \frac{1}{4.88 * 90} = 1.19 * 10^{-3} \text{ kg/s m}$$

$$N_r = \frac{2}{3} * 13.20 = 8.8$$

Using Kern's method, the mean coefficient for a tube bundle is given by:

$$(h_c)_b = 0.95 k_L \left[\frac{\rho_L (\rho_L - \rho_v) g}{\mu_L \Gamma_h} \right]^{1/3} N_r^{-1/6}$$

$$\begin{aligned} h_c &= 0.95 * 0.16 \left[\frac{554.2 * (554.2 - 10.28) * 9.81}{0.861 * 10^{-3} * 1.19 * 10^{-3}} \right]^{1/3} * 8.8^{-1/6} \\ &= 1506.15 \text{ W/m}^2\text{C} \end{aligned}$$

Close enough to assumed value of 1500 W/m²°C, so no correction to T_w needed.

Tube-Side Coefficient

$$\text{Tube cross - sectional area} = \frac{3.14}{4} (16.8 * 10^{-3})^2 * \frac{90}{4} = 0.0049 \text{ m}^2$$

$$\text{Density of water, at } 35^\circ\text{C} = 993 \text{ kg/m}^3$$

$$\text{Tube velocity} = \frac{45.34}{993} * \frac{1}{0.0049} = 9.31 \text{ m/s}$$

$$h_i = \frac{4200(1.35 + 0.02t)u_t^{0.8}}{d_i^{0.2}}$$

$$\begin{aligned} h_i &= \frac{4200(1.35 + 0.02 * 35)9.31^{0.8}}{16.8^{0.2}} \\ &= 29181.05 \text{ W/m}^2\text{C} \end{aligned}$$

Fouling factors: as neither fluid is heavily fouling, use $6000 \text{ W/m}^2\text{C}$ for each side.

$$k_w = 50 \text{ W/m}^2\text{C}$$

Overall Coefficient

$$\frac{1}{U_o} = \frac{1}{h_o} + \frac{1}{h_{do}} + \frac{d_o \cdot \ln\left(\frac{d_o}{d_i}\right)}{2 \cdot K_w} + \frac{d_o}{d_i \cdot h_{id}} + \frac{d_o}{d_i \cdot h_i}$$

$$\frac{1}{U} = \frac{1}{1506.15} + \frac{1}{6000} + \frac{20 \cdot 10^{-3} \ln\left(\frac{20}{16.8}\right)}{2 \cdot 50} + \frac{20}{16.8} \cdot \left(\frac{1}{6000} + \frac{1}{29181.05}\right)$$

$$\frac{1}{U} = 1.10 \cdot 10^{-3}$$

$$U = 905.23 \text{ W/m}^2\text{C}$$

Close enough to estimate, firm up design.

Shell-Side Pressure Drop

Use pull-through floating head, no need for close clearance.

Select baffle spacing D shell diameter, 45 percent cut.

From Figure 12.10, clearance = 95 mm.

$$\text{Shell i.d.} = 330.14 + 95 = 425.14 \text{ mm}$$

Use Kern's method to make an approximate estimate.

$$A_s = \frac{(p_t - d_o) D_s l_B}{p_t}$$

$$\text{Cross-flow area } A_s = \frac{(25 - 20)}{25} \cdot 425.14 \cdot 425.14 \cdot 10^{-6}$$

$$= 0.036 \text{ m}^2$$

Mass flow-rate, based on inlet conditions

$$G_s = \frac{1372.26}{3600} \cdot \frac{1}{0.036} = 10.58 \text{ kg/s m}^2$$

For a square pitch arrangement:

$$d_e = \frac{1.27}{d_o} (p_t^2 - 0.785 d_o^2)$$

$$\text{Equivalent diameter, } d_e = \frac{1.27}{20} (25^2 - 0.785 \cdot 20^2)$$

$$= 19.8 \text{ mm}$$

$$\text{Vapour viscosity} = 0.008 \text{ mNs/m}^2$$

$$Re = \frac{10.58 * 19.8 * 10^{-3}}{0.008 * 10^{-3}} = 26185.5$$

$$\text{From Figure 12.30, } j_f = 2.2 * 10^{-2}$$

$$u_s = \frac{G_s}{\rho_v} = \frac{10.58}{10.28} = 1.02 \text{ m/s}$$

Take pressure drop as 50 per cent of that calculated using the inlet flow; neglect viscosity correction.

$$\begin{aligned} \Delta P_s &= 8j_f \left(\frac{D_s}{d_e} \right) * \left(\frac{L}{l_B} \right) * \frac{\rho u_s^2}{2} * \left(\frac{\mu}{\mu_w} \right)^{-0.14} \\ \Delta P_s &= \frac{1}{2} [8 * 2.2 * 10^{-2} * \left(\frac{425.14}{19.8} \right) * \left(\frac{4.88}{0.425} \right) * \frac{10.28(1.02)^2}{2}] \\ &= 116 \text{ N/m}^2 \\ &= 0.11 \text{ kPa} \end{aligned}$$

Negligible; more sophisticated method of calculation not justified.

Tube-Side Pressure Drop

$$\text{Viscosity of water} = 0.6 \text{ mN s/m}^2$$

$$Re = \frac{u_t \rho d_i}{\mu} = \frac{9.31 * 993 * 16.8 * 10^{-3}}{0.6 * 10^{-3}} = 258855.24$$

$$\text{From Figure 12.24, } j_f = 3.5 * 10^{-3}.$$

Neglect viscosity correction.

$$\begin{aligned} \Delta P_t &= N_p \left[8j_f * \left(\frac{L}{d_i} \right) * \left(\frac{\mu}{\mu_w} \right)^{-m} + 2.5 \right] \frac{\rho u_t^2}{2} \\ \Delta P_t &= 4 \left[8 * 3.5 * 10^{-3} * \left(\frac{4.88}{16.8 * 10^{-3}} \right) + 2.5 \right] \frac{993 * 9.31^2}{2} \\ &= 1830409 \text{ N/m}^2 \\ &= 1830 \text{ kPa (265 psi)}, \end{aligned}$$

acceptable.

Condenser B5

Physical Properties

Heat Duty = 1363.01 kg/h

Pressure = 0.8 bar

Vapour Enter = 108 °C

Condensation Complete = 125 °C

Average Molecular Weight of Vapours = 85

Enthalpy of Vapour = 5977.67 kJ/kg

Enthalpy of condensate = 247.0 kJ/kg

Cooling Water Available = 30 °C

Temperature limited = 40 °C

O.D. of tubes = 20 mm

I.D. of tubes = 16.8 mm

Length = 4.88 m (16 ft)

The physical properties of the mixture will be taken as the mean, at the average temperature.

$$\text{Heat transferred from vapour} = \frac{1363.01}{3600} (5977.67 - 247.0) = 2169.71 \text{ kW}$$

$$\text{Cooling water flow} = \frac{2169.71}{(40 - 30) * 4.18} = 51.90 \text{ kg/s}$$

$$\text{Assumed overall coefficient (Table 12.1)} = 900 \text{ W/m}^2\text{°C}$$

Mean temperature difference: the condensation range is small and the change in saturation temperature will be linear, so the corrected logarithmic mean temperature difference can be used.

$$R = \frac{(T_1 - T_2)}{(t_2 - t_1)}$$

$$R = \frac{(125 - 108)}{(40 - 30)} = 1.7$$

$$S = \frac{(t_2 - t_1)}{(T_1 - t_1)}$$

$$S = \frac{(40 - 30)}{(125 - 30)} = 0.105$$

Try a horizontal exchanger, condensation in the shell, four tube passes. For one shell pass, four tube passes, from Figure 12.19, $F_t = 0.92$

$$\Delta T_{lm} = \frac{(125 - 40) - (108 - 30)}{\ln \frac{(125 - 40)}{(108 - 30)}} = 81.44 \text{ } ^\circ\text{C}$$

$$\Delta T_m = 0.92 * 81.44 = 74.92 \text{ } ^\circ\text{C}$$

$$\text{Trial area} = \frac{2169.71 * 10^3}{900 * 74.92} = 32.17 \text{ m}^2$$

$$\begin{aligned} \text{Surface area of one tube} &= 20 * 10^{-3} * 3.14 * 4.88 \\ &= 0.305 \text{ m}^2 \text{ (ignore tube sheet thickness)} \end{aligned}$$

$$\text{Number of tubes} = \frac{32.17}{0.305} = 105$$

Use square pitch, $P_t = 1.25 * 20 \text{ mm} = 25 \text{ mm}$.

Tube bundle diameter

$$D_b = d_o \left(\frac{N_t}{K_1} \right)^{1/n_1}$$

$$D_b = 20 \left(\frac{105}{0.158} \right)^{1/2.263} = 353.41 \text{ mm}$$

$$\text{Number of tubes in center row } N_r = \frac{D_b}{P_t} = \frac{353.41}{25} = 14.13$$

Shell-Side Coefficient

Estimate tube wall temperature, T_w ; assume condensing coefficient of $1500 \text{ W/m}^2\text{ } ^\circ\text{C}$,

Mean temperature

$$\text{Shell} - \text{Side} = \frac{125 + 108}{2} = 116.5 \text{ } ^\circ\text{C}$$

$$\text{Tube} - \text{Side} = \frac{40 + 30}{2} = 35 \text{ } ^\circ\text{C}$$

$$(116.5 - T_w)1500 = (116.5 - 35)900$$

$$T_w = 67.6 \text{ } ^\circ\text{C}$$

$$\text{Mean temperature condensate} = \frac{116.5 + 67.6}{2} = 92 \text{ } ^\circ\text{C}$$

Physical properties at 92 °C

$$\mu_L = 0.36 \text{ mNs/m}^2$$

$$\rho_L = 457 \text{ kg/m}^3$$

$$k_L = 0.120 \text{ W/m}^\circ\text{C}$$

vapor density at mean vapor temperature

$$\rho_v = \frac{85}{22.4} * \frac{273}{(273 + 116.5)} * \frac{10}{1} = 26.59 \text{ kg/m}^3$$

$$\Gamma_h = \frac{W_c}{LN_t} = \frac{1363.01}{3600} * \frac{1}{4.88 * 105} = 7.38 * 10^{-4} \text{ kg/s m}$$

$$N_r = \frac{2}{3} * 14.13 = 9.42$$

Using Kern's method, the mean coefficient for a tube bundle is given by:

$$(h_c)_b = 0.95 k_L \left[\frac{\rho_L (\rho_L - \rho_v) g}{\mu_L \Gamma_h} \right]^{1/3} N_r^{-1/6}$$

$$\begin{aligned} h_c &= 0.95 * 0.120 \left[\frac{457 * (457 - 26.59) * 9.81}{0.36 * 10^{-3} * 7.38 * 10^{-4}} \right]^{1/3} * 9.42^{-1/6} \\ &= 1519.14 \text{ W/m}^2\text{C} \end{aligned}$$

Close enough to assumed value of 1500 W/m²°C, so no correction to T_w needed.

Tube-Side Coefficient

$$\text{Tube cross - sectional area} = \frac{3.14}{4} (16.8 * 10^{-3})^2 * \frac{105}{4} = 0.0058 \text{ m}^2$$

$$\text{Density of water, at } 35^\circ\text{C} = 993 \text{ kg/m}^3$$

$$\text{Tube velocity} = \frac{51.90}{993} * \frac{1}{0.0058} = 9.01 \text{ m/s}$$

$$h_i = \frac{4200(1.35 + 0.02t)u_t^{0.8}}{d_i^{0.2}}$$

$$\begin{aligned} h_i &= \frac{4200(1.35 + 0.02 * 35)9.01^{0.8}}{16.8^{0.2}} \\ &= 28426.35 \text{ W/m}^\circ\text{C} \end{aligned}$$

Fouling factors: as neither fluid is heavily fouling, use 6000 W/m²°C for each side.

$$k_w = 50 \text{ W/m}^\circ\text{C}$$

Overall Coefficient

$$\frac{1}{U_o} = \frac{1}{h_o} + \frac{1}{h_{do}} + \frac{d_o \cdot \ln\left(\frac{d_o}{d_i}\right)}{2 \cdot K_W} + \frac{d_o}{d_i \cdot h_{id}} + \frac{d_o}{d_i \cdot h_i}$$
$$\frac{1}{U} = \frac{1}{1519.14} + \frac{1}{6000} + \frac{20 \cdot 10^{-3} \ln\left(\frac{20}{16.8}\right)}{2 \cdot 50} + \frac{20}{16.8} \cdot \left(\frac{1}{6000} + \frac{1}{28426.35}\right)$$
$$\frac{1}{U} = 1.10 \cdot 10^{-3}$$
$$U = 909.01 \text{ W/m}^2\text{°C}$$

Close enough to estimate, firm up design.

Shell-Side Pressure Drop

Use pull-through floating head, no need for close clearance.

Select baffle spacing D shell diameter, 45 percent cut.

From Figure 12.10, clearance = 95 mm.

$$\text{Shell i.d.} = 353.41 + 95 = 448.41 \text{ mm}$$

Use Kern's method to make an approximate estimate.

$$A_s = \frac{(p_t - d_o) D_s l_B}{p_t}$$
$$\text{Cross-flow area } A_s = \frac{(25 - 20)}{25} \cdot 448.41 \cdot 448.41 \cdot 10^{-6}$$
$$= 0.040 \text{ m}^2$$

Mass flow-rate, based on inlet conditions

$$G_s = \frac{1363.01}{3600} \cdot \frac{1}{0.040} = 9.46 \text{ kg/s m}^2$$

For a square pitch arrangement:

$$d_e = \frac{1.27}{d_o} (p_t^2 - 0.785 d_o^2)$$
$$\text{Equivalent diameter, } d_e = \frac{1.27}{20} (25^2 - 0.785 \cdot 20^2)$$
$$= 19.8 \text{ mm}$$

$$\text{Vapour viscosity} = 0.008 \text{ mNs/m}^2$$

$$Re = \frac{9.46 * 19.8 * 10^{-3}}{0.008 * 10^{-3}} = 23413.5$$

$$\text{From Figure 12.30, } j_f = 2.2 * 10^{-2}$$

$$u_s = \frac{G_s}{\rho_v} = \frac{9.46}{26.59} = 0.35 \text{ m/s}$$

Take pressure drop as 50 per cent of that calculated using the inlet flow; neglect viscosity correction.

$$\begin{aligned} \Delta P_s &= 8j_f \left(\frac{D_s}{d_e} \right) * \left(\frac{L}{l_B} \right) * \frac{\rho u_s^2}{2} * \left(\frac{\mu}{\mu_w} \right)^{-0.14} \\ \Delta P_s &= \frac{1}{2} [8 * 2.2 * 10^{-2} * \left(\frac{448.41}{19.8} \right) * \left(\frac{4.88}{0.448} \right) * \frac{26.59(0.35)^2}{2}] \\ &= 35.35 \text{ N/m}^2 \\ &= 0.035 \text{ kPa} \end{aligned}$$

Negligible; more sophisticated method of calculation not justified.

Tube-Side Pressure Drop

$$\text{Viscosity of water} = 0.6 \text{ mN s/m}^2$$

$$Re = \frac{u_t \rho d_i}{\mu} = \frac{9.01 * 993 * 16.8 * 10^{-3}}{0.6 * 10^{-3}} = 250514.04$$

$$\text{From Figure 12.24, } j_f = 3.5 * 10^{-3}.$$

Neglect viscosity correction.

$$\begin{aligned} \Delta P_t &= N_p \left[8j_f * \left(\frac{L}{d_i} \right) * \left(\frac{\mu}{\mu_w} \right)^{-m} + 2.5 \right] \frac{\rho u_t^2}{2} \\ \Delta P_t &= 4 \left[8 * 3.5 * 10^{-3} * \left(\frac{4.88}{16.8 * 10^{-3}} \right) + 2.5 \right] \frac{993 * 9.01^2}{2} \\ &= 1714345 \text{ N/m}^2 \\ &= 1714 \text{ kPa (249 psi)}, \end{aligned}$$

acceptable.

Reboiler B14

Physical Properties:

Temperature = 183.87 °C

Pressure = 0.5 bar

Boiling point = 170.2 °C

Latent heat = 254.4 kJ/kg

Mean specific heat, liquid = 0.809 kJ/kg°C

Critical pressure, P_c = 44.8 bar

Heat loads:

$$\text{Sensible heat (maximum)} = (183.87 - 0) * 0.809 = 148.75 \text{ kJ/kg}$$

$$\text{Total heat load} = (148.75 + 254.4) * \frac{2360.43}{3600} = 264.33 \text{ kW},$$

add 5 % for heat losses

$$\text{maximum heat load (duty)} = 1.05 * 264.33$$

$$= 277.55 \text{ kW}$$

From Figure 12.1 assume $U = 1000 \text{ W/m}^2\text{°C}$.

Mean temperature difference; both sides isothermal, steam saturation temperature at 0.5 bar = 236.4 °C

$$\Delta T_m = 236.4 - 170.2 = 66.2\text{°C}$$

$$\text{Area (outside)required} = \frac{277.55 * 10^3}{1000 * 66.2} = 4.19 \text{ m}^2$$

Select 25 mm i.d., 30 mm o.d. plain U-tubes,

Nominal length 4.8 m (one U-tube)

$$\text{Number of U tubes} = \frac{4.19}{(30 * 10^{-3}) * 3.14 * 4.8} = 10$$

Use square pitch arrangement, pitch = 1.5* tube o.d.

$$= 1.5 * 30 = 45 \text{ mm}$$

Draw a tube layout diagram, take minimum bend radius

$$1.5 * \text{tube o.d.} = 45 \text{ mm}$$

Proposed layout gives 10 U-tubes, tube outer limit diameter 420 mm.

Boiling coefficient

Use Mostinski's equation:

heat flux, based on estimated area,

$$q = \frac{277.55}{4.19} = 46.24 \text{ kW/m}^2$$

$$h_{nb} = 0.104 * (P_c)^{0.69} * (q)^{0.7} \left[1.8 * \left(\frac{P}{P_c} \right)^{0.17} + 4 * \left(\frac{P}{P_c} \right)^{1.2} + 10 * \left(\frac{P}{P_c} \right)^{10} \right]$$

$$\begin{aligned} h_{nb} &= 0.104 * (44.8)^{0.69} \\ &\quad * (66.24 * 10^3)^{0.7} \left[1.8 * \left(\frac{0.5}{44.8} \right)^{0.17} + 4 * \left(\frac{0.5}{44.8} \right)^{1.2} + 10 * \left(\frac{0.5}{44.8} \right)^{10} \right] \\ &= 3397.81 * 0.85 \\ &= 2909.98 \text{ W/m}^2\text{°C} \end{aligned}$$

Take steam condensing coefficient as $8000 \text{ W/m}^2\text{°C}$, fouling coefficient $5000 \text{ W/m}^2\text{°C}$; fouling coefficient, essentially clean, $10,000 \text{ W/m}^2\text{°C}$.

Tube material will be plain carbon steel, $k_w = 55 \text{ W/m}^2\text{°C}$

$$\begin{aligned} \frac{1}{U_o} &= \frac{1}{h_o} + \frac{1}{h_{do}} + \frac{d_o \cdot \ln \left(\frac{d_o}{d_i} \right)}{2 \cdot K_w} + \frac{d_o}{d_i \cdot h_{id}} + \frac{d_o}{d_i \cdot h_i} \\ \frac{1}{U_o} &= \frac{1}{2909.98} + \frac{1}{10,000} + \frac{30 * 10^{-3} * \ln \left(\frac{30}{25} \right)}{2 * 55} + \frac{30}{25} \left(\frac{1}{5000} + \frac{1}{8000} \right) \\ U_o &= 1132.02 \text{ W/m}^2\text{°C} \end{aligned}$$

Close enough to original estimate of $1000 \text{ W/m}^2\text{°C}$ for the design to stand.

Myers and Katz (Chem. Eng. Prog. Sym. Ser. 49(5) 107 114) give some data on the boiling on banks of tubes. To compare the value estimate with their values an estimate of the boiling film temperature difference is required:

$$= \frac{1132.02}{2909.98} * 66.2 = 25.75 \text{ °C (74.75 °F)}$$

Myers data, extrapolated, gives a coefficient of around $3000 \text{ Btu/h ft}^2\text{°F}$ at a 75 °F temperature difference = $17,100 \text{ W/m}^2\text{°C}$, so the estimated value of 2909.98 is certainly on the safe side.

Check maximum allowable heat flux. Use modified Zuber equation

$$\text{Surface tension (estimated)} = 9.7 * 10^{-3} \text{ N/m}$$

$$\rho_L = 550 \text{ kg/m}^3$$

$$\rho_v = \frac{123.11}{22.4} * \frac{273}{(273 + 170.2)} * 0.5 = 1.69 \text{ kg/m}^3$$

$$N_t = 52$$

For square arrangement $K_b = 0.44$

$$q_{cb} = K_b * \left(\frac{P_t}{d_o}\right) * \left(\frac{\lambda}{\sqrt{N_t}}\right) [\sigma g * (\rho_L - \rho_v) * \rho_v^2]^{0.25}$$

$$q_{cb} = 0.44 * 1.5 * \frac{254.4 * 10^3}{\sqrt{52}} [9.7 * 10^{-3} * 9.81 * (550 - 1.69) * 1.69^2]^{0.25}$$

$$= 81,352.22 \frac{W}{m^2}$$

$$= 82 \text{ k W/m}^2$$

Applying a factor of 0.7, maximum flux should not exceed $82 * 0.7 = 56.94 \text{ kW/m}^2$.

Actual flux of 46.24 kW/m^2 is well below maximum allowable.

Layout

From tube sheet layout $D_b = 420 \text{ mm}$.

Take shell diameter as twice bundle diameter

$$D_s = 2 * 420 = 840 \text{ mm}.$$

Take liquid level as 500 mm from base,

Freeboard = $840 - 500 = 340 \text{ mm}$, satisfactory.

From sketch, width at liquid level = 0.8 m .

Surface area of liquid = $0.8 * 2.4 = 1.9 \text{ m}^2$.

$$\text{Vapour velocity at surface} = \frac{5000}{3600} * \frac{1}{1.69} * \frac{1}{1.9} = 0.432 \text{ m/s}$$

Maximum allowable velocity

$$v_v < 0.2 * \left[\frac{\rho_L - \rho_v}{\rho_v}\right]^{\frac{1}{2}}$$

$$v_v = 0.2 * \left[\frac{550 - 1.69}{1.69}\right]^{\frac{1}{2}} = 3.60 \text{ m/s}$$

So, actual velocity is well below maximum allowable velocity. A smaller shell diameter could be considered.

Reboiler B5

Physical Properties:

Temperature = 245.72 °C

Pressure = 0.8 bar

Boiling point = 170 °C

Latent heat = 241.5 kJ/kg

Mean specific heat, liquid = 1.38 kJ/kg°C

Critical pressure, P_c = 58 bar

Heat loads:

$$\text{Sensible heat (maximum)} = (170 - 0) * 1.38 = 234.6 \text{ kJ/kg}$$

$$\text{Total heat load} = (234.6 + 241.5) * \frac{3165.24}{3600} = 418.60 \text{ kW},$$

add 5 % for heat losses

$$\text{maximum heat load (duty)} = 1.05 * 418.60$$

$$= 436.53 \text{ kW}$$

From Figure 12.1 assume $U = 1000 \text{ W/m}^2\text{°C}$.

Mean temperature difference; both sides isothermal, steam saturation temperature at 0.8 bar = 236.4°C

$$\Delta T_m = 236.4 - 170 = 66.4 \text{ °C}$$

$$\text{Area (outside) required} = \frac{436.53 * 10^3}{1000 * 66.4} = 6.57 \text{ m}^2$$

Select 25 mm i.d., 30 mm o.d. plain U-tubes,

Nominal length 4.8 m (one U-tube)

$$\text{Number of U tubes} = \frac{6.57}{(30 * 10^{-3}) * 3.14 * 4.8} = 15$$

Use square pitch arrangement, pitch = 1.5 * tube o.d.

$$= 1.5 * 30 = 45 \text{ mm}$$

Draw a tube layout diagram, take minimum bend radius

$$1.5 * \text{tube o.d.} = 45 \text{ mm}$$

Proposed layout gives 15 U-tubes, tube outer limit diameter 420 mm.

Boiling coefficient

Use Mostinski's equation:

heat flux, based on estimated area,

$$q = \frac{436.53}{6.57} = 66.44 \text{ kW/m}^2$$

$$h_{nb} = 0.104 * (P_c)^{0.69} * (q)^{0.7} \left[1.8 * \left(\frac{P}{P_c} \right)^{0.17} + 4 * \left(\frac{P}{P_c} \right)^{1.2} + 10 * \left(\frac{P}{P_c} \right)^{10} \right]$$

$$h_{nb} = 0.104 * (58)^{0.69} * (66.44 * 10^3)^{0.7} \left[1.8 * \left(\frac{0.8}{58} \right)^{0.17} + 4 * \left(\frac{0.8}{58} \right)^{1.2} + 10 * \left(\frac{0.8}{58} \right)^{10} \right]$$

$$= 4069.10 * 0.892$$

$$= 3631.32 \text{ W/m}^2\text{°C}$$

Take steam condensing coefficient as $8000 \text{ W/m}^2\text{°C}$, fouling coefficient $5000 \text{ W/m}^2\text{°C}$; fouling coefficient, essentially clean, $10,000 \text{ W/m}^2\text{°C}$.

Tube material will be plain carbon steel, $k_w = 55 \text{ W/m}^2\text{°C}$

$$\frac{1}{U_o} = \frac{1}{h_o} + \frac{1}{h_{do}} + \frac{d_o \cdot \ln \left(\frac{d_o}{d_i} \right)}{2 \cdot K_w} + \frac{d_o}{d_i \cdot h_{id}} + \frac{d_o}{d_i \cdot h_i}$$

$$\frac{1}{U_o} = \frac{1}{3631.32} + \frac{1}{10,000} + \frac{30 * 10^{-3} * \ln \left(\frac{30}{25} \right)}{2 * 55} + \frac{30}{25} \left(\frac{1}{5000} + \frac{1}{8000} \right)$$

$$U_o = 1226.83 \text{ W/m}^2\text{°C}$$

Close enough to original estimate of $1000 \text{ W/m}^2\text{°C}$ for the design to stand.

Myers and Katz (Chem. Eng. Prog. Sym. Ser. 49(5) 107 114) give some data on the boiling on banks of tubes. To compare the value estimate with their values an estimate of the boiling film temperature difference is required:

$$= \frac{1226.83}{3631.32} * 66.4 = 22.43 \text{ °C (72.37°F)}$$

Myers data, extrapolated, gives a coefficient of around $3000 \text{ Btu/h ft}^2\text{°F}$ at a 73°F temperature difference = $17,100 \text{ W/m}^2\text{°C}$, so the estimated value of 3631.32 is certainly on the safe side.

Check maximum allowable heat flux. Use modified Zuber equation

$$\text{Surface tension (estimated)} = 9.7 * 10^{-3} \text{ N/m}$$

$$\rho_L = 550 \text{ kg/m}^3$$

$$\rho_v = \frac{151.16}{22.4} * \frac{273}{(273 + 170)} * 0.8 = 3.32 \text{ kg/m}^3$$

$$N_t = 52$$

For square arrangement $K_b = 0.44$

$$q_{cb} = K_b * \left(\frac{P_t}{d_o}\right) * \left(\frac{\lambda}{\sqrt{N_t}}\right) [\sigma g * (\rho_L - \rho_v) * \rho_v^2]^{0.25}$$

$$q_{cb} = 0.44 * 1.5 * \frac{241.5 * 10^3}{\sqrt{52}} [9.7 * 10^{-3} * 9.81 * (550 - 3.32) * 3.32^2]^{0.25}$$

$$= 1,08,161.30 \frac{W}{m^2}$$

$$= 108 \text{ k W/m}^2$$

Applying a factor of 0.7, maximum flux should not exceed $108 * 0.7 = 75.6 \text{ kW/m}^2$.

Actual flux of 66.44 kW/m^2 is well below maximum allowable.

Layout

From tube sheet layout $D_b = 420 \text{ mm}$.

Take shell diameter as twice bundle diameter

$$D_s = 2 * 420 = 840 \text{ mm}.$$

Take liquid level as 500 mm from base,

Freeboard = $840 - 500 = 340 \text{ mm}$, satisfactory.

From sketch, width at liquid level = 0.8 m .

Surface area of liquid = $0.8 * 2.4 = 1.9 \text{ m}^2$.

$$\text{Vapour velocity at surface} = \frac{5000}{3600} * \frac{1}{3.32} * \frac{1}{1.9} = 0.220 \text{ m/s}$$

Maximum allowable velocity

$$v_v < 0.2 * \left[\frac{\rho_L - \rho_v}{\rho_v}\right]^{\frac{1}{2}}$$

$$v_v = 0.2 * \left[\frac{550 - 3.32}{3.32}\right]^{\frac{1}{2}} = 2.56 \text{ m/s}$$

So, actual velocity is well below maximum allowable velocity. A smaller shell diameter could be considered.